

Module -2 – CMOS inverter

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Modeling of MOS Transistor using SPICE

SPICE (Simulation Program with Integrated Circuit Emphasis) is a general-purpose circuit simulator which is used very widely both in the microelectronics industry and in educational institutions as an essential computer-aided design (CAD) tool for circuit design.

- ▶ In the fast-advancing field of VLSI design, a sound knowledge of physical models used to describe the behavior of transistors and knowledge of various device parameters are essential for performing detailed circuit simulations and for optimizing design.
- ▶ SPICE has three built-in MOSFET models –
- ▶ LEVEL 1 (MOS1) is described by a square-law. current-voltage characteristic.
- ▶ LEVEL 2 (MOS2) is a detailed analytical MOSFET model.
- ▶ LEVEL 3 (MOS3) is a semi-empirical model.
- ▶ Both MOS2 and MOS3 include second-order effects such as the short-channel threshold voltage, subthreshold conduction, scattering-limited velocity saturation, and charge-controlled capacitances.

Basic Concepts

The source-substrate and the drain-substrate junctions, which are reverse-biased under normal operating conditions, are represented by ideal diodes in this equivalent circuit.

- ▶ The parasitic source and drain resistances
- ▶ are represented by the resistors R_D and R_S ,
- ▶ respectively, connected between the drain
- ▶ current source and the respective terminals.

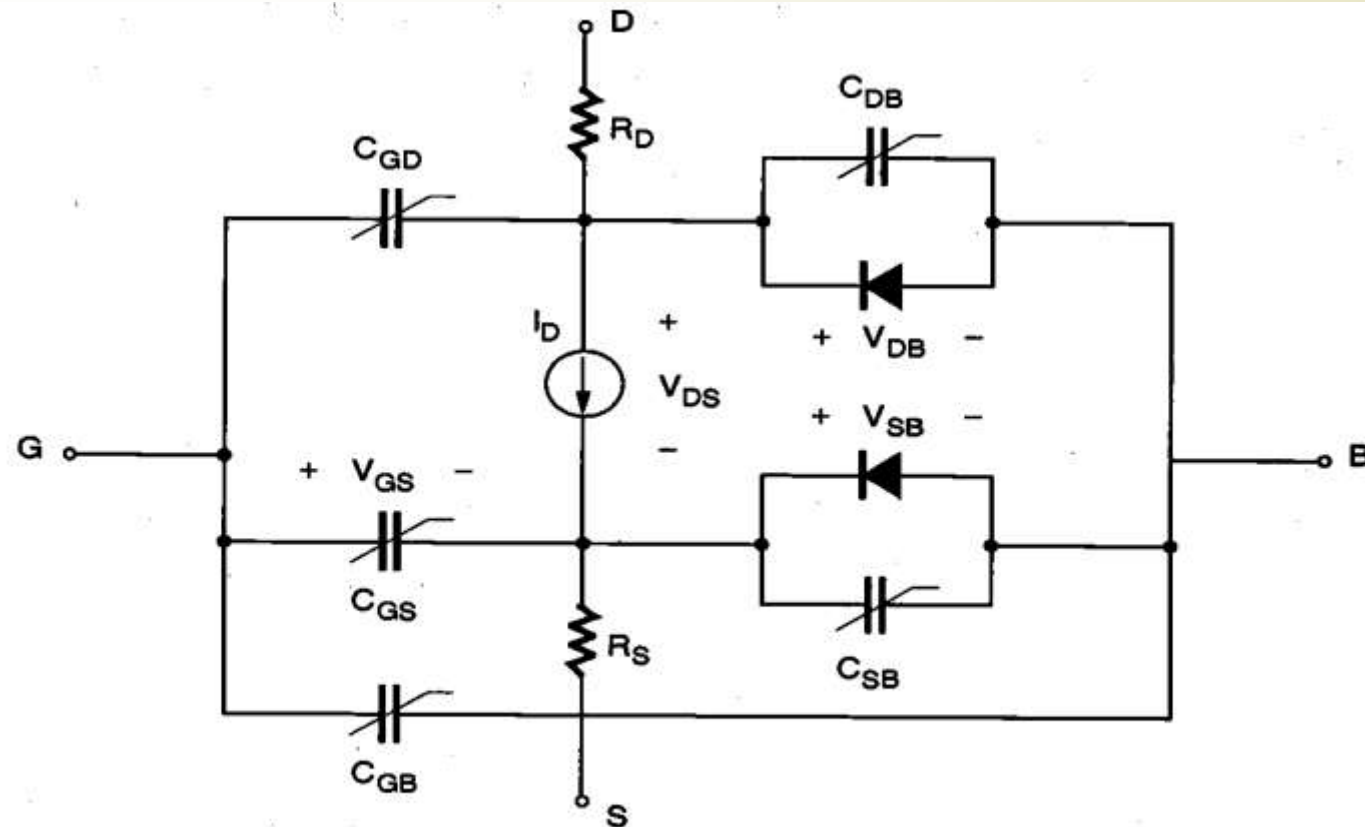


Figure Equivalent circuit structure of the LEVEL 1 MOSFET model in SPICE.

The LEVEL 1 Model Equations

- ▶ The LEVEL 1 model is the simplest current-voltage description of the MOSFET, which is basically the GCA-based quadratic model.
- ▶ The equations used for the LEVEL 1 n-channel MOSFET model in SPICE are as follows.

Linear Region

$$I_D = \frac{k'}{2} \cdot \frac{W}{L_{eff}} \cdot [2 \cdot (V_{GS} - V_T) V_{DS} - V_{DS}^2] \cdot (1 + \lambda V_{DS}) \quad \begin{array}{l} \text{for } V_{GS} \geq V_T \\ \text{and } V_{DS} < V_{GS} - V_T \end{array}$$

Saturation Region

$$I_D = \frac{k'}{2} \cdot \frac{W}{L_{eff}} \cdot (V_{GS} - V_T)^2 \cdot (1 + \lambda \cdot V_{DS}) \quad \begin{array}{l} \text{for } V_{GS} \geq V_T \\ \text{and } V_{DS} \geq V_{GS} - V_T \end{array}$$

The LEVEL 2 Model Equations

- ▶ To obtain a more accurate model for the drain current, it is necessary to eliminate some of the simplifying assumptions made in the original GCA analysis.
- ▶ Specifically, the bulk depletion charge must be calculated by taking into account its dependence on the channel voltage. current-voltage characteristics in model 2 can be obtained as –

$$I_D = \frac{k'}{(1 - \lambda \cdot V_{DS})} \cdot \frac{W}{L_{eff}} \cdot \left\{ \left(V_{GS} - V_{FB} - |2\phi_F| - \frac{V_{DS}}{2} \right) \cdot V_{DS} - \frac{2}{3} \cdot \gamma \cdot \left[(V_{DS} - V_{BS} + |2\phi_F|)^{3/2} - (-V_{BS} + |2\phi_F|)^{3/2} \right] \right\}$$

- ▶ The saturation voltage V_{DSAT} can be calculated as

$$V_{DSAT} = V_{GS} - V_{FB} - |2\phi_F| + \gamma^2 \cdot \left(1 - \sqrt{1 + \frac{2}{\gamma^2} \cdot (V_{GS} - V_{FB})} \right)$$

- ▶ Saturation mode current

$$I_D = I_{Dsat} \cdot \frac{1}{(1 - \lambda \cdot V_{DS})}$$

The LEVEL 3 Model Equations

- ▶ The majority of the LEVEL 3 model equations are empirical. The aims of using empirical equations instead of analytical models are both to improve the accuracy of the model and to limit the complexity of the calculations.
- ▶ The drain current in the linear region
- ▶ is given by the following equation:

$$I_D = \mu_s \cdot C_{ox} \cdot \frac{W}{L_{eff}} \cdot \left(V_{GS} - V_T - \frac{1 + F_B}{2} \cdot V_{DS} \right) \cdot V_{DS}$$

where

$$F_B = \frac{\gamma \cdot F_s}{4 \cdot \sqrt{|2\phi_F| + V_{SB}}} + F_n$$

- ▶ the empirical parameter F_B expresses the dependence of the bulk depletion charge on the three-dimensional geometry of the MOSFET. Here, the parameters V_T , F_s , and μ_s are influenced by the short-channel effects, while the parameter F_n is influenced by the narrow-channel effects.

$$\mu_s = \frac{\mu}{1 + \theta \cdot (V_{GS} - V_T)}$$

CMOS Inverter

- ▶ The inverter is truly the nucleus of all digital designs. Once its operation and properties are clearly understood, designing more intricate structures such as NAND gates, adders, multipliers, and microprocessors is greatly simplified.
- ▶ The analysis of inverters can be extended to explain the behavior of more complex gates such as NAND, NOR, or XOR, which in turn form the building blocks for modules such as multipliers and processors.
- ▶ **Static CMOS Inverter**
- ▶ Inverter operation is readily understood with the aid of the simple switch model of the MOS transistor.
- ▶ The transistor is nothing more than a switch with an infinite OFF resistance (for $V_{GS} < V_T$) and a finite on-resistance (for $V_{GS} > V_T$) .

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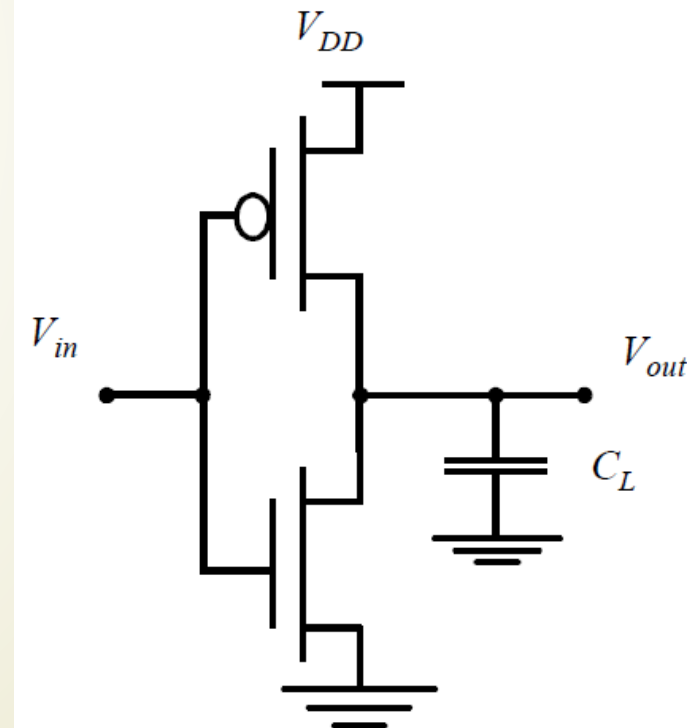
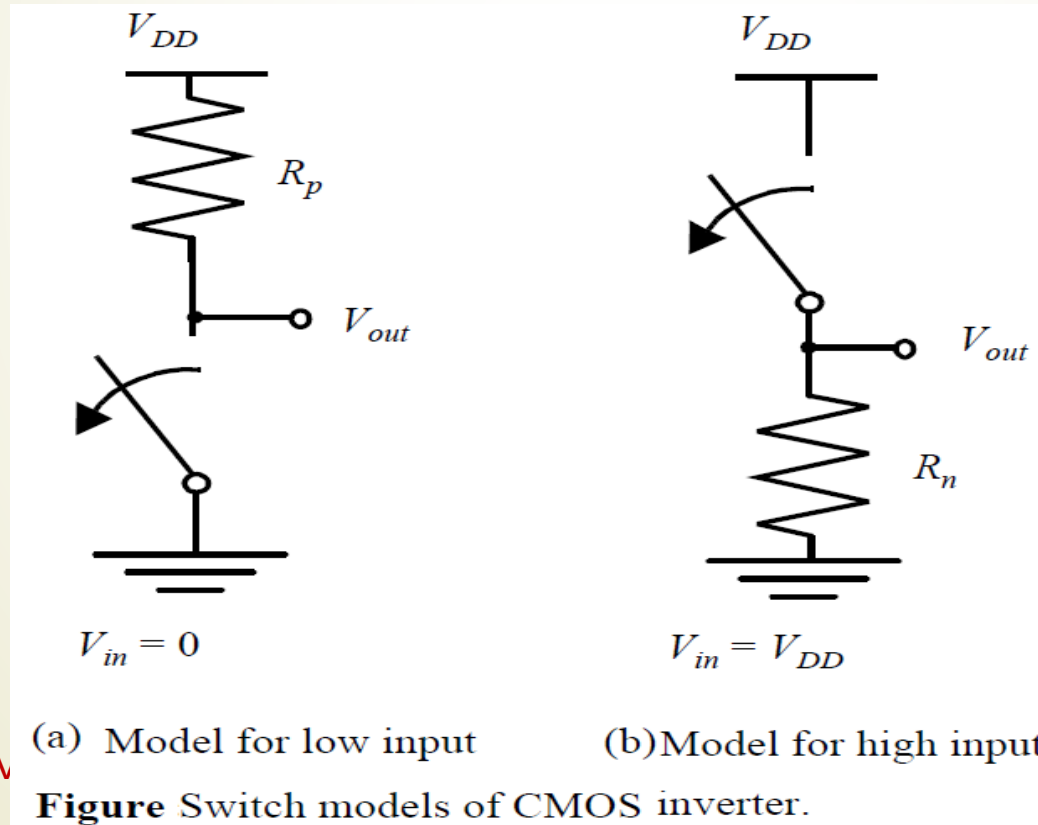


Figure Static CMOS inverter

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- When V_{in} is low (0 V), NMOS and PMOS transistors are OFF and ON, respectively. The equivalent circuit of Figure (a) shows that a path exists between V_{DD} and V_{out} , yielding a high output voltage.
- When V_{in} is high and equal to V_{DD} , the NMOS transistor is ON, while the PMOS is OFF. This yields the equivalent circuit of Figure (b). A direct path exists between V_{out} and the ground node,



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➤ Important properties of static CMOS inverter

- **High noise margins** -The high and low output levels equal V_{DD} and **GND**, respectively; in other words, the voltage swing is equal to the supply voltage. This results in high noise margins.
- **Ratioless Logic** -The logic levels are not dependent upon the relative device sizes, so that the transistors can be minimum size. Gates with this property are called ratioless.
- **High Input Impedance** - The input impedance of the CMOS inverter is extremely high, as the gate of an MOS transistor is a virtually perfect insulator and draws no dc input current. Since the input node of the inverter only connects to transistor gates. A single inverter can theoretically drive an infinite number of gates in static case. however, increasing the fan-out also increases the propagation delay in transient case.
- **Low output Impedance** -In steady state, there always exists a path with finite resistance between the output and either V_{DD} or **GND**. A well-designed CMOS inverter, therefore, has a low output impedance, which makes it less sensitive to noise and disturbances.
- **static power** -No direct path exists between the supply and ground rails under steady-state operating conditions (this is, when the input and outputs remain constant). The absence of current flow (ignoring leakage currents) means that the gate does not consume much static power.

CMOS Inverter voltage-transfer characteristic

- ▶ The nature and the form of the voltage-transfer characteristic (VTC) can be graphically deduced by superimposing the current characteristics of the NMOS and the PMOS devices.
- ▶ It requires that the I-V curves of the NMOS and PMOS devices are transformed onto a common coordinate set. The PMOS I-V relations can be translated into this variable space by the following relations.

$$I_{DSp} = -I_{DSn}$$

$$V_{GSn} = V_{in} \ ; \ V_{GSp} = V_{in} - V_{DD} \quad \text{.....(1)}$$

$$V_{DSn} = V_{out} \ ; \ V_{DSp} = V_{out} - V_{DD}$$

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- ▶ The load-line curves of the PMOS device are obtained by a mirroring around the x -axis and a horizontal shift over V_{DD} .

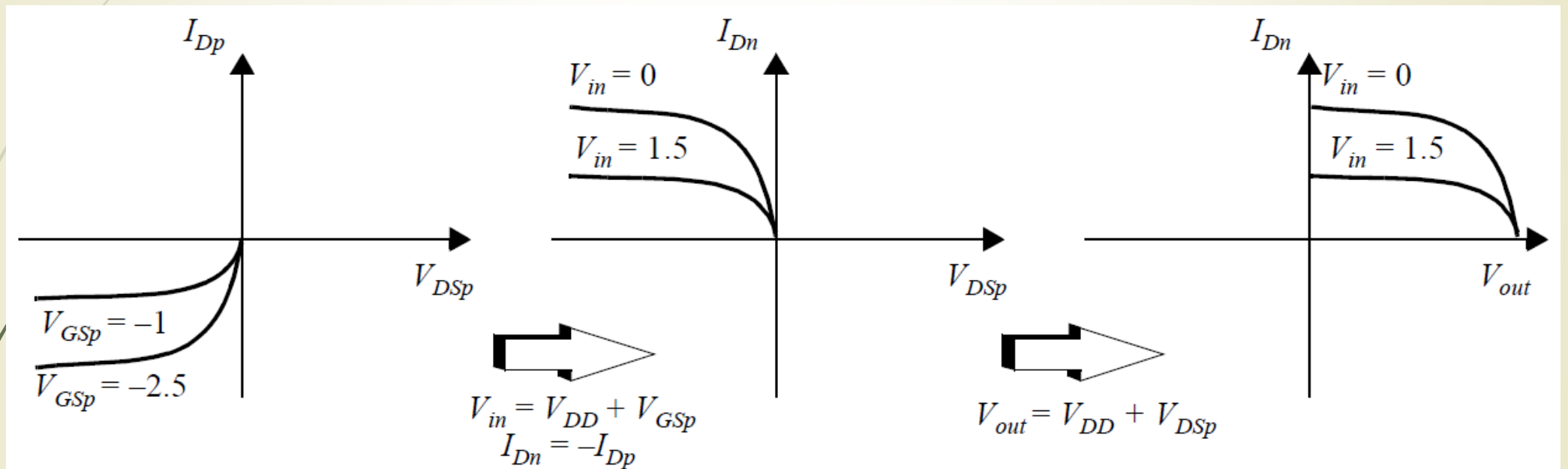


Figure Transforming PMOS I - V characteristic to a common coordinate set (assuming $V_{DD} = 2.5$ V).

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- ▶ The resulting load lines are shown in figure. For a dc operating points to be valid, the currents through the NMOS and PMOS devices must be equal.
- ▶ Graphically, this means that the dc points must be located at the intersection of corresponding load lines.
- ▶ A number of those points (for $V_{in} = 0, 0.5, 1, 1.5, 2$, and 2.5 V) are marked on the graph.
- ▶ As can be observed, all operating points are
- ▶ located either at the high or low output levels.

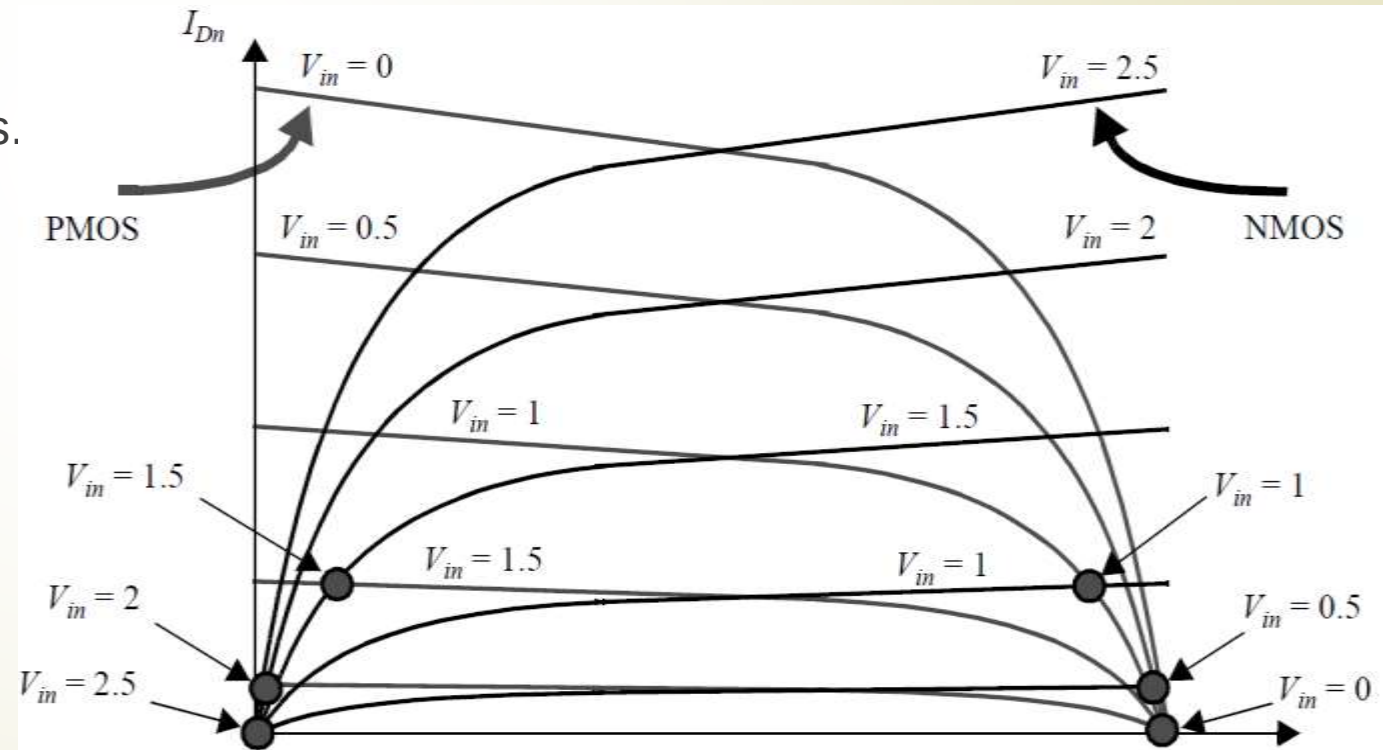


Figure Load curves for NMOS and PMOS transistors of the static CMOS inverter ($V_{DD} = 2.5$ V). The dots represent the dc operation points for various input voltages.

Continue....

- ▶ In that operation region, a small change in the input voltage results in a large output variation. All these observations translate into the VTC doped silicon substrate.
- ▶ Before going into the analytical details of the operation of the CMOS inverter, a qualitative analysis of the transient behavior of the gate.
- ▶ The gate response time is simply determined by the
- ▶ time it takes to charge the capacitor C_L through
- ▶ The resistor R_p . we learned that the propagation delay of
- ▶ such a network is proportional to the its time constant
- ▶ $R_p C_L$.
- ▶ Hence, a fast gate is built either by keeping the output
- ▶ capacitance small or by decreasing the on-resistance
- ▶ of the transistor. The latter is achieved by
- ▶ increasing the W/L ratio of the device.

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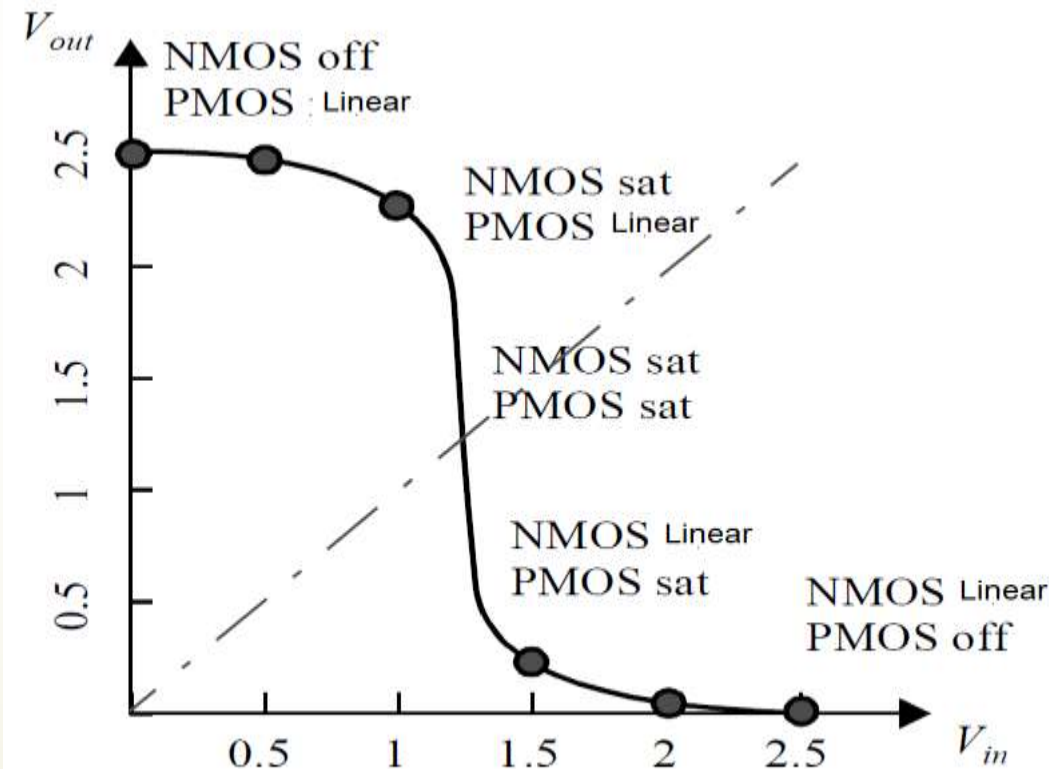


Figure VTC of static CMOS inverter,

Switching Threshold

- ▶ The switching threshold, V_M , is defined as the point where $V_{in} = V_{out}$. Its value can be obtained graphically from the intersection of the VTC with the line given by $V_{in} = V_{out}$. In this region, both PMOS and NMOS are always saturated.
- ▶ We solve the case where the supply voltage is high so that the devices can be assumed to be velocity-saturated ($V_{DSAT} < V_M - V_T$). We furthermore ignore the channel length modulation effects.

$$k_n V_{DSATn} \left(V_M - V_{Tn} - \frac{V_{DSATn}}{2} \right) + k_p V_{DSATp} \left(V_M - V_{DD} - V_{Tp} - \frac{V_{DSATp}}{2} \right) = 0 \quad \text{.....(2)}$$

- ▶ Solving for V_M yields

$$V_M = \frac{\left(V_{Tn} + \frac{V_{DSATn}}{2} \right) + r \left(V_{DD} + V_{Tp} + \frac{V_{DSATp}}{2} \right)}{1 + r} \quad \text{with} \quad r = \frac{k_p V_{DSATp}}{k_n V_{DSATn}} = \frac{v_{satp} W_p}{v_{satn} W_n} \quad \text{.....(3)}$$

continue.....

➤ For large values of V_{DD} Eq. (3) can be simplified $V_M \approx \frac{rV_{DD}}{1+r}$ (4)

➤ we can derive the required ratio of PMOS versus NMOS transistor sizes such that the switching threshold is set to a desired value V_M . When using this expression, please make sure that the assumption that both devices are velocity-saturated still holds for the chosen operation point.

➤
$$\frac{(W/L)_p}{(W/L)_n} = \frac{k'_n V_{DSATn} (V_M - V_{Tn} - V_{DSATn}/2)}{k'_p V_{DSATp} (V_{DD} - V_M + V_{Tp} + V_{DSATp}/2)}$$
(5)

CMOS Inverter - Noise Margin

- **Definition** - The amount of noise voltage which the inverter can reject without effecting the output.
- If higher is the value of noise margin better is the capability of inverter.
- In given figure this output signal of first inverter is being transmitted to the second inverter input via an interconnect, which is prone to signal noise.
- Consequently, If the input voltage of the second inverter is smaller than V_{OL} this signal will be interpreted correctly as a logic "0" input by the second inverter. On the other hand, if the input voltage becomes larger than V_{IL} as a result of noise, then it may not be interpreted correctly by the inverter. Thus, we conclude that V_{IL} is the maximum allowable voltage at the input of the second inverter, which is low enough to ensure a logic "1" output.
- Same way it is applicable
- For second and third
- Inverter.

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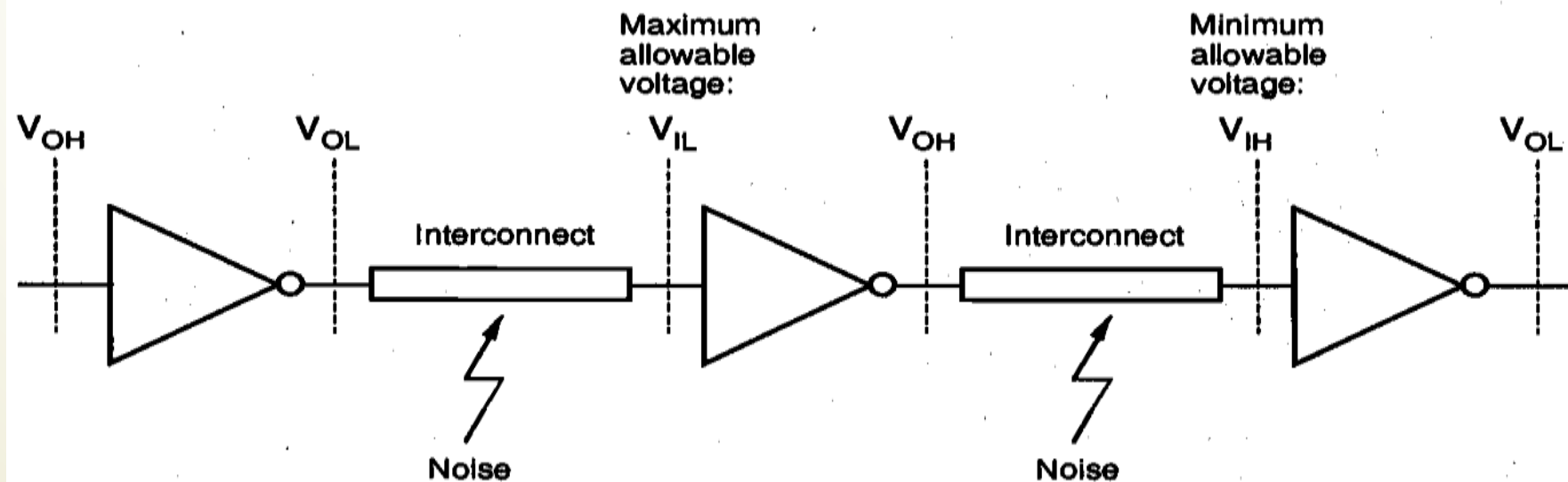


Figure Propagation of digital signals under the influence of noise

Continued...

- So from previous observation noise tolerances for digital circuits, called noise margins and denoted by **NM**.
- Two noise margins will be defined: the noise margin for low signal levels (**NM_L**) and the noise margin for high signal levels (**NM_H**) .

$$NM_L = V_{IL} - V_{OL}$$

$$NM_H = V_{OH} - V_{IH}$$

- The shaded areas indicate the valid regions of the input and output voltages, and the noise margins are shown as the amount of variation in the signal levels that can be allowed while the signal is transmitted from the output of one gate to the input of the next gate.

- Note that there is a voltage range between **V_{IL}** and **V_{IH}** corresponding to input voltage values. The region between **V_{IL}** and **V_{IH}** is called uncertain region or alternatively the transition region.

.....(6)

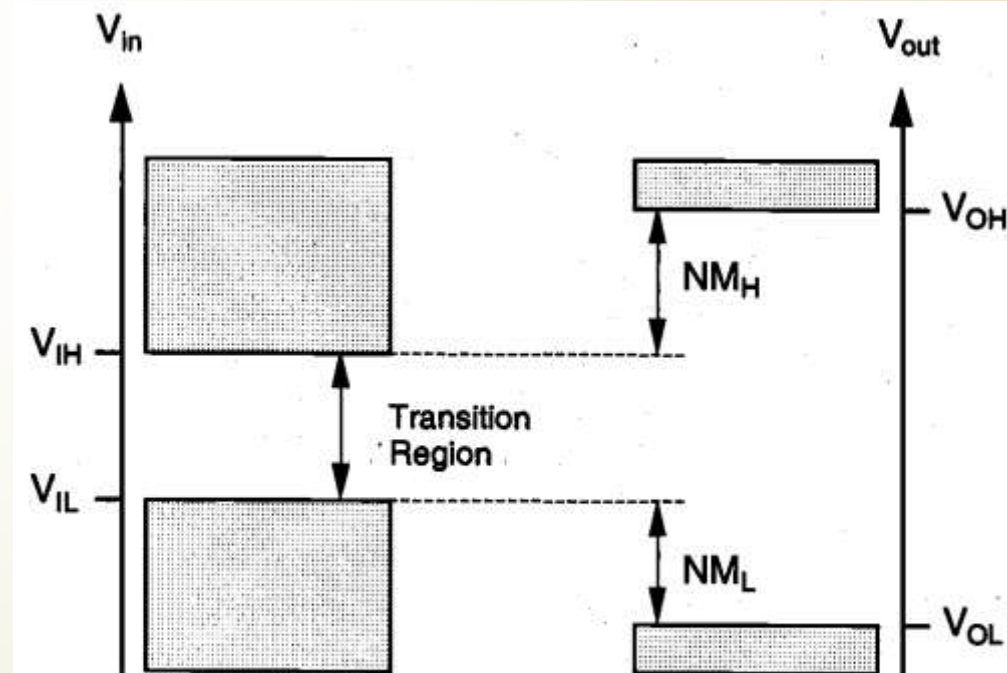


Figure Definition of noise margins NM_L and NM_H .

Calculation of Voltages in CMOS Inverter

- CMOS Inverter circuit voltage configuration is shown in figure. According to circuit diagram voltages can be written as –

$$V_{GS,n} = V_{in} \quad \dots\dots (7)$$

$$V_{DS,n} = V_{out}$$

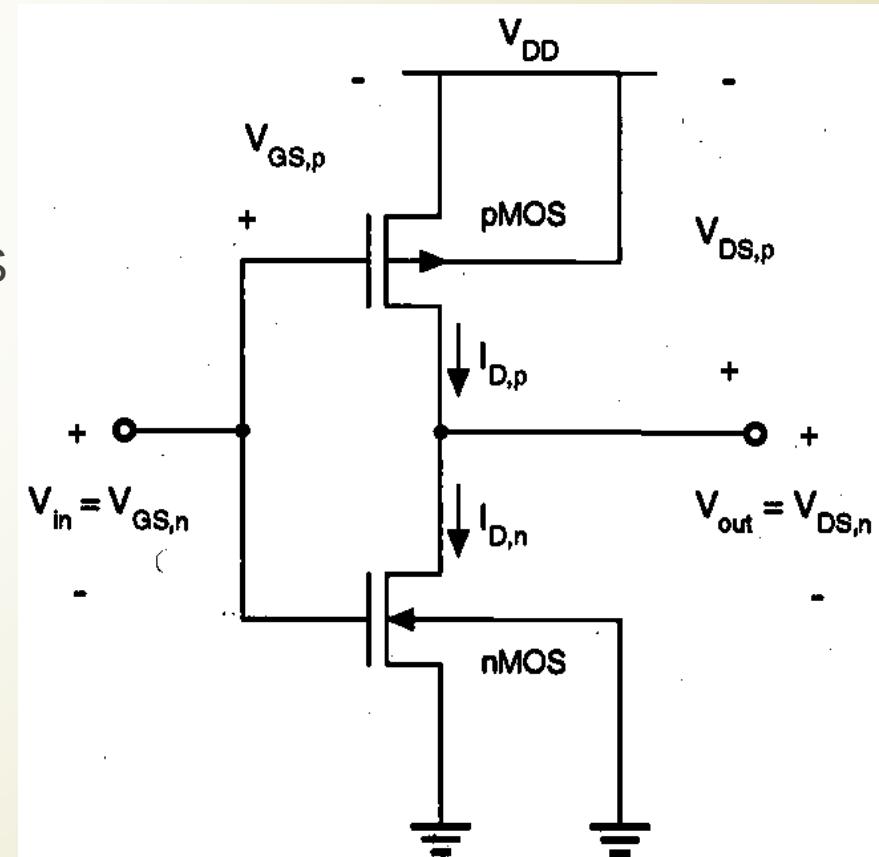
$$V_{GS,p} = -(V_{DD} - V_{in})$$

$$V_{DS,p} = -(V_{DD} - V_{out}) \quad \dots\dots(8)$$

- When $V_{in} < V_{T0,n}$ the nMOS transistor is cut-OFF. The PMOS transistor is on, operating in the linear region. Since the drain currents of both transistors are approximately equal to zero.

$$I_{D,n} = I_{D,p} = 0 \quad \dots\dots(9)$$

$$V_{out} = V_{OH} = V_{DD} \quad \dots\dots(10)$$



Continued....

- when the input voltage exceeds $(V_{DD} + V_{T0,p})$ the PMOS transistor is turned off and NMOS transistor is operating in the linear region. Consequently, the output voltage of the circuit is

$$V_{out} = V_{OL} = 0 \quad \text{.....(11)}$$

- The NMOS transistor operates in saturation if $V_{in} > V_{T0,n}$ if the following condition is satisfied

$$V_{DS,n} \geq V_{GS,n} - V_{T0,n} \Leftrightarrow V_{out} \geq V_{in} - V_{T0,n} \quad \text{.....(12)}$$

- The PMOS transistor operates in saturation if $V_{in} < (V_{DD} + V_{T0,p})$

$$V_{DS,p} \leq V_{GS,p} - V_{T0,p} \Leftrightarrow V_{out} \leq V_{in} - V_{T0,p} \quad \text{.....(13)}$$

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- Here, we identify five distinct regions, labeled A through E, each corresponding to a different set of operating conditions

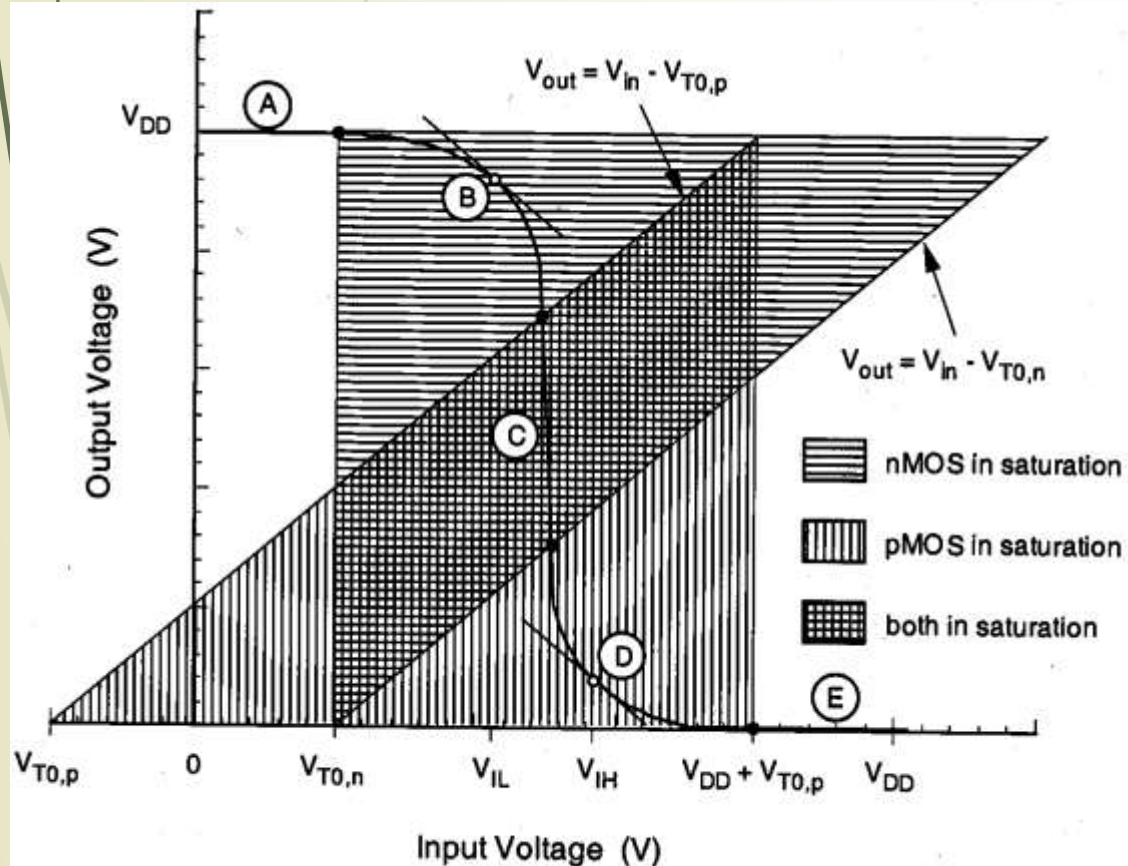


Figure Operating regions of the nMOS and the pMOS transistors.

Region	V_{in}	V_{out}	nMOS	pMOS
A	$< V_{T0,n}$	V_{OH}	cut-off	linear
B	V_{IL}	high $\approx V_{OH}$	saturation	linear
C	V_{th}	V_{th}	saturation	saturation
D	V_{IH}	low $\approx V_{OL}$	linear	saturation
E	$> (V_{DD} + V_{T0,p})$	V_{OL}	linear	cut-off

Continue....

- We already know that the drain current $I_{D,n}$ of the NMOS transistor is a function of the voltages $V_{GS,n}$ and $V_{DS,n}$. The NMOS drain current is also a function of the inverter input and output voltages V_{in} and V_{out} .

$$I_{D,n} = f(V_{in}, V_{out})$$

- Similarly for PMOS

$$I_{D,p} = f(V_{in}, V_{out})$$

- **Calculation of V_{IL}**

- By definition, the slope of the VTC is equal to (-1), $dV_{out}/dV_{in} = -1$. when the input voltage is $V_{in} = V_{IL}$.
- Note that in this case, the NMOS transistor operates in saturation while the PMOS transistor operates in the linear region. From $I_{D,n} = I_{D,p}$ we obtain the following current equation:

$$\frac{k_n}{2} \cdot (V_{GS,n} - V_{T0,n})^2 = \frac{k_p}{2} \cdot [2 \cdot (V_{GS,p} - V_{T0,p}) \cdot V_{DS,p} - V_{DS,p}^2] \quad \dots(14)$$

- Using equation (7) and (8), this expression can be rewritten as –

$$\frac{k_n}{2} \cdot (V_{in} - V_{T0,n})^2 = \frac{k_p}{2} \cdot [2 \cdot (V_{in} - V_{DD} - V_{T0,p}) \cdot (V_{out} - V_{DD}) - (V_{out} - V_{DD})^2] \quad \dots\dots\dots(15)$$

Continue.....

- Differentiate equation (15) with respect to V_{in} .

$$k_n \cdot (V_{in} - V_{T0,n}) = k_p \cdot \left[(V_{in} - V_{DD} - V_{T0,p}) \cdot \left(\frac{dV_{out}}{dV_{in}} \right) + (V_{out} - V_{DD}) - (V_{out} - V_{DD}) \cdot \left(\frac{dV_{out}}{dV_{in}} \right) \right] \quad \dots(16)$$

- Substitute $V_{in} = V_{IL}$ and $dV_{out}/dV_{in} = -1$ in equation (16)

$$k_n \cdot (V_{IL} - V_{T0,n}) = k_p \cdot (2V_{out} - V_{IL} + V_{T0,p} - V_{DD}) \quad \dots(17)$$

- Voltage V_{IL} can now found as a function V_{out}

$$V_{IL} = \frac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R} \quad \dots(18)$$

Where

$$k_R = \frac{k_n}{k_p}$$

Calculation of V_{IH}

- When the input voltage is equal to V_{IH} , the NMOS transistor operates in the linear region, and the PMOS transistor operates in saturation. Apply KCL to the output node $I_{D,n} = I_{D,p}$.

- $$\frac{k_n}{2} \cdot [2 \cdot (V_{GS,n} - V_{T0,n}) \cdot V_{DS,n} - V_{DS,n}^2] = \frac{k_p}{2} \cdot (V_{GS,p} - V_{T0,p})^2 \quad \dots\dots(19)$$

- Using equation (7) and (8), this expression can be rewritten as

- $$\frac{k_n}{2} \cdot [2 \cdot (V_{in} - V_{T0,n}) \cdot V_{out} - V_{out}^2] = \frac{k_p}{2} \cdot (V_{in} - V_{DD} - V_{T0,p})^2 \quad \dots(20)$$

- Now, differentiate both sides of (20) with respect to V_{in}

$$k_n \cdot \left[(V_{in} - V_{T0,n}) \cdot \left(\frac{dV_{out}}{dV_{in}} \right) + V_{out} - V_{out} \cdot \left(\frac{dV_{out}}{dV_{in}} \right) \right] = k_p \cdot (V_{in} - V_{DD} - V_{T0,p}) \quad \dots\dots(21)$$

- Substitute $V_{in} = V_{IH}$ and $dV_{out}/dV_{in} = -1$ in equation (20)

$$k_n \cdot (-V_{IH} + V_{T0,n} + 2V_{out}) = k_p \cdot (V_{IH} - V_{DD} - V_{T0,p}) \quad \dots\dots(22)$$

Continued....

- ▶ The critical voltage V_{IH} can now be found as a function of V_{out} , as follows –

$$V_{IH} = \frac{V_{DD} + V_{T0,p} + k_R \cdot (2 V_{out} + V_{T0,n})}{1 + k_R} \quad \text{.....(23)}$$

- ▶ Calculation of V_{lth}

- ▶ The inverter threshold voltage is defined as $V_{th} = V_{in} = V_{out}$. For $V_{in} = V_{out}$, both transistors are expected to be in saturation mode. Hence we can write KCL equation.

$$\frac{k_n}{2} \cdot (V_{GS,n} - V_{T0,n})^2 = \frac{k_p}{2} \cdot (V_{GS,p} - V_{T0,p})^2 \quad \text{.....(23)}$$

- ▶ Using equation (7) and (8) , this expression can be rewritten as

$$\frac{k_n}{2} \cdot (V_{in} - V_{T0,n})^2 = \frac{k_p}{2} \cdot (V_{in} - V_{DD} - V_{T0,p})^2 \quad \text{..... (24)}$$

Continued....

- ▶ The correct solution for V_{in} for this equation is

$$V_{in} \cdot \left(1 + \sqrt{\frac{k_p}{k_n}} \right) = V_{T0,n} + \sqrt{\frac{k_p}{k_n}} \cdot (V_{DD} + V_{T0,p}) \quad \dots\dots(25)$$

- ▶ Finally, the inverter threshold (switching threshold) voltage V_{th} is found as

$$V_{th} = \frac{V_{T0,n} + \sqrt{\frac{1}{k_R}} \cdot (V_{DD} + V_{T0,p})}{\left(1 + \sqrt{\frac{1}{k_R}} \right)} \quad \dots\dots(26)$$

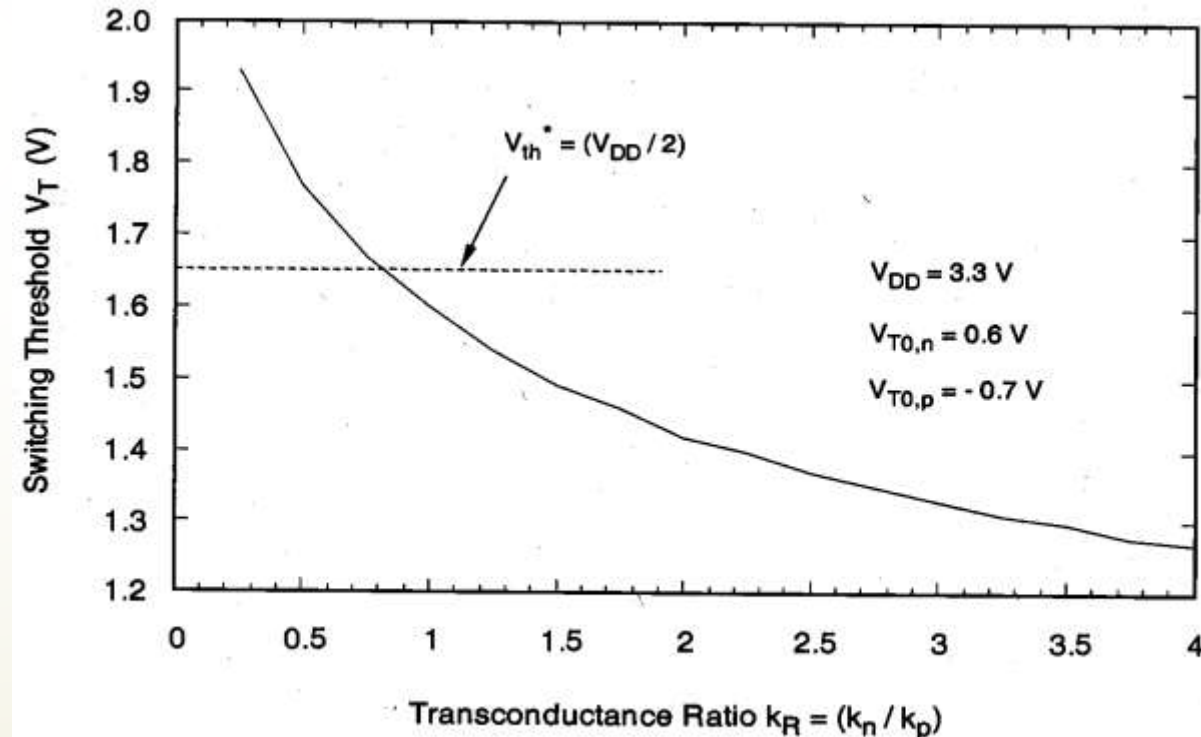


Figure Variation of the inversion threshold voltage as a function of k_R .

Design of CMOS Inverters

- ▶ The inverter threshold voltage V_{th} was identified as one of the most important parameters that characterize the steady-state input-output behavior of the CMOS inverter circuit.
- ▶ The problem of designing a CMOS inverter can be reduced to setting the inverter threshold to a desired voltage value. The corresponding ratio k_R can be found as follows. reorganizing (26) yields.

$$\sqrt{\frac{1}{k_R}} = \frac{V_{th} - V_{T0,n}}{V_{DD} + V_{T0,p} - V_{th}} \quad \dots\dots(27)$$

- ▶ Solve the k_R that is required to achieve the given V_{th} .

$$k_R = \frac{k_n}{k_p} = \left(\frac{V_{DD} + V_{T0,p} - V_{th}}{V_{th} - V_{T0,n}} \right)^2 \quad \dots\dots(28)$$

- ▶ Recall that the switching threshold voltage of an ideal inverter is defined as

$$V_{th,ideal} = \frac{1}{2} \cdot V_{DD} \quad \dots\dots(29)$$

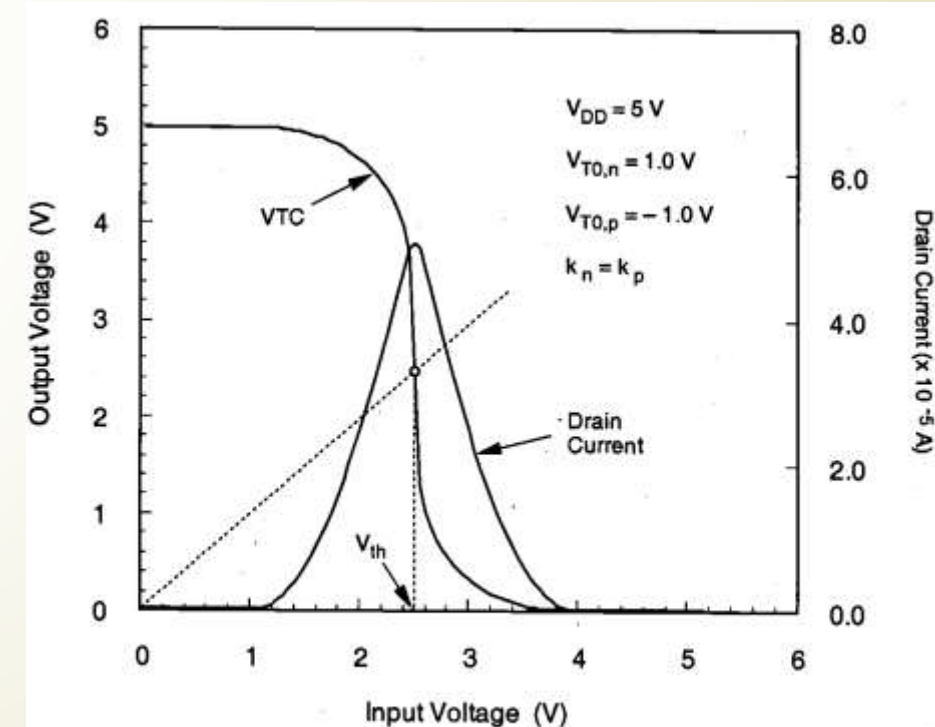


Figure Typical VTC and the power supply current of a CMOS inverter circuit.

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- Substitute the value from (29) in (28).

$$\left(\frac{k_n}{k_p} \right)_{ideal} = \left(\frac{0.5 V_{DD} + V_{T0,p}}{0.5 V_{DD} - V_{T0,n}} \right)^2 \quad \dots(30)$$

- We can achieve completely symmetric input-output characteristics by setting the threshold voltages as
- $V_{T0} = V_{T0,n} = |V_{T0,p}|$. This reduces equation (30)

$$\left(\frac{k_n}{k_p} \right)_{symmetric\ inverter} = 1 \quad \dots (31)$$

- Note that the ratio k_R is defined as

$$\frac{k_n}{k_p} = \frac{\mu_n C_{ox} \cdot \left(\frac{W}{L} \right)_n}{\mu_p C_{ox} \cdot \left(\frac{W}{L} \right)_p} = \frac{\mu_n \cdot \left(\frac{W}{L} \right)_n}{\mu_p \cdot \left(\frac{W}{L} \right)_p} \quad \dots\dots(32)$$

Continued.....

- Assume the same t_{ox} and C_{ox} . The unity-ratio condition (32) for the ideal symmetric inverter.

$$\frac{\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p} = \frac{\mu_p}{\mu_n} \approx \frac{230 \text{ cm}^2/\text{V}\cdot\text{s}}{580 \text{ cm}^2/\text{V}\cdot\text{s}} \quad \dots (32)$$

$$\left(\frac{W}{L}\right)_p \approx 2.5 \left(\frac{W}{L}\right)_n \quad \dots (33)$$

The VTCs of three CMOS inverter circuits with different k_R ratios are shown in Fig. It can be seen clearly that the inverter threshold voltage V_{th} shifts to lower values with increasing k_R ratio.

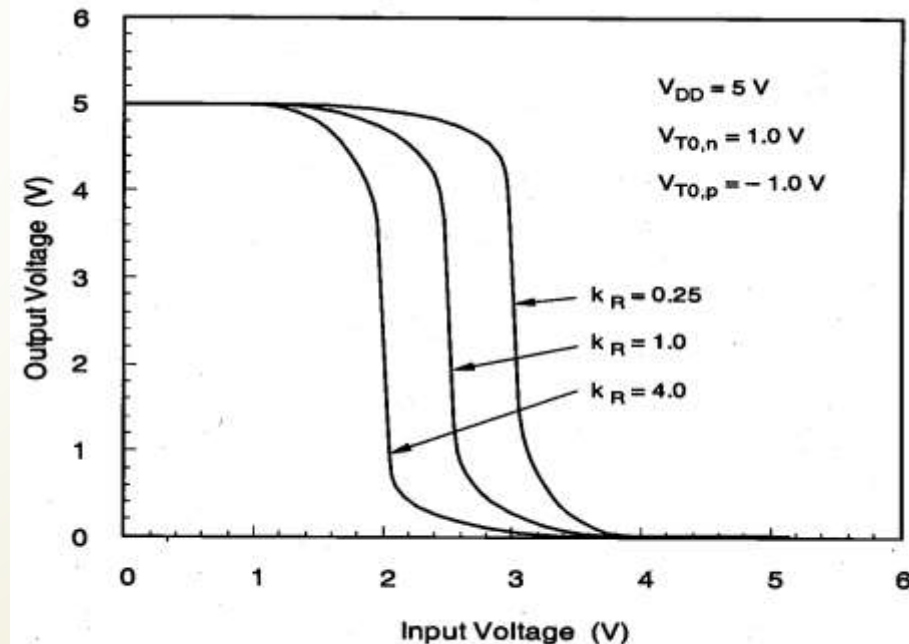
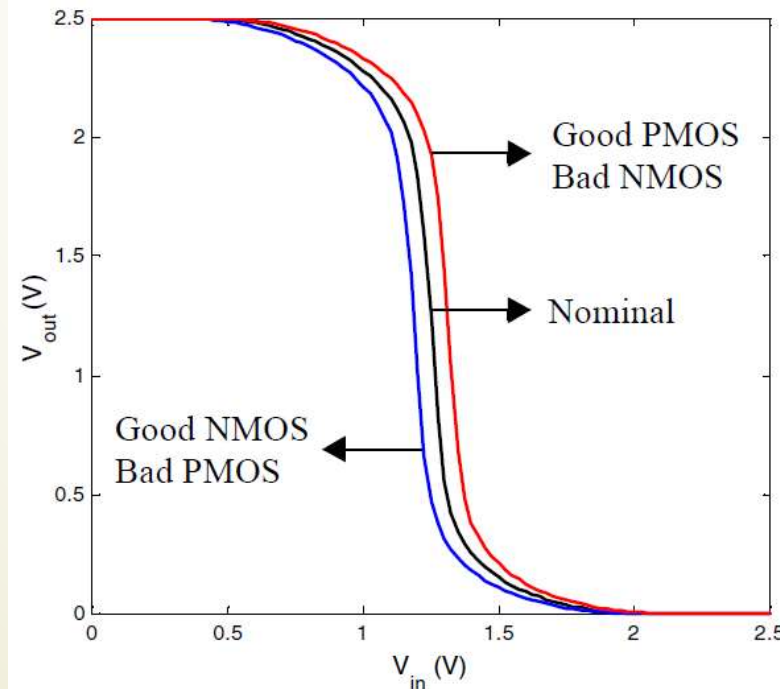


Figure Voltage transfer characteristics of three CMOS inverters, with different nMOS-to- pMOS ratios.

Problem 1

- Consider a CMOS inverter circuit with the following parameters:

$$\begin{aligned}V_{DD} &= 3.3 \text{ V} \\V_{T0,n} &= 0.6 \text{ V} \\V_{T0,p} &= -0.7 \text{ V} \\k_n &= 200 \mu\text{A/V}^2 \\k_p &= 80 \mu\text{A/V}^2\end{aligned}$$

- Calculate the noise margins of the circuit. Notice that the CMOS inverter being considered here has $k_R \neq 2.5$ and $V_{T0,n} \neq |V_{T0,p}|$. hence, it is not a symmetric inverter.

Solution

➤ From equation (18)

$$V_{IL} = \frac{2V_{out} + V_{T0,p} - V_{DD} + k_R V_{T0,n}}{1 + k_R}$$
$$= \frac{2V_{out} - 0.7 - 3.3 + 1.5}{1 + 2.5} = 0.57V_{out} - 0.71$$

➤ Now substitute this in equation (17)

$$k_n \cdot (V_{IL} - V_{T0,n}) = k_p \cdot (2V_{out} - V_{IL} + V_{T0,p} - V_{DD})$$

$$2.5(0.57V_{out} - 0.71 - 0.6)^2 = 2(0.57V_{out} - 0.71 - 3.3 + 0.7)(V_{out} - 3.3) - (V_{out} - 3.3)^2$$

$$0.66V_{out}^2 + 0.05V_{out} - 6.65 = 0$$

$$V_{out} = 3.14 \text{ V}$$

➤ Now we can calculate the value in V_{IL}

$$V_{IL} = 0.57 \cdot 3.14 - 0.71 = \underline{\underline{1.08 \text{ V}}}$$

Continue.....

- To calculate V_{IH} in terms of the output voltage, use

$$\begin{aligned} V_{IH} &= \frac{V_{DD} + V_{T0,p} + k_R \cdot (2 V_{out} + V_{T0,n})}{1 + k_R} \\ &= \frac{3.3 - 0.7 + 2.5(2 V_{out} + 0.6)}{1 + 2.5} = 1.43 V_{out} + 1.17 \end{aligned}$$

- Use equation (20)

$$\begin{aligned} \frac{k_n}{2} \cdot [2 \cdot (V_{in} - V_{T0,n}) \cdot V_{out} - V_{out}^2] &= \frac{k_p}{2} \cdot (V_{in} - V_{DD} - V_{T0,p})^2 \\ 2.5 [2(1.43 V_{out} + 1.17 - 0.6) V_{out} - V_{out}^2] &= (1.43 V_{out} - 1.43)^2 \\ 2.61 V_{out}^2 + 6.94 V_{out} - 2.04 &= 0 \\ V_{out} &= 0.27 \text{ V} \end{aligned}$$

- we can calculate the critical voltage V_{IH} $V_{IH} = 1.43 \cdot 0.37 + 1.17 = \underline{\underline{1.55 \text{ V}}}$

- we can find the noise margin

$$\begin{aligned} NM_L &= V_{IL} - V_{OL} = 1.08 \text{ V} \\ NM_H &= V_{OH} - V_{IH} = 1.75 \text{ V} \end{aligned}$$

Supply Voltage scaling in CMOS

- ▶ In the following, we will briefly examine the effects of supply voltage scaling, i.e., reduction of V_{DD} , upon the static voltage transfer characteristics of CMOS inverters. overall power
- ▶ dissipation of any digital circuit is a strong function of the supply voltage V_{DD} .

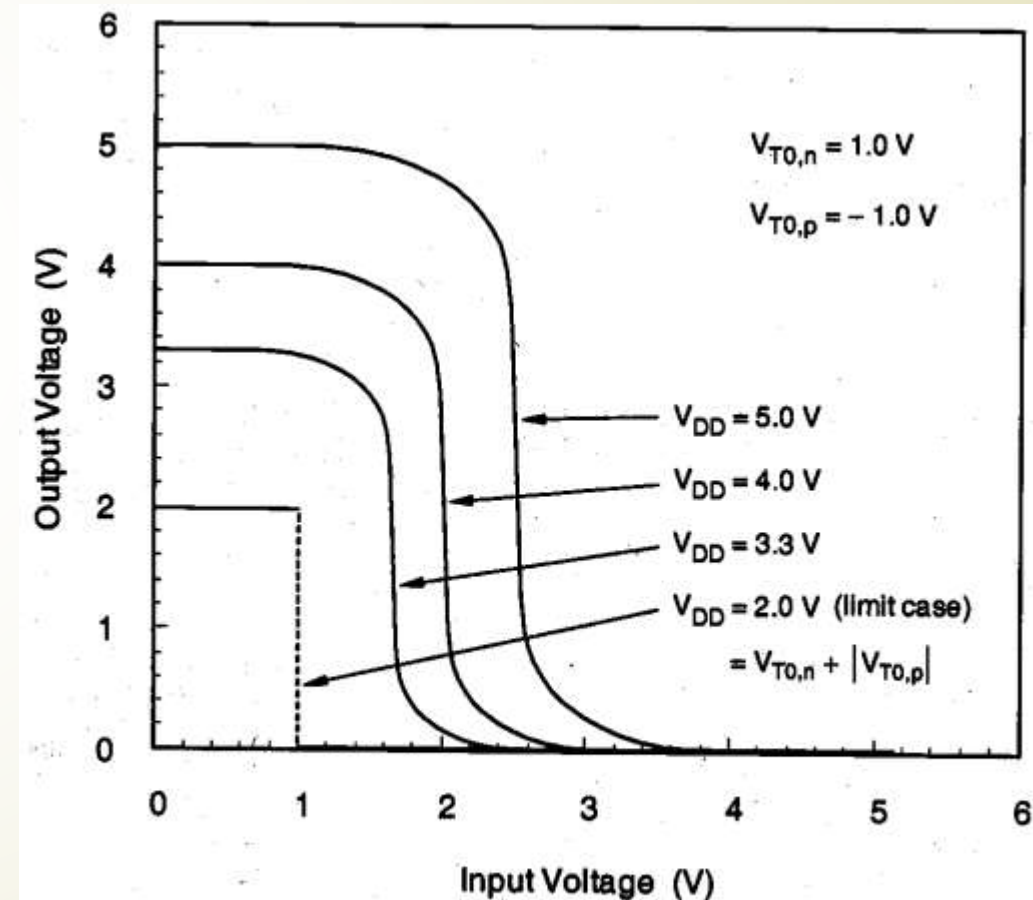


Figure Voltage transfer characteristics of a CMOS inverter, obtained with different power supply voltage levels.

MOS Inverters : Dynamic Characteristics

- ▶ We will investigate the dynamic (time-domain) behavior of the inverter circuits. The switching characteristics of digital integrated circuits and, in particular, of inverter circuits, essentially determine the overall operating speed of digital systems.

- ▶ **Introduction**

- ▶ The capacitances C_{gd} and C_{gs} are primarily due to gate overlap with diffusion, while C_{db} and C_{sb} are voltage-dependent junction capacitances. we also consider the lumped interconnect capacitance C_{int} , which represents the parasitic capacitance contribution of the metal or polysilicon.

- ▶ we first combine the capacitances into an equivalent
- ▶ lumped linear capacitance, connected between the
- ▶ output node of the inverter and the ground.

- ▶ This combined capacitance at the output node
- ▶ will be called the load capacitance C_{load} .

- ▶
$$C_{load} = C_{gd,n} + C_{gd,p} + C_{db,n} + C_{db,p} + C_{int} + C_g \quad \dots(1)$$

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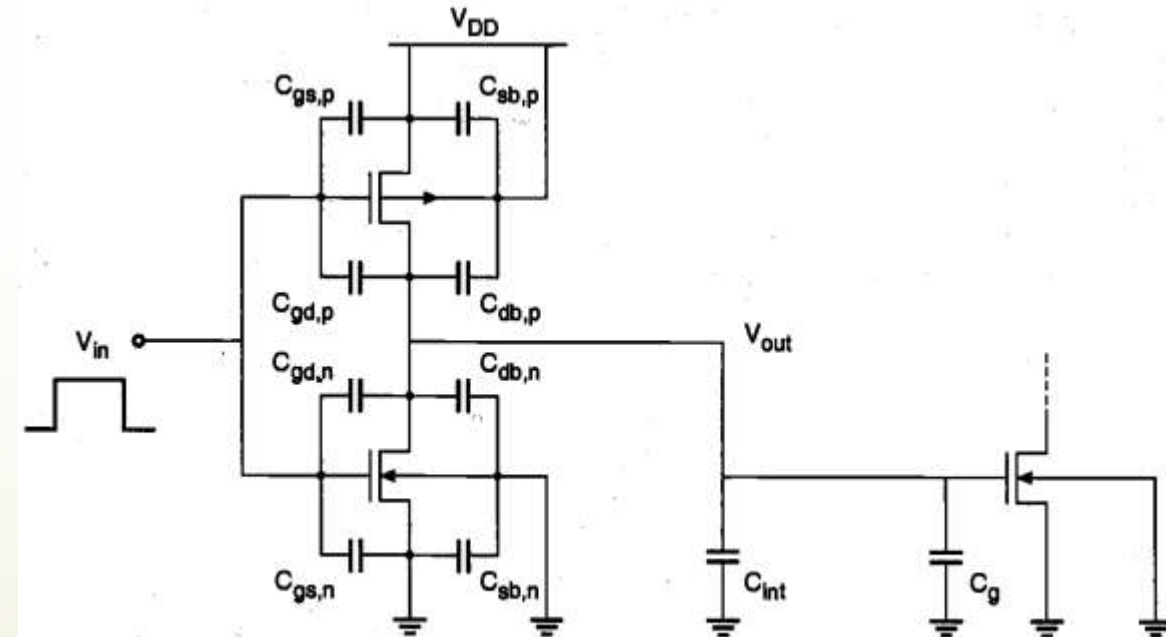


Figure Cascaded CMOS inverter stages.

Continue....

- ▶ In particular, $C_{sb,n}$ and $C_{sb,p}$ have no effect on the transient behavior. The capacitances $C_{gs,n}$ and $C_{gs,p}$ are also not included in (1) because they are connected to the input node. The capacitance terms $C_{db,n}$ and $C_{db,p}$ in (1) are the equivalent junction capacitances calculated for a particular output voltage transition.
- ▶ The first-stage CMOS inverter is shown with the single
- ▶ lumped output load capacitance C_{load} . Now, the problem
- ▶ of analyzing the switching behavior can be handled more
- ▶ easily.
- ▶ In fact, the question of inverter transient response is
- ▶ reduced to finding the charge-up and charge-down times
- ▶ of a single capacitance which is charged
- ▶ and discharged through one transistor.

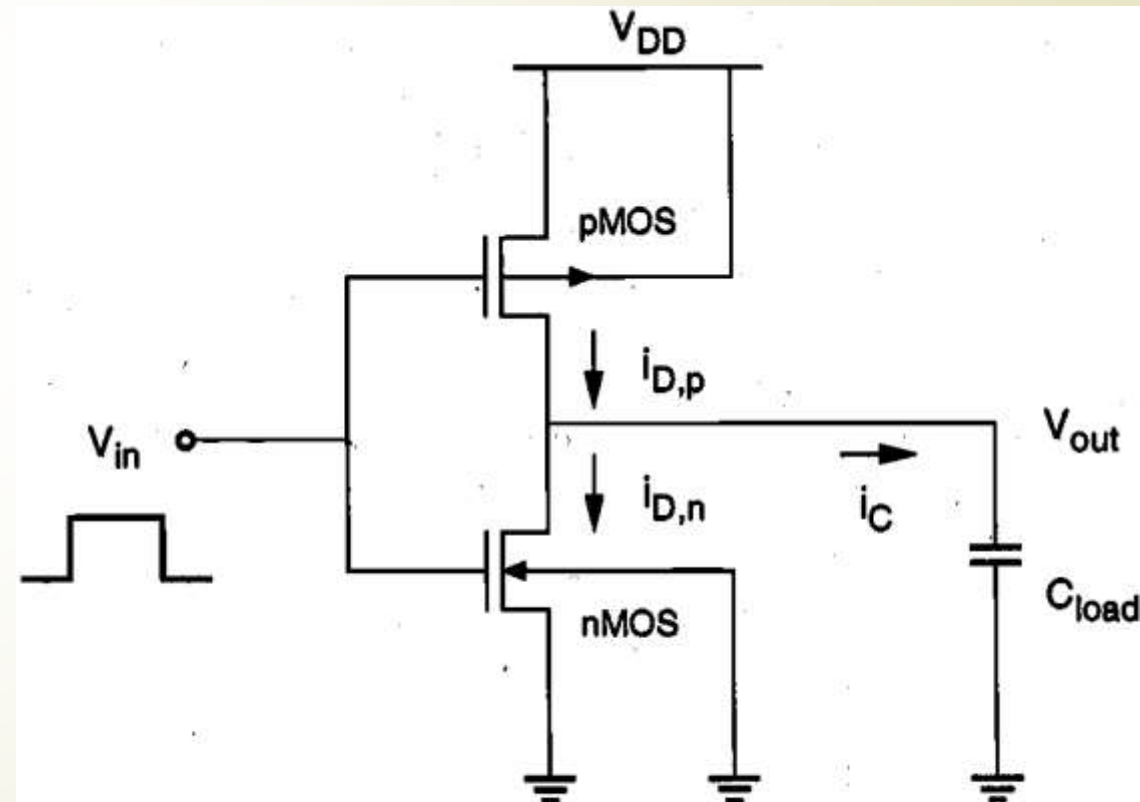


Figure First-stage CMOS inverter with lumped output load capacitance.

Delay-Time Definitions

- ▶ The propagation delay times τ_{PHL} and τ_{PLH} determine the input-to-output signal delay during the high-to-low and low-to-high transitions of the output, respectively.
- ▶ By definition, τ_{PHL} is the time delay between the $V_{50\%}$ -transition of the rising input voltage and the $V_{50\%}$ -transition of the falling output voltage.
- ▶ Similarly τ_{PLH} is defined as the time delay between the $V_{50\%}$ -transition of the falling input voltage and the $V_{50\%}$ -transition of the rising output voltage.
- ▶ To simplify the analysis τ_{PHL} becomes the time required for the output voltage to fall from V_{OH} to the $V_{50\%}$ level.
- ▶ τ_{PLH} becomes the time required for the output voltage to rise from V_{OL} to the $V_{50\%}$ level. The voltage point $V_{50\%}$ is defined as follows

$$V_{50\%} = V_{OL} + \frac{1}{2}(V_{OH} - V_{OL}) = \frac{1}{2}(V_{OL} + V_{OH}) \quad \dots(2)$$

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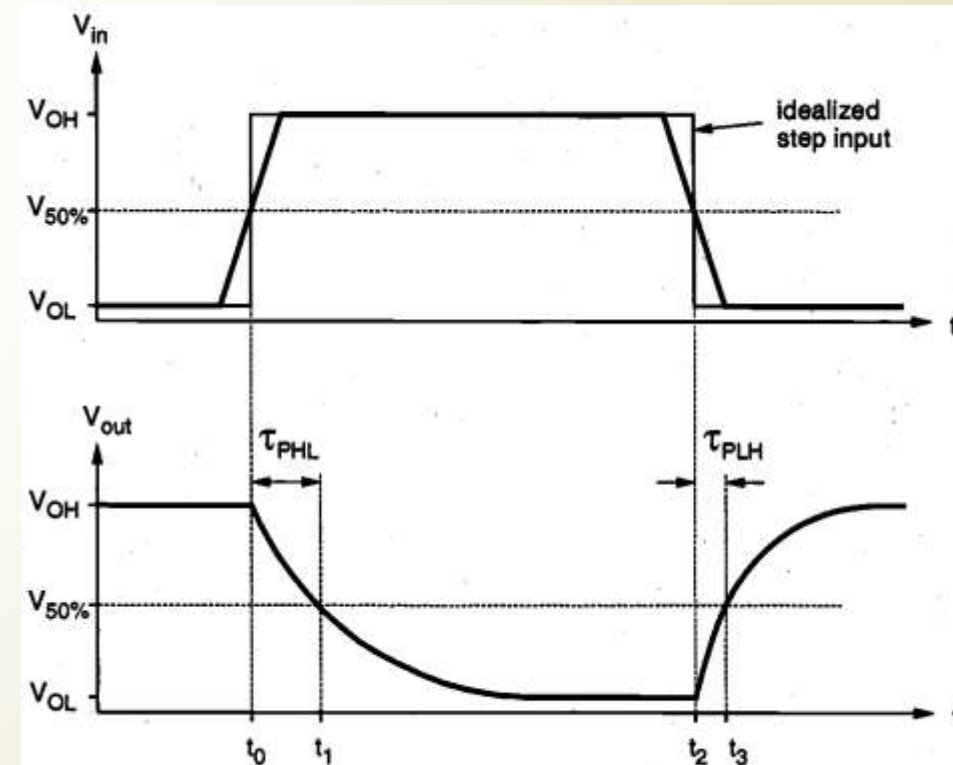


Figure Input and output voltage waveforms of a typical inverter,

Continue....

- Thus, the propagation delay times

$$\begin{aligned}\tau_{PHL} &= t_1 - t_0 \\ \tau_{PLH} &= t_3 - t_2\end{aligned}\quad \text{.....(3)}$$

- The average propagation delay τ_P of the inverter characterizes the average time required for the input signal to propagate through the inverter.

$$\tau_P = \frac{\tau_{PHL} + \tau_{PLH}}{2} \quad \text{.....(4)}$$

- The rise time τ_{rise} is defined here as the time required for the output voltage to rise from the $V_{10\%}$ level to $V_{90\%}$ level.

- The fall time τ_{fall} is defined here as the time required for the output voltage to drop from the $V_{90\%}$ level to $V_{10\%}$ level.

$$\begin{aligned}V_{10\%} &= V_{OL} + 0.1 \cdot (V_{OH} - V_{OL}) \\ V_{90\%} &= V_{OL} + 0.9 \cdot (V_{OH} - V_{OL})\end{aligned}\quad \text{... (5)}$$

$$\begin{aligned}\tau_{fall} &= t_B - t_A \\ \tau_{rise} &= t_D - t_C\end{aligned}\quad \text{.....(6)}$$

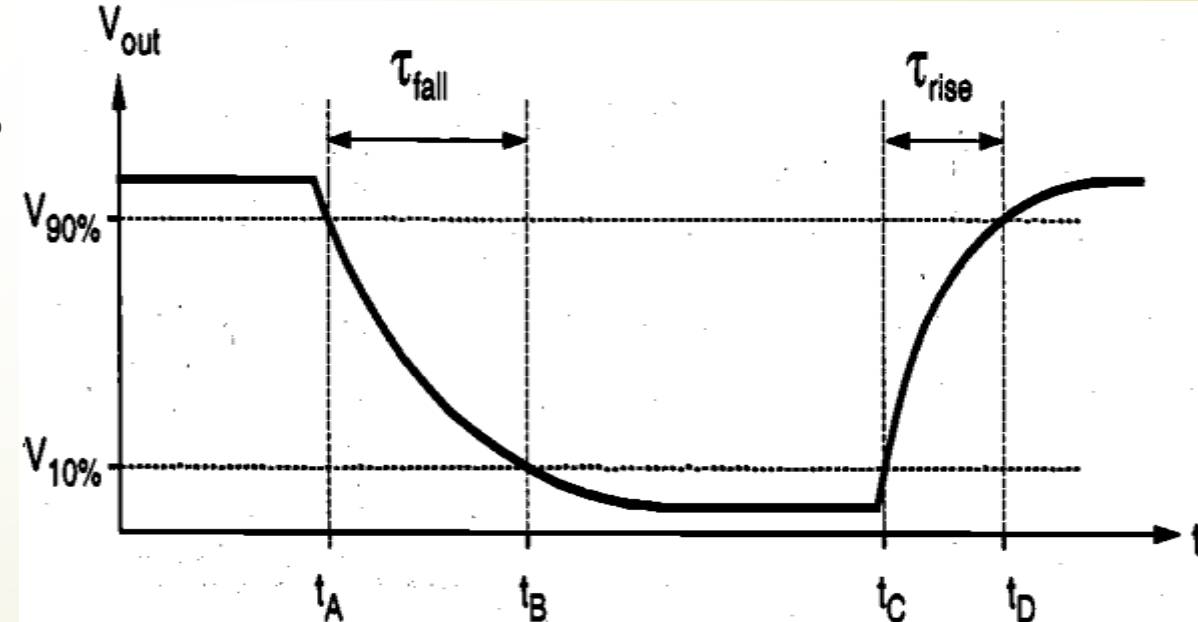


Figure Output voltage rise and fall times.

Performance of CMOS Inverter

- ▶ The qualitative analysis presented earlier concluded that the propagation delay of the CMOS inverter is determined by the time it takes to charge and discharge the load capacitor C_L through the PMOS and NMOS transistors.
- ▶ This observation suggests that getting C_L as small as possible is crucial to the realization of high-performance CMOS circuits.
- ▶ It is hence worthwhile to first study the major components of the load capacitance before embarking onto an in-depth analysis of the propagation delay.
- ▶ **Computing the Capacitances**
- ▶ Manual analysis of MOS circuits where each capacitor is considered individually is virtually impossible.
- ▶ To make the analysis tractable, we assume that all capacitances are lumped together into one single capacitor C_L , located between V_{out} and GND.
- ▶ It includes all the capacitances influencing the transient response of node V_{out} .

Continue.....

- Gate-Drain Capacitance C_{gd12}
- **M1** and **M2** are either in cut-off or in the saturation mode during the first half (up to 50% point) of the output transient. Under these circumstances, the only contributions to C_{gd12} are the overlap capacitances.
- During a low-high or high-low transition, the terminals of the gate drain capacitor are moving in opposite directions. The voltage change over the floating capacitor is hence twice the actual output voltage swing.
- We use the following equation for the gate-drain capacitors. – $C_{gd} = 2C_{GD0}W$

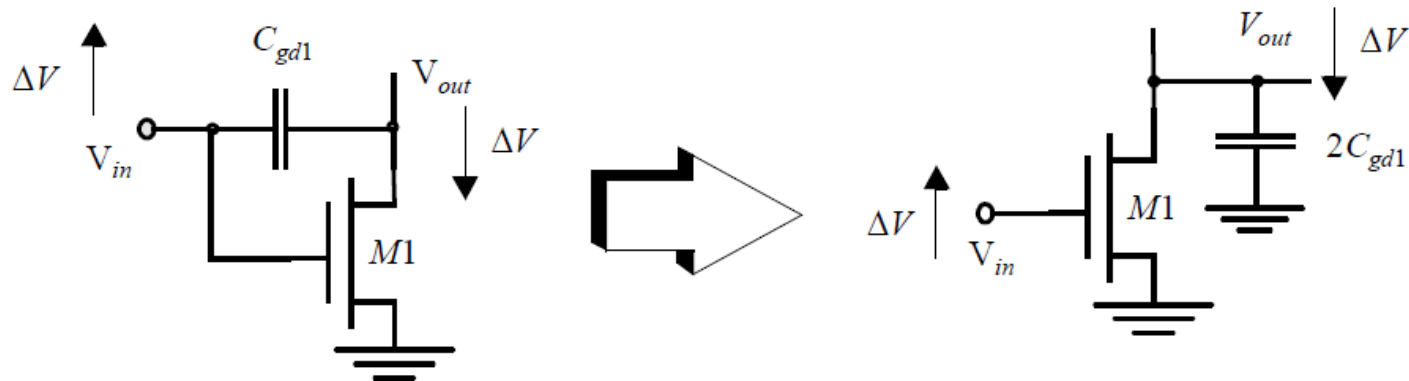


Figure The Miller effect—A capacitor experiencing identical but opposite voltage swings at both its terminals

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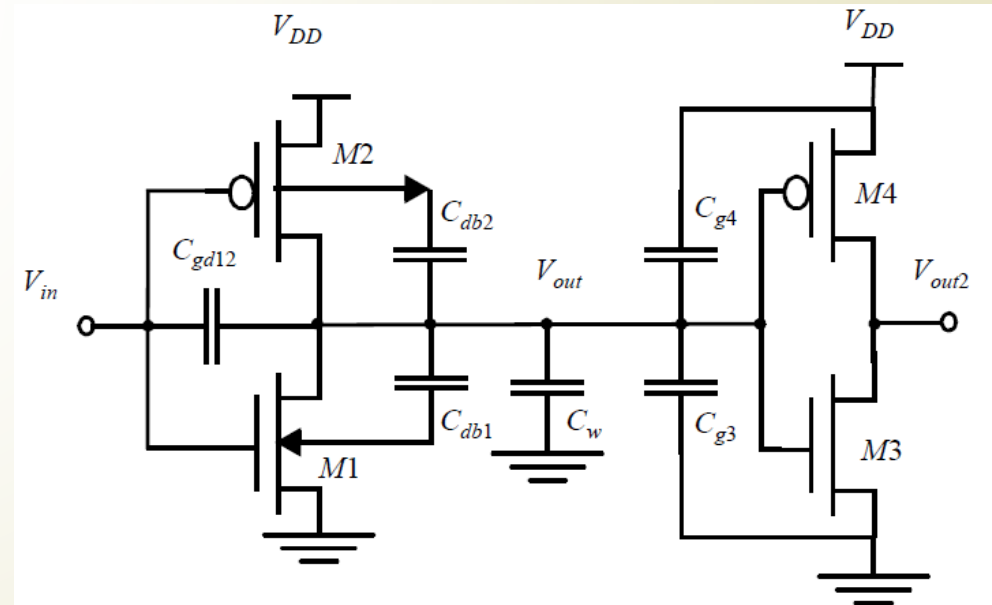


Figure Parasitic capacitances, influencing the transient behavior of the cascaded inverter pair.

Diffusion Capacitances C_{db1} and C_{db2}

- ▶ The capacitance between drain and bulk is due to the reverse-biased **pn**-junction. Such a capacitor is, unfortunately, quite nonlinear and depends heavily on the applied voltage.
- ▶ A multiplication factor K_{eq} is introduced to relate the linearized capacitor to the value of the junction capacitance under zero-bias conditions

$$C_{eq} = K_{eq} C_{j0} \quad \text{.....(7)}$$

- ▶ **Wiring Capacitance C_w**

- ▶ The capacitance due to the wiring depends upon the length and width of the connecting wires, and is a function of the distance of the fanout from the driving gate and the number of fanout gates.

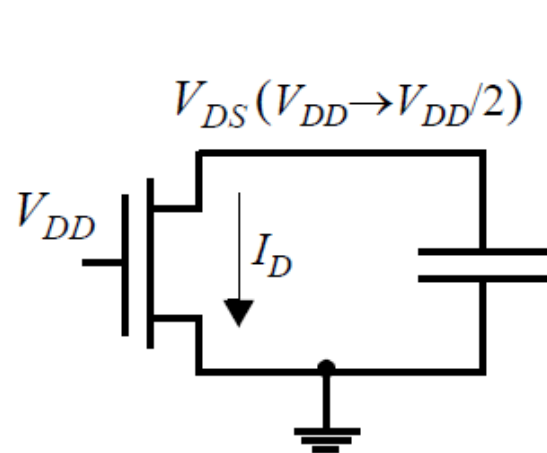
- ▶ **Gate Capacitance of Fanout C_{g3} and C_{g4}**

- ▶ We assume that the fanout capacitance equals the total gate capacitance of the loading gates M3 and M4.

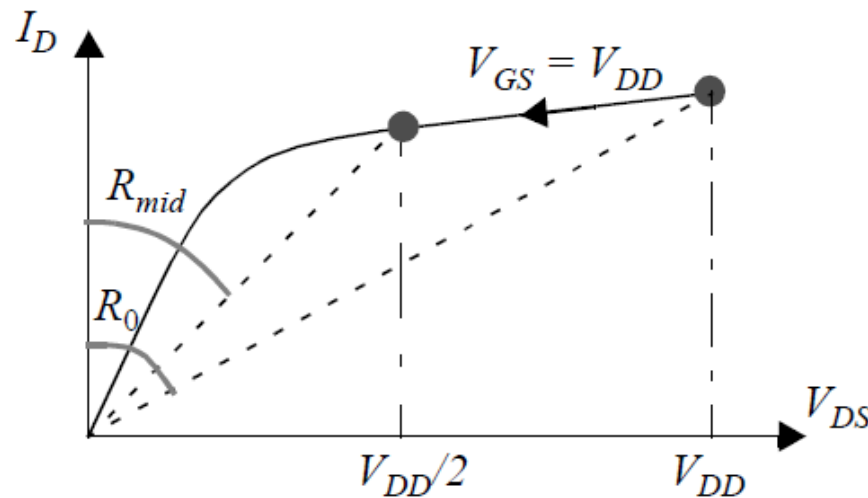
$$\begin{aligned} C_{fanout} &= C_{gate}(\text{NMOS}) + C_{gate}(\text{PMOS}) \\ &= (C_{GSON} + C_{GDO_n} + W_n L_n C_{ox}) + (C_{GSOP} + C_{GDO_p} + W_p L_p C_{ox}) \end{aligned} \quad \text{.....(8)}$$

Propagation Delay: First-Order Analysis

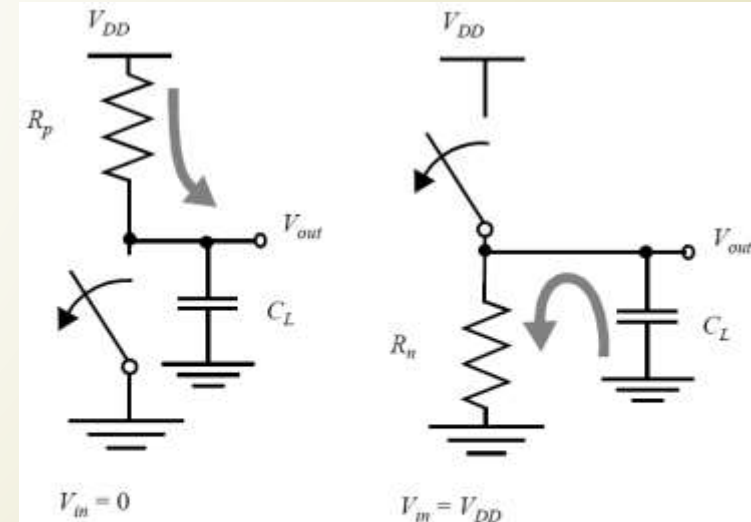
- One way to compute the propagation delay of the inverter is to integrate the capacitor (dis)charge current. with i is the (dis)charging current, V the voltage over the capacitor, and v_1 and v_2 the
 - Initial and final voltage.
- $$t_p = \int_{v_1}^{v_2} \frac{C_L(v)}{i(v)} dv \dots (9)$$
- An exact computation of this equation is intractable, as both $C_L(v)$ and $i(v)$ are nonlinear functions of V .
 - An expression for the average on-resistance of the MOS transistor.
 - This scenario is plotted in for the case of an NMOS discharging a capacitor from V_{DD} to $V_{DD}/2$.



(a) schematic



(b) trajectory traversed on I_D - V_{DS} curve.



(a) Low-to-high

(b) High-to-low

Figure Discharging a capacitor through an NMOS transistor: and I - V trajectory

Figure Switch model of dynamic behavior of static CMOS inverter.

Continue.....

- we can derive the value of the equivalent resistance, which averages the resistance of the device over the interval

$$R_{eq} = \frac{1}{V_{DD}/2} \int_{V_{DD}/2}^{V_{DD}} \frac{V}{I_{DSAT}(1 + \lambda V)} dV \approx \frac{3}{4} \frac{V_{DD}}{I_{DSAT}} \left(1 - \frac{7}{9} \lambda V_{DD} \right) \quad \text{.....(10)}$$

- Deriving the propagation delay of the resulting circuit is now straightforward, and is nothing more than the analysis of a first-order linear RC-network. the propagation delay of such a network for a voltage step at the input is proportional to the time-constant of the network.

- $$t_{pHL} = \ln(2) R_{eqn} C_L = 0.69 R_{eqn} C_L \quad \text{.....(11)}$$

- Similarly, we can obtain the propagation delay for the low-to-high transition

- $$t_{pLH} = 0.69 R_{eqp} C_L \quad \text{.....(12)}$$

- R_{eqp} and R_{eqn} the equivalent on-resistance of the PMOS and NMOS transistor. The overall propagation delay of the inverter is defined as the average of the two values.

- $$t_p = \frac{t_{pHL} + t_{pLH}}{2} = 0.69 C_L \left(\frac{R_{eqn} + R_{eqp}}{2} \right) \quad \text{.....(13)}$$

Continue.....

- Combining Eq. (10) and Eq. (11), and assuming for the time being that the channel-length modulation factor λ is ignorable, yields the following expression for t_{pHL} .

$$t_{pHL} = 0.69 \frac{3 C_L V_{DD}}{4 I_{DSATn}} = 0.52 \frac{C_L V_{DD}}{(W/L)_n k'_n V_{DSATn} (V_{DD} - V_{Tn} - V_{DSATn}/2)} \quad \text{.....(14)}$$

- In the majority of designs, the supply voltage is chosen high enough so that $V_{DD} \gg V_{Tn} + V_{DSATn}/2$. Under these conditions, the delay becomes virtually independent of the supply voltage.

$$t_{pHL} \approx 0.52 \frac{C_L}{(W/L)_n k'_n V_{DSATn}} \quad \text{.....(15)}$$

- From the above analysis we can plot the propagation delay of the inverter as a function of the supply voltage.

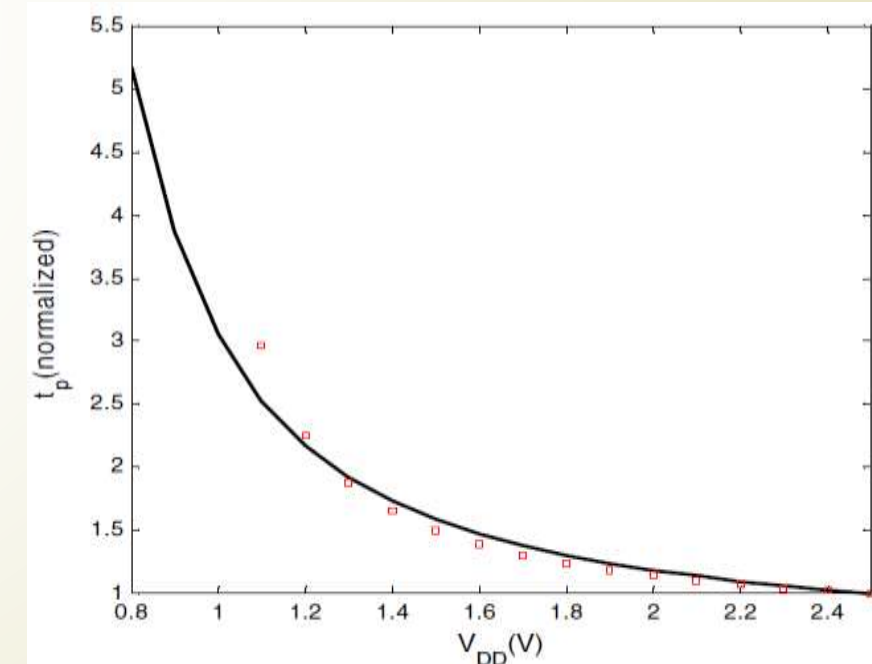


Figure Propagation delay of CMOS inverter as a function of supply voltage

Problem 2

- Q) For a supply voltage of 2.5 V, the normalized on-resistances of NMOS and PMOS transistors equal 13kΩ and 31kΩ, respectively. From the layout, we determine the (W/L) ratios of the transistors to be 1.5 for the NMOS, and 4.5 for the PMOS. capacitor 6.1 fF (nmos) and 6.0 fF (PMOS). Calculate delay?

- Solution

$$t_{pHL} = 0.69 \times \left(\frac{13\text{k}\Omega}{1.5} \right) \times 6.1\text{fF} = 36 \text{ psec}$$

$$t_{pLH} = 0.69 \times \left(\frac{31\text{k}\Omega}{4.5} \right) \times 6.0\text{fF} = 29 \text{ psec}$$

and

$$t_p = \left(\frac{36 + 29}{2} \right) = 32.5 \text{ psec}$$

Propagation Delay from a Design Perspective

- ▶ We deduce that the propagation delay of a gate can be minimized in the following ways
- ▶ **Reduce C_L** : Remember that three major factors contribute to the load capacitance: the internal diffusion capacitance of the gate itself, the interconnect capacitance, and the fanout. Careful layout helps to reduce the diffusion and interconnect capacitances. Good design practice requires keeping the drain diffusion areas as small as possible.
- ▶ **Increase the W/L ratio of the transistors** : Increasing the transistor size also raises the diffusion capacitance and hence C_L . In fact, once the intrinsic capacitance (i.e. the diffusion capacitance) starts to dominate the extrinsic load formed by wiring and fanout, increasing the gate size does not longer help in reducing the delay. In addition, wide transistors have a larger gate capacitance, which increases the fan-out factor of the driving gate and adversely affects its speed.
- ▶ **Increase V_{DD}** : The delay of a gate can be modulated by modifying the supply voltage. This flexibility allows the designer to trade-off energy dissipation for performance. However, increasing the supply voltage above a certain level yields only very minimal improvement. Also, reliability concerns (oxide breakdown, hot-electron effects) enforce firm upper-bounds on the supply voltage in deep sub-micron processes

NMOS/PMOS Ratio

- ▶ We have consistently widened the PMOS transistor so that its resistance matches. The motivation behind this approach is to create an inverter with symmetrical VTC. However, this does not imply that this ratio also yields the minimum overall propagation delay.
- ▶ If symmetry and noise margins are not of prime concern it is actually possible to speed up the inverter by reducing the width of the PMOS device.
- ▶ This optimum ratio can be derived through the following simple analysis. Consider two identical, cascaded CMOS inverters. The load capacitance of the first gate equals approximately

$$C_L = (C_{dp1} + C_{dn1}) + (C_{gp2} + C_{gn2}) + C_W \quad \text{.....(16)}$$

- ▶ When the PMOS devices are made β times larger than the NMOS ones ($\beta = (W/L)_p / (W/L)_n$) all transistor capacitances will scale in approximately the same way or $C_{dp1} \approx \beta C_{dn1}$, and $C_{gp2} \approx \beta C_{gn2}$. Eqn (16) then be rewritten

$$C_L = (1 + \beta)(C_{dn1} + C_{gn2}) + C_W \quad \text{..... (17)}$$

- ▶ An expression for the propagation delay can be derived, based on Eq (13)

$$\begin{aligned} t_p &= \frac{0.69}{2} ((1 + \beta)(C_{dn1} + C_{gn2}) + C_W) \left(R_{eqn} + \frac{R_{eqp}}{\beta} \right) \quad \text{.....(18)} \\ &= 0.345 ((1 + \beta)(C_{dn1} + C_{gn2}) + C_W) R_{eqn} \left(1 + \frac{r}{\beta} \right) \end{aligned}$$

Continue.....

- ▶ $r (= R_{eqp}/R_{eqn})$ represents the resistance ratio of identically-sized PMOS and NMOS transistors. The optimal value of β can be found by setting $\frac{\partial t_p}{\partial \beta}$ to 0, which yields.

$$\beta_{opt} = \sqrt{r \left(1 + \frac{C_w}{C_{dn1} + C_{gn2}} \right)} \quad \text{.....(19)}$$

- ▶ This means that when the wiring capacitance is negligible
- ▶ $(C_{dn1} + C_{gn2} \gg C_w)$, β_{opt} equals $\sqrt{r}$. The surprising result of this
- ▶ analysis is that smaller device sizes (and hence smaller design
- ▶ area) yield a faster design.

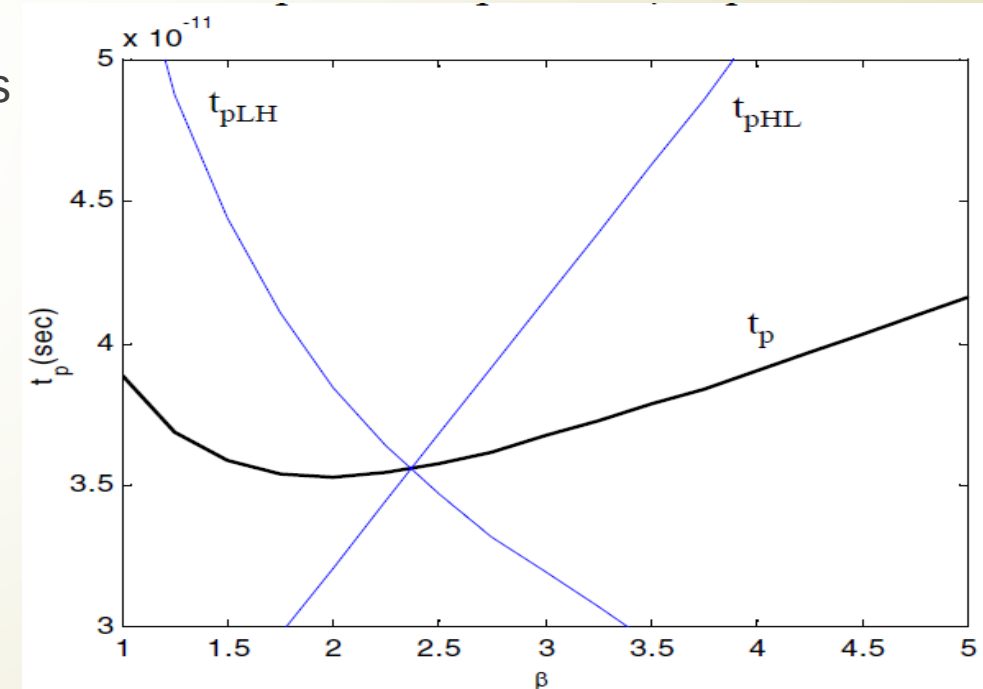


Figure Propagation delay of CMOS inverter as a function of the PMOS/NMOS transistor ratio β .

Sizing Inverters for Performance

- we assume a symmetrical inverter, this is an inverter where PMOS and NMOS are sized such that the rise and fall delays are identical.
- The load capacitance of the inverter can be divided into an intrinsic and an extrinsic component.
 $C_L = C_{int} + C_{ext}$.

- we can express the propagation delay as follows

$$\begin{aligned} t_p &= 0.69 R_{eq} (C_{int} + C_{ext}) \\ &= 0.69 R_{eq} C_{int} (1 + C_{ext}/C_{int}) = t_{p0} (1 + C_{ext}/C_{int}) \end{aligned} \quad \dots(20)$$

- $t_{p0} = 0.69 R_{eq} \cdot C_{int}$ represents the delay of the inverter only loaded by its own intrinsic capacitance ($C_{ext} = 0$) and is called the intrinsic or unloaded delay. The sizing factor S , we consider for inverter to a reference gate—typically a minimum-sized inverter. Hence $C_{int} = S C_{iref}$. The resistance of the gate relates to the reference gate as $R_{eq} = R_{ref}/S$. We can now rewrite the equation (20).

$$\begin{aligned} t_p &= 0.69 (R_{ref}/S) (S C_{iref}) (1 + C_{ext}/(S C_{iref})) \\ &= 0.69 R_{ref} C_{iref} \left(1 + \frac{C_{ext}}{S C_{iref}} \right) = t_{p0} \left(1 + \frac{C_{ext}}{S C_{iref}} \right) \end{aligned} \quad \dots(21)$$

Continue....

- ▶ This leads to two important conclusions:
- ▶ (1) The intrinsic delay of the inverter t_{p0} is independent of the sizing of the gate, and is purely determined by technology and inverter layout.
- ▶ (2) Making S infinitely large yields the maximum obtainable performance gain, eliminating the impact of any external load, and reducing the delay to the intrinsic one.

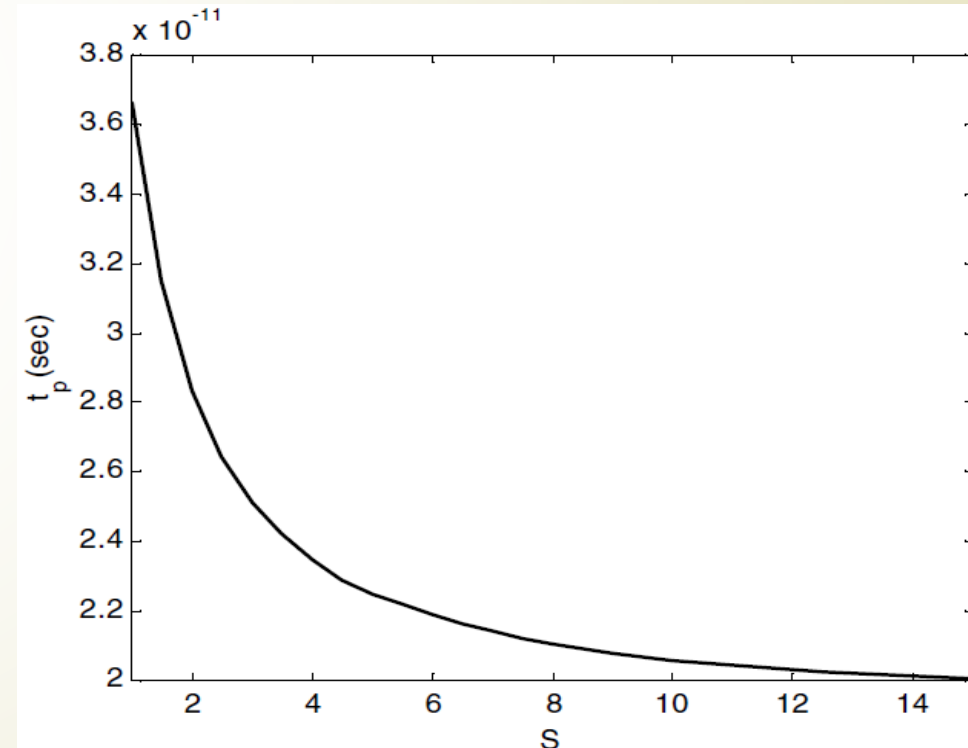
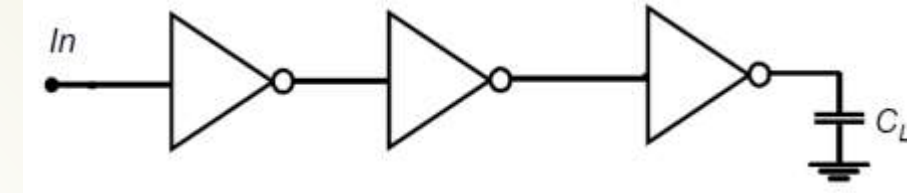


Figure Increasing inverter performance by sizing the NMOS and PMOS transistor with an identical factor S for a fixed fanout

Sizing of inverter chain

- For an inverter the load capacitance can be divided into an intrinsic and extrinsic capacitive component.

$$C_L = C_{int} + C_{ext} \quad \dots\dots (22)$$



C_{int} = intrinsic output capacitance of the inverter, associated with diffusion capacitance

C_{ext} = fanout and wire capacitance

- The input gate capacitance and output gate capacitance are function of transistor gate size.

$$C_{int} = \gamma C_g \quad \dots\dots(23)$$

- γ is a proportionality factor, which is only a function of technology. Now equation (20) can be rewritten as

$$t_p = t_{po} \left(1 + \frac{C_{ext}}{\gamma C_g} \right) = t_{po} \left(1 + \frac{f}{\gamma} \right) \quad \dots\dots(24)$$

- The delay of the inverter is only a function of the ratio between its external load capacitance and input capacitance. This ratio is called the effective fanout f .

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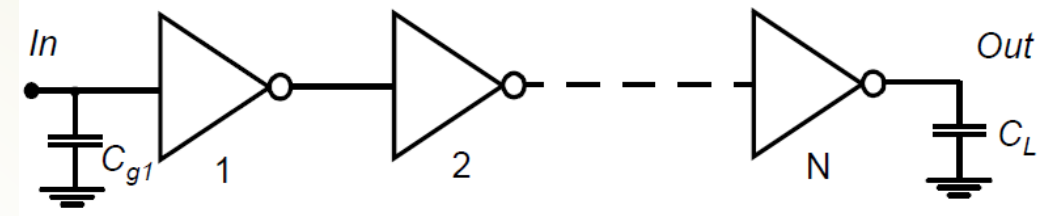
- ▶ Let us consider the circuit. The goal is to minimize the delay through the inverter chain, with the input capacitance of the first inverter C_{g1} typically a minimally-sized device and the load capacitance C_L fixed .
- ▶ Given the delay expression for the j-th inverter stage

$$t_{p,j} = t_{p0} \left(1 + \frac{C_{g,j+1}}{\gamma C_{g,j}} \right) = t_{p0} (1 + f_j / \gamma) \quad \text{.....(24)}$$

we can derive the total delay of the chain

$$t_p = \sum_{j=1}^N t_{p,j} = t_{p0} \sum_{j=1}^N \left(1 + \frac{C_{g,j+1}}{\gamma C_{g,j}} \right), \text{ with } C_{g,N+1} = C_L \quad \text{.....(25)}$$

- ▶ This equation has N-1 unknowns, being $C_{g,2}, C_{g,3}, \dots, C_{g,N}$. The minimum delay can be found by taking N-1 partial derivatives, and equating them to 0 or $\partial t_p / \partial C_{g,j} = 0$



Continue.....

- ▶ In other words, the optimum size of each inverter is the geometric mean of its neighbors sizes

$$C_{g,j} = \sqrt{C_{g,j-1} C_{g,j+1}} \quad \text{.....(26)}$$

- ▶ With $C_{g,1}$ and C_L given, we can derive the sizing factor.

$$f = \sqrt[N]{C_L / C_{g,1}} = \sqrt[N]{F} \quad \text{.....(27)}$$

and the minimum delay through the chain,

$$t_p = N t_{p0} (1 + \sqrt[N]{F} / \gamma) \quad \text{.....(28)}$$

F represents the overall effective fanout of the circuit

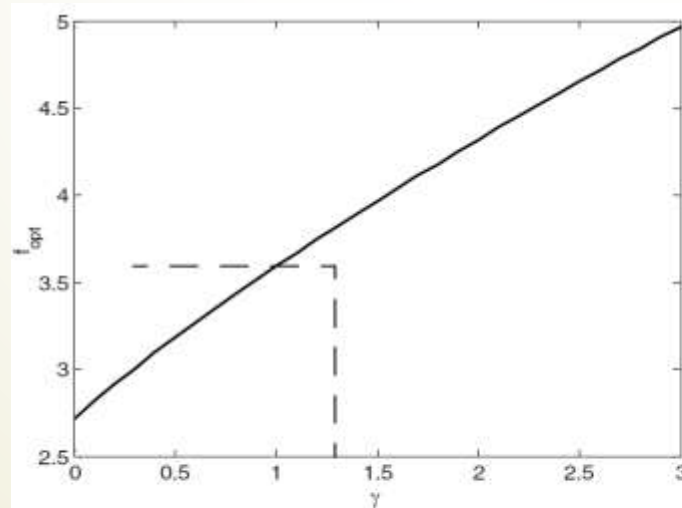
Choosing the Right Number of Stages in an Inverter Chain

- ▶ The optimum value can be found by differentiating the minimum delay expression by the number of stages, and setting the result to 0.

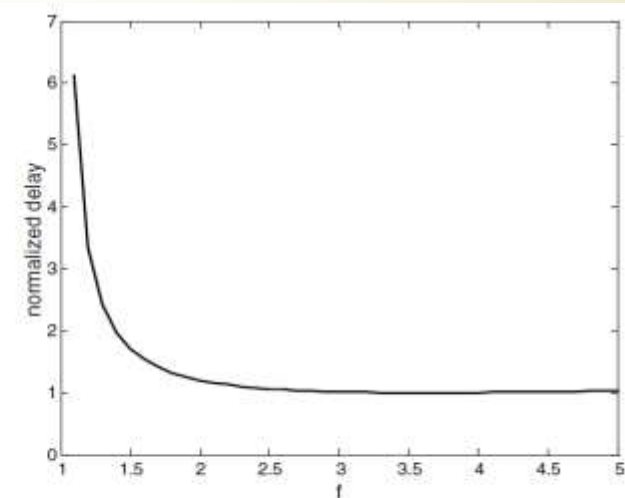
$$\gamma + \sqrt[N]{F} - \frac{\sqrt[N]{F} \ln F}{N} = 0 \quad \text{..... (29)}$$

or equivalently $f = e^{(1+\gamma/f)}$ (30)

- ▶ This equation only has a closed-form solution for $\gamma = 0$, this is when the self-loading is ignored and the load capacitance only consists of the fanout. A common practice is to select an optimum fanout of 4.



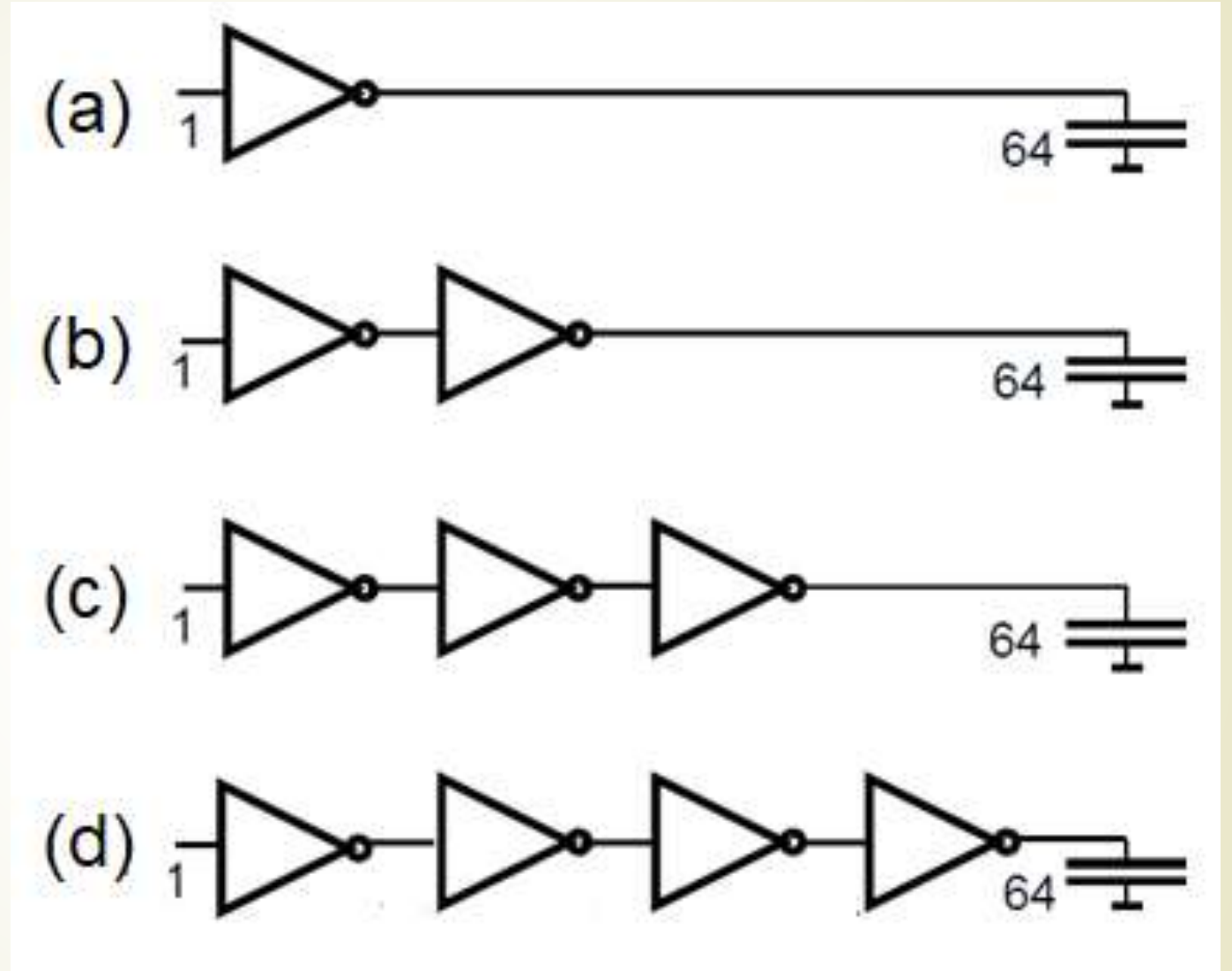
(a) Optimum effective fanout f (or inverter scaling factor) as a function of the self-loading factor γ in an inverter chain.



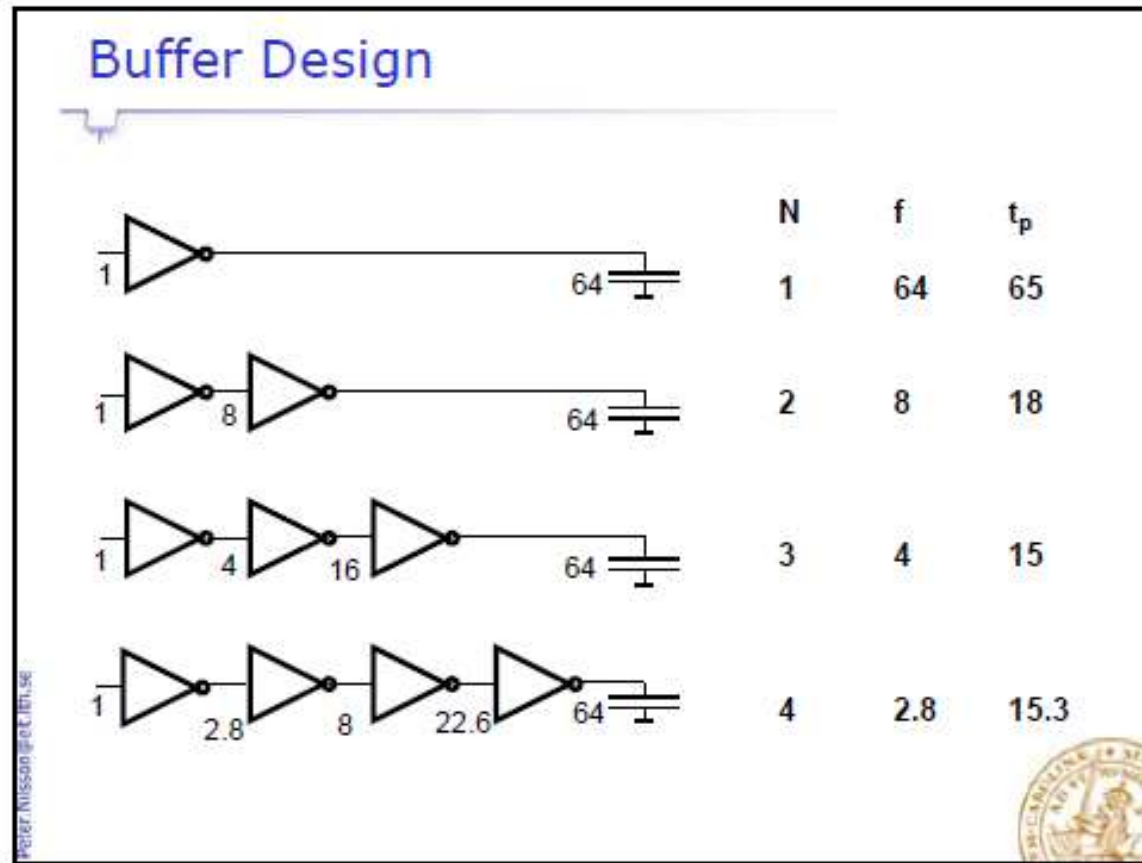
(b) Normalized propagation delay ($t_p/(t_{popt})$) as a function of the effective fanout f for $\gamma=1$.

Problem

- **Q)** Assume a circuit designed with a chain of inverters as shown in Figure 2. The unloaded delay is 1 and $\gamma = 1$ in each case of inverter chain. Evaluate the value of effective fanout (small f) and t_p for given inverter chains?



Solution





Thank You

Rohit Lorenzo, VIT-AP

Power

By

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Talk Outline

- **Motivation - Why Low Power**
- **Sources of power dissipation**
- **Why Leakage Power is Important**
- **Impact of Process parameter variation**
- **Mechanism Of Leakage components**

Power Dissipation

- Today: power dissipation is a main problem
- High Power dissipation leads to
 - ⊖ Higher weight (batteries)
 - ⊖ Reduced mobility
 - ⊖ High efforts for cooling
 - ⊖ Increasing operational costs
 - ⊖ Reduced reliability



Motivation - Why Low Power

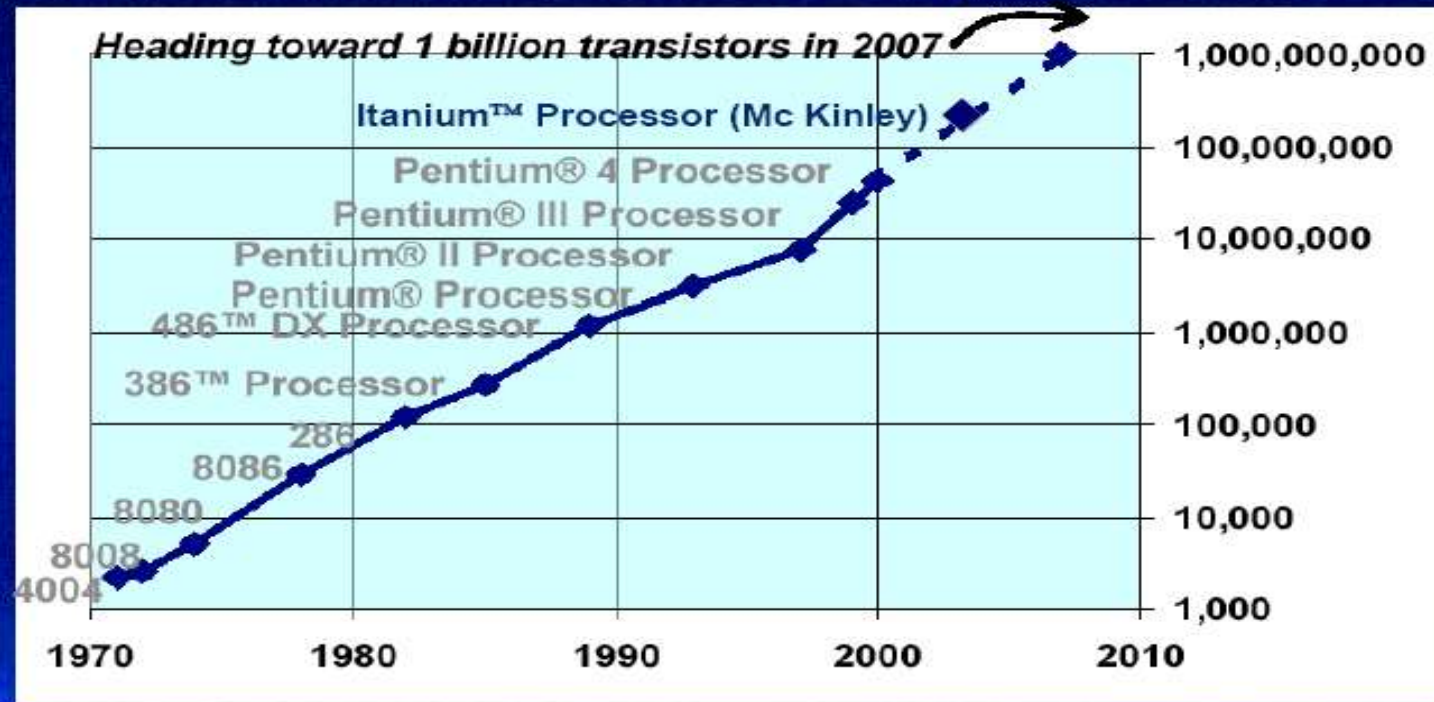
□ Changing Trend

- Until recently performance has been synonyms with circuit speed or processing Power e.g. MIPS
- Earlier power consumption were of secondary concern
- In Nanometer technology power has become a most important issue because of
 - Increasing transistor count
 - Higher speed of operation
 - Greater device leakage current
- Increased process parameter variability due to aggressive scaling has created problem in reliability & Testing

Why Low Power



Moore's Law Continues Transistors doubling every 2 years toward the billion-transistor microprocessor



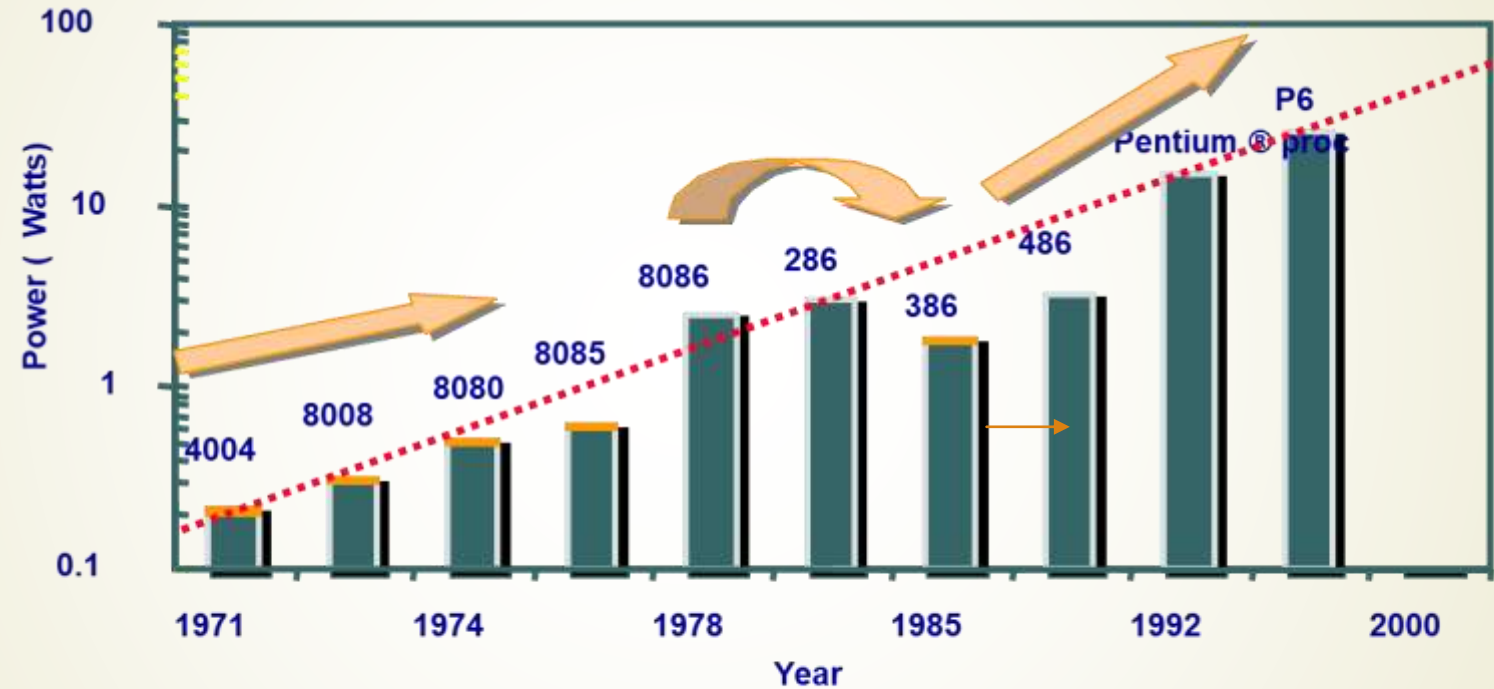
Source: Intel



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Why Low Power



Lead Microprocessors power continues to increase

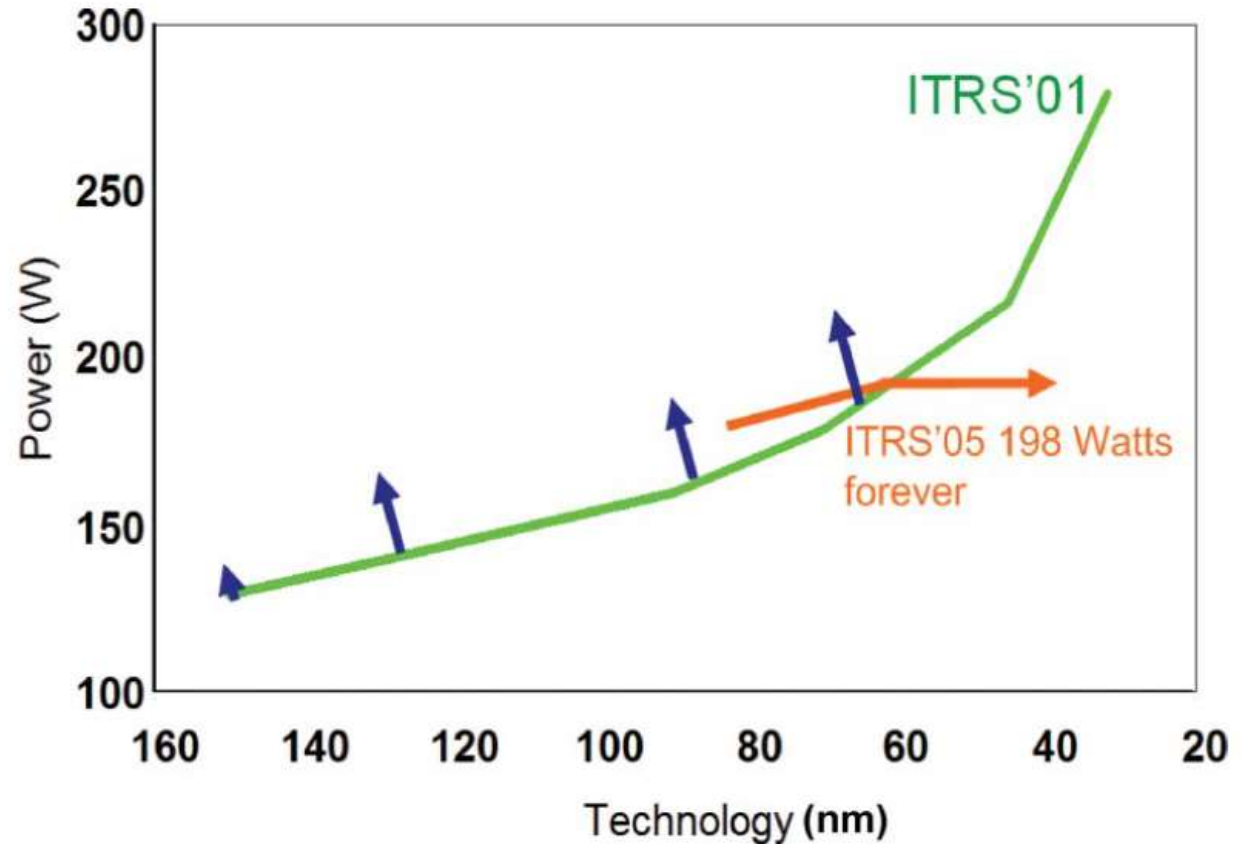
Latest Processor

Brand	Desktop				Mobile			
	Code-named	Cores	Fab	Date released	Code-named	Cores	Fab	Date released
Core i7	Bloomfield	4	45 nm	November 2008	Clarksfield	4	45 nm	September 2009
	Lynnfield	4	45 nm	September 2009	Arrandale	2	32 nm	January 2010
	Gulftown	6	32 nm	July 2010	Sandy Bridge	4	32 nm	January 2011
	Sandy Bridge	4	32 nm	January 2011	Sandy Bridge	2	32 nm	February 2011
	Sandy Bridge-E	6	32 nm	November 2011	Ivy Bridge	2-4	22 nm	May 2012
	Sandy Bridge-E	4	32 nm	February 2012	Haswell	2-4	22 nm	June 2013
	Ivy Bridge	4	22 nm	April 2012	Broadwell	2	14 nm	January 2015
	Haswell	4	22 nm	June 2013	Broadwell	4	14 nm	June 2015
	Ivy Bridge-E	4-6	22 nm	September 2013	Skylake	2-4	14 nm	September 2015
	Broadwell	4	14 nm	June 2015	Kaby Lake	2	14 nm	August 2016
	Skylake	4	14 nm	August 2015	Kaby Lake	4	14 nm	January 2017
	Kaby Lake	4	14 nm	January 2017	Coffee Lake	4-6	14 nm	April 2018
	Coffee Lake	6	14 nm	October 2017	Whiskey Lake	4	14 nm	Aug. 2018 & April 2019
	Coffee Lake	8	14 nm	October 2018	Ice Lake	4	10 nm	May & Aug. 2019
					Comet Lake	4-6	14 nm	September 2019

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Increased Power Dissipation in DSM

- As the device dimension shrink down power goes on increasing in deep sub micron technology (DSM).



Packaging and Cooling Cost

- High performance processor Consumes heavy power .
- Cost associated with packaging and cooling is prohibitive.
- Low power methodology to be Used to reduce the cost of Packaging & cooling.

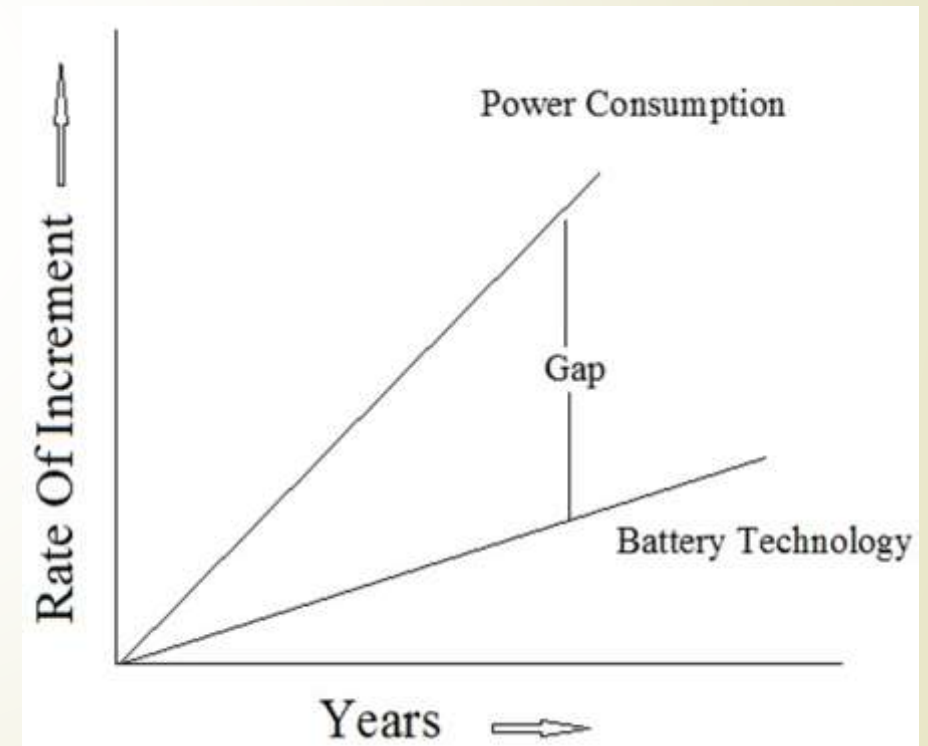


Portable System

- Increasing customer demand for handheld battery operated devices such as cell phone, tablet and laptop etc.
- Mobile phone with limitless functionality & limitless power supply is the requirement
- Growth rate of these portable equipment is very high
- All these devices are battery operated , battery life is of primary concern
- Unfortunately, battery technology has not kept up the energy requirement of the portable equipment

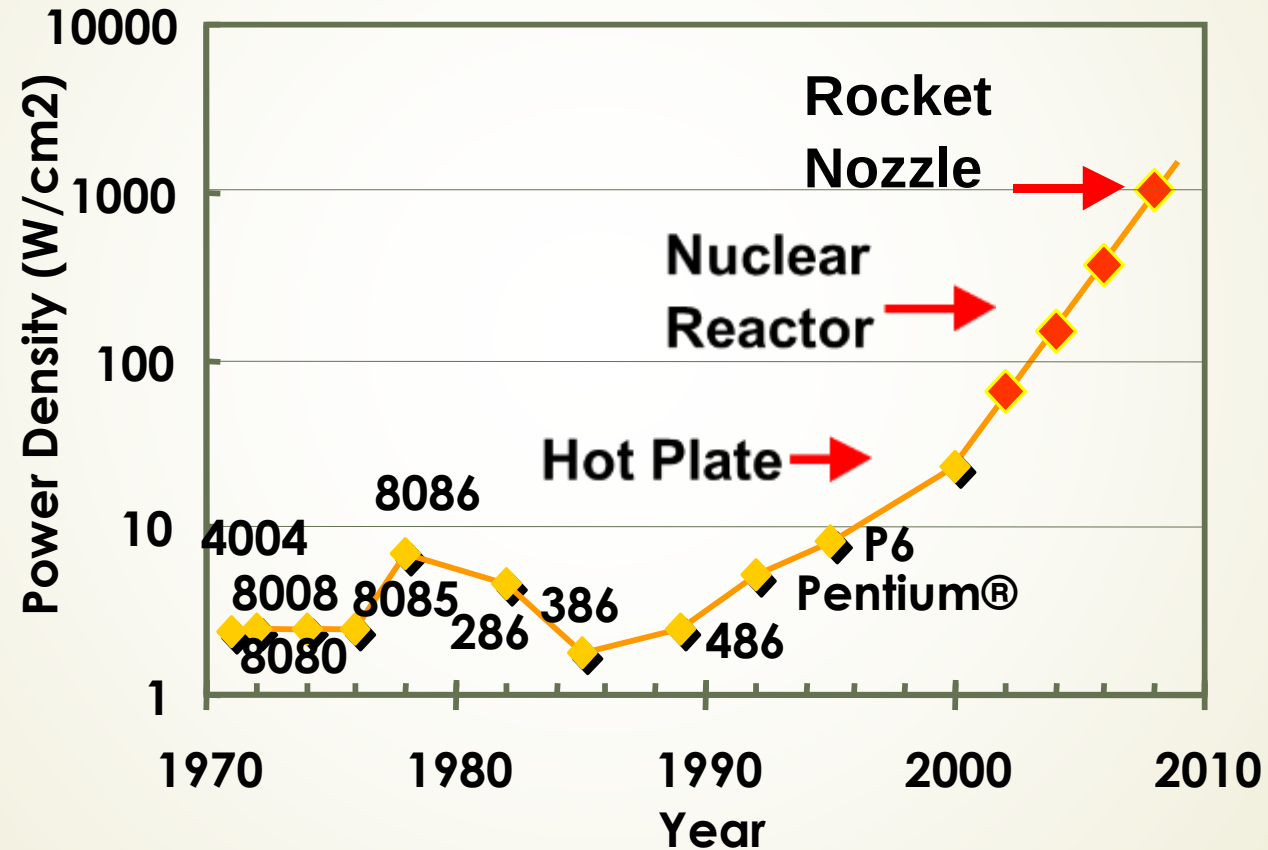
Advantages & Disadvantages

- Commercial success of these products depends upon on size ,weight , cost , computing power and battery life.
- Low Power design Methodology Is very important to make them Commercially viable.



Increasing Power Density

Source: Intel®



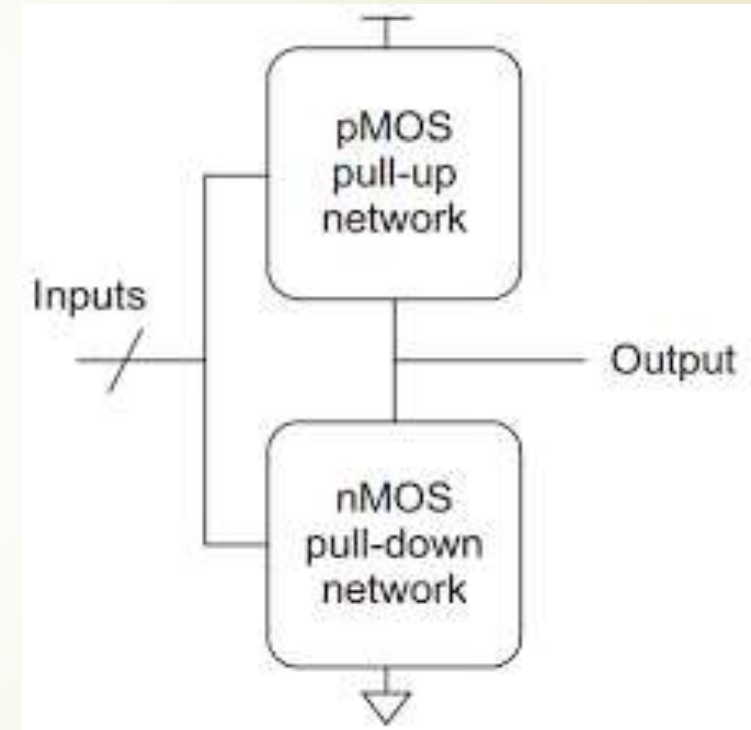
Environment

- ▶ According to an estimate of the U.S environmental protection agency (EPA) , 80% of power consumption by office equipment are due to computing equipment and a large part from unused equipment
- ▶ Power is dissipated mostly in the form of heat .The cooling techniques , such as AC transfers the heat to the environment
- ▶ To reduce adverse effect on environment efforts such as EPA's energy star program leading to power management standard for desktops and laptops has emerged

CMOS Technology of Choice

➤ Advantages

- Ease of Fabrication
- Good noise Margin
- Robust
- Lower Switching Activity
- Good input and output decoupling
- No charge Sharing problem
- Availability of matured logic and synthesis tool.



Continue....

- Disadvantage
 - Larger number of transistor (larger chip area & delay)
 - Weak Output driving capability
 - Larger number of standard cell required substantial engineering effort for technology mapping
 - Glitching power dissipation
 - Short circuit power dissipation

Sources of Power Dissipation

- **Dynamic Power (Active Mode)**

- Switching power
- Glitching power
- Short circuit power

- **Static Power (Stand by Mode)**

- Diode leakage current
- Sub-threshold leakage current
- Gate leakage current

Power Dissipation CMOS Equation

$$P = \alpha f C_L V_{DD}^2 + V_{DD} I_{peak} (P_{0 \rightarrow 1} + P_{1 \rightarrow 0}) + V_{DD} I_{leak}$$

Dynamic
power
($\approx 30 - 50\%$
today and
decreasing
relatively)

Short-circuit
power
($\approx 10\%$ today and
decreasing
absolutely)

Leakage (Static)
power
($\approx 20 - 70\%$
today and
increasing)

Definitions and Basics

- ▶ The *instantaneous power* $P(t)$ consumed or supplied by a circuit element is the product of the current through the element and the voltage across the element.

$$P(t) = I(t)V(t) \quad \dots\dots\dots(1)$$

- ▶ The *energy* consumed or supplied over some time interval T is the integral of the instantaneous power

$$E = \int_0^T P(t) dt \quad \dots\dots\dots(2)$$

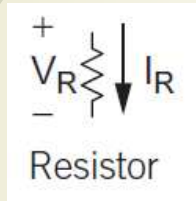
The *average power* over this interval is

$$P_{\text{avg}} = \frac{E}{T} = \frac{1}{T} \int_0^T P(t) dt \quad \dots\dots\dots(3)$$

- ▶ Power is expressed in units of Watts (W). Energy in circuits is usually expressed in Joules (J)

Examples

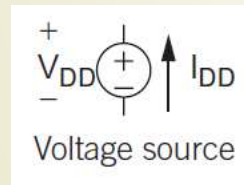
- ▶ The voltage and current are related by Ohm's Law, $V = IR$, so the instantaneous power dissipated in the resistor is



$$P_R(t) = \frac{V_R^2(t)}{R} = I_R^2(t) R \quad \dots\dots\dots(4)$$

This power is converted from electricity to heat

If voltage source V_{DD} . It supplies power proportional to its current



$$P_{VDD}(t) = I_{DD}(t) V_{DD} \quad \dots\dots\dots(5)$$

Continue.....

- When the capacitor is charged from 0 to V_C , it stores energy E_C .

$$\begin{array}{c} + \\ V_C \end{array} \parallel C \downarrow I_C = C \frac{dV}{dt}$$

Capacitor

$$E_C = \int_0^{\infty} I(t)V(t)dt = \int_0^{\infty} C \frac{dV}{dt} V(t)dt = C \int_0^{V_C} V(t)dV = \frac{1}{2} CV_C^2 \quad \text{.....(6)}$$

The capacitor releases this energy when it discharges back to 0

Dynamic Power Dissipation

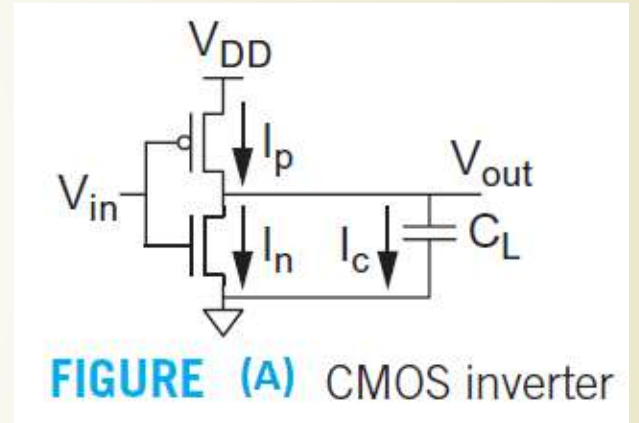
- Figure A shows a CMOS inverter driving a load capacitance. When the input switches from 1 to 0, the pMOS transistor turns ON and charges the load to V_{DD} . According to EQ (6), the energy stored in the capacitor is

$$E_C = \frac{1}{2} C_L V_{DD}^2 \quad \dots\dots\dots(7)$$

The energy delivered from the power supply is

$$E_C = \int_0^\infty I(t) V_{DD} dt = \int_0^\infty C \frac{dV}{dt} V_{DD} dt = C V_{DD} \int_0^{V_{DD}} dV = C V_{DD}^2 \quad \dots\dots(8)$$

- Observe that only half of the energy from the power supply is stored in the capacitor. The other half is dissipated (converted to heat) in the pMOS transistor. The power dissipated depends only on the load capacitance, not on the size of the transistor or the speed at which the gate switches



Continue.....

- Suppose that the gate switches at some average frequency f_{sw} . Over some interval T , the load will be charged and discharged Tf_{sw} times. Then, according to EQ (3), the average power dissipation is

$$P_{\text{switching}} = \frac{E}{T} = \frac{Tf_{sw} CV_{DD}^2}{T} = CV_{DD}^2 f_{sw} \dots\dots\dots(9)$$

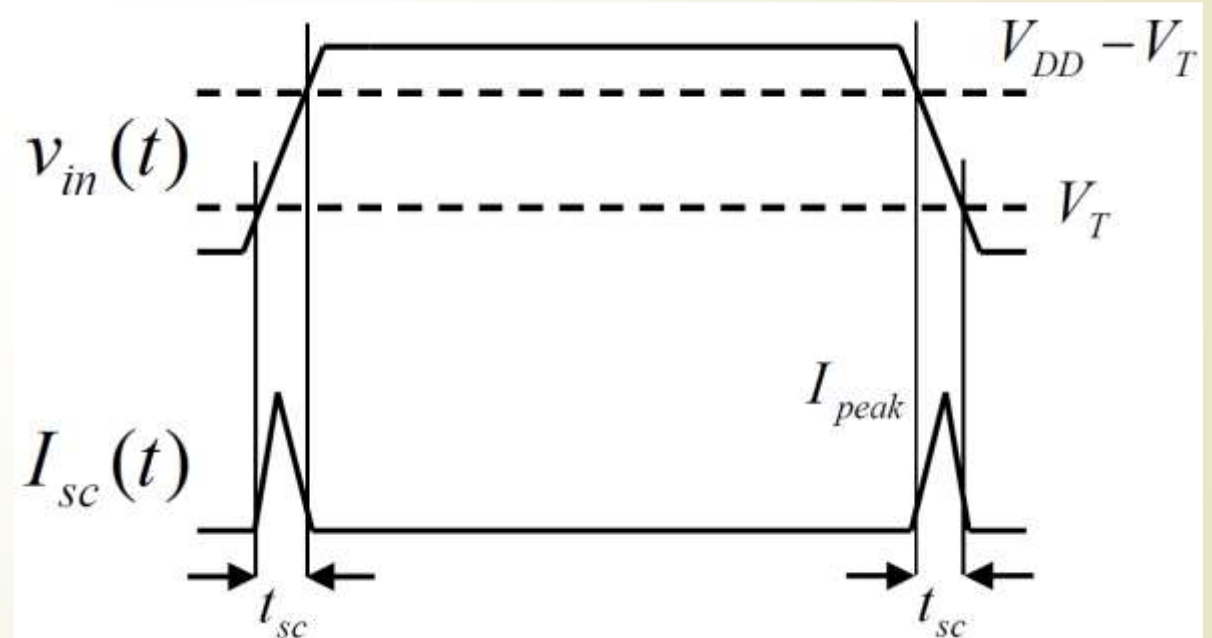
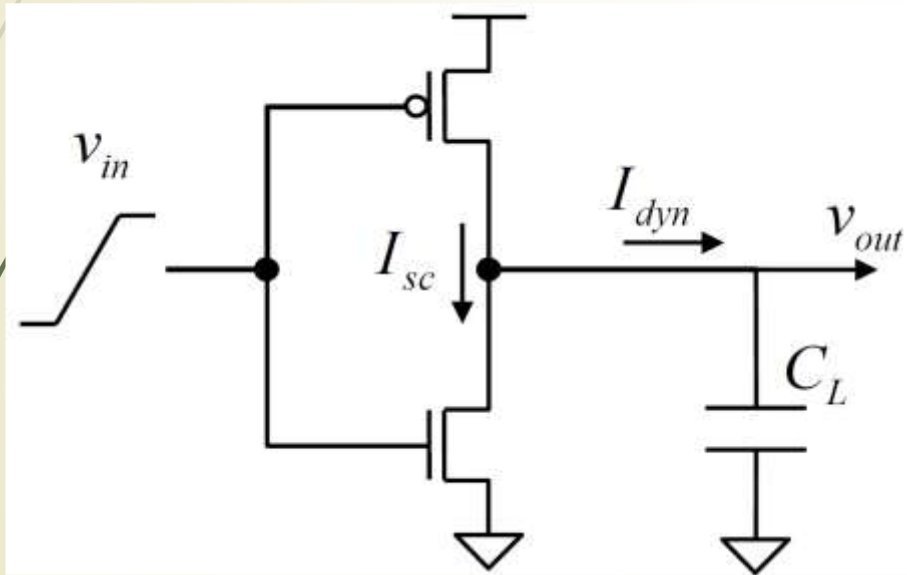
- This is called the *dynamic power* because it arises from the switching of the load. Because most gates do not switch every clock cycle, it is often more convenient to express switching frequency f_{sw} as an *activity factor* α times the clock frequency f . Now, the dynamic power dissipation may be rewritten as

$$P_{\text{switching}} = \alpha CV_{DD}^2 f \dots\dots\dots(10)$$

- The activity factor is the probability that the circuit node transitions from 0 to 1.

Short Circuit Power Dissipation

- Short-circuit power dissipation occurs as both pull-up and pulldown networks are partially ON while the input switches. It increases as the input edge rates become slower because both networks are ON for more time.
- However, it decreases as load capacitance increases because with large loads the output only switches a small amount during the input transition.



Continue.....

- Under the (reasonable) assumption that the resulting current spikes can be approximated as triangles and that the inverter is symmetrical in its rising and falling responses, we can compute the energy consumed per switching period.

- $$E_{dp} = V_{DD} \frac{I_{peak} t_{sc}}{2} + V_{DD} \frac{I_{peak} t_{sc}}{2} = t_{sc} V_{DD} I_{peak} \quad \text{.....(11)}$$

- as well as the average power consumption

- $$P_{dp} = t_{sc} V_{DD} I_{peak} f = C_{sc} V_{DD}^2 f \quad \text{.....(12)}$$

Problem 1

- Calculate the amount of energy and power needed to charge and discharge the capacitance. The value of load capacitance is equal to 6 fF for a supply voltage of 2.5 V. Consider time $t_p = 32.5$ ps.

- Solution

$$E_{dyn} = C_{int} V_{DD}^2 = 37.5 \text{ fF}$$

$$P_{dyn} = E_{dyn} / 2t_p = 580 \text{ } \mu W$$

Problem2

- ▶ A digital system-on-chip in a 1V 65 nm process (with 50 nm drawn channel lengths and $\lambda = 25$ nm) has 1 billion transistors, of which 50 million are in logic gates and the remainder in memory arrays. The average logic transistor width is 12λ and the average memory transistor width is 4λ . The memory arrays are divided into banks and only the necessary bank is activated so the memory activity factor is 0.02. The static CMOS logic gates have an average activity factor of 0.1. Assume each transistor contributes $1\text{fF}/\mu\text{m}$ of gate capacitance and $0.8\text{ fF}/\mu\text{m}$ of diffusion capacitance. Neglect wire capacitance for now (though it could account for a large fraction of total power). Estimate the switching power when operating at 1 GHz.

- ▶ **Solution**

- ▶ There are $(50 \times 10^6 \text{ logic transistors})(12\lambda)(0.025 \mu\text{m}/\lambda)((1 + 0.8) \text{ fF}/\mu\text{m}) = 27 \text{ nF}$ of logic transistors and $(950 \times 10^6 \text{ memory transistors})(4\lambda)(0.025 \mu\text{m}/\lambda)((1 + 0.8) \text{ fF}/\mu\text{m}) = 171 \text{ nF}$ of memory transistors. The switching power consumption is $[(0.1)(27 \times 10^{-9}) + (0.02)(171 \times 10^{-9})](1.0 \text{ V})^2(10^9 \text{ Hz}) = 6.1 \text{ W}$.

Switching Activity (α)

➤ $P_0 \rightarrow_1 = P_0 \cdot P_1$

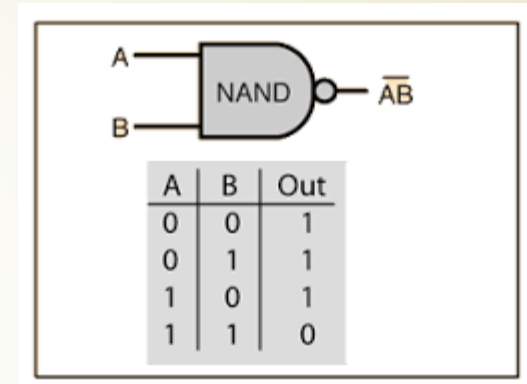
➤ $P_0 = \frac{1}{4}$, $P_1 = \frac{3}{4}$

➤ $P_0 \rightarrow_1 = \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$

➤ $P_0 = \frac{n_0}{2^n}$, n_0 = No. of zeros

➤ $P_1 = \frac{n_1}{2^n}$, $n_1 = (2^n - n_0)$

➤ $P_0 \rightarrow_1 = P_0 \cdot P_1 = \frac{n_0}{2^n} \times \frac{(2^n - n_0)}{2^n} = \frac{n_0(2^n - n_0)}{2^{2n}}$





Continue....

2 input Gates	inv	NAND/AND	NOR/OR
Activity	1/4	3/16	3/16

Continue....

- ▶ The activity factor of a node is the probability that it switches from 0 to 1. This probability depends on the logic function. By analyzing the probability that each node is 1, we can estimate the activity factors.
- ▶ Define P_i to be the probability that node i is 1. $\overline{P_i} = 1 - P_i$ is the probability that node i is 0. α_i , the activity factor of node i ,

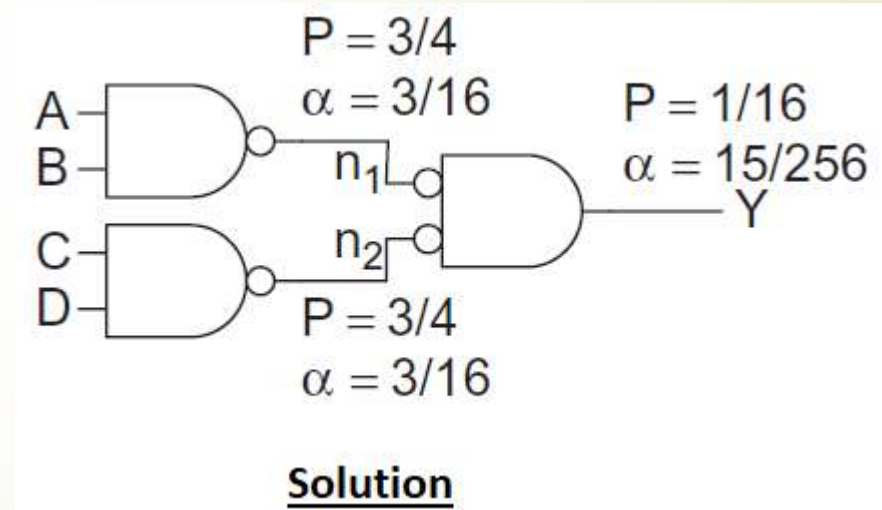
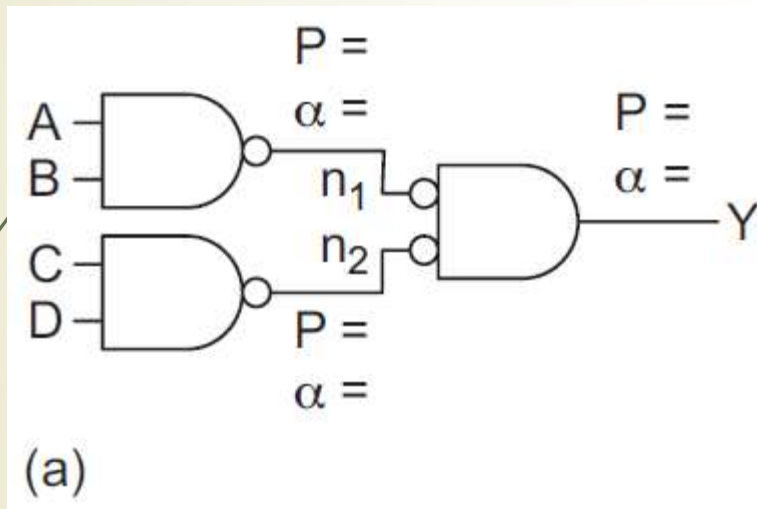
$$\alpha_i = \overline{P_i}P_i$$

TABLE Switching probabilities

Gate	P_y
AND2	$P_A P_B$
AND3	$P_A P_B P_C$
OR2	$1 - \overline{P_A} \overline{P_B}$
NAND2	$1 - P_A P_B$
NOR2	$\overline{P_A} \overline{P_B}$
XOR2	$P_A \overline{P_B} + \overline{P_A} P_B$

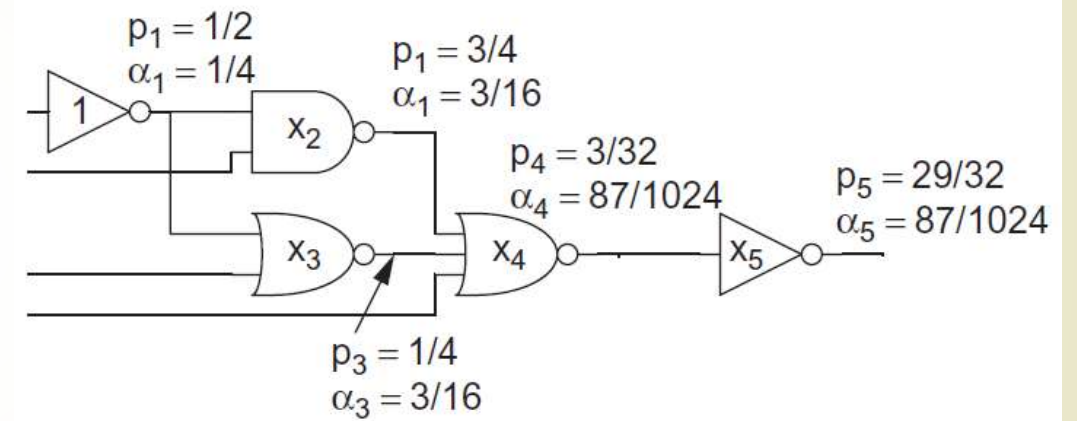
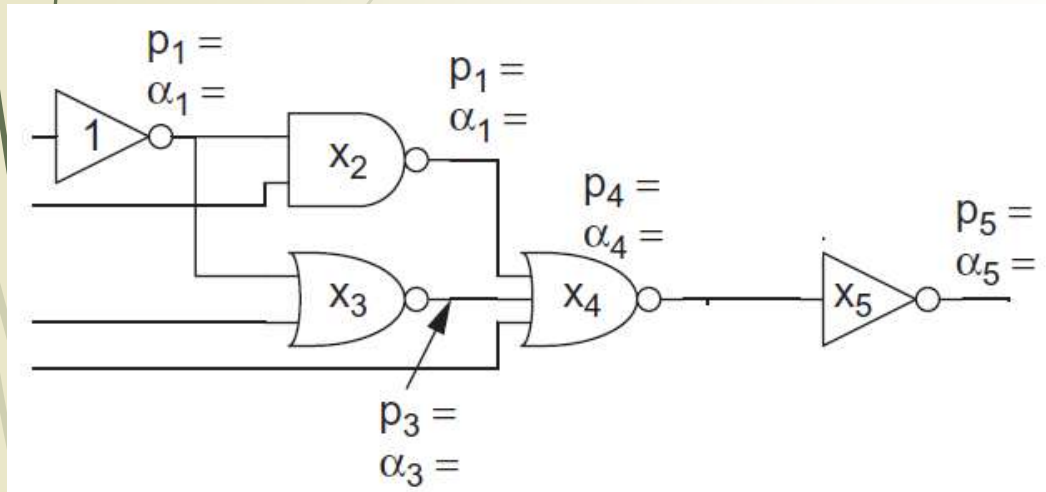
Problem3

- Figure shows a 4-input AND gate built using a tree (a) and a chain (b) of gates. Determine the activity factors at each node in the circuit assuming the input probabilities $P_A = P_B = P_C = P_D = 0.5$.



Problem4

- Calculate probability and switching activity factor



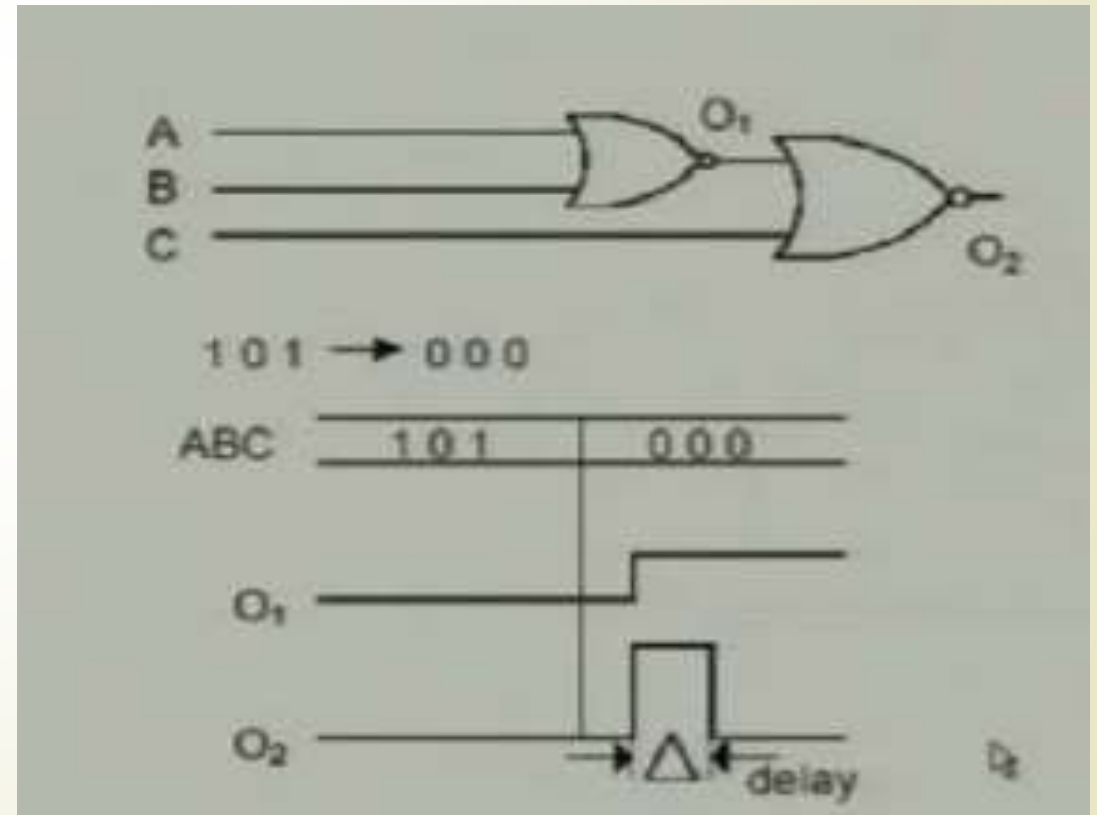
Solution

Glitching power Dissipation

- ▶ The third source of dynamic power or dissipation is glitching power this glitching power dissipation occurs due to the finite delay of the gate. Because, of the finite delay of these gates there is a time the output is becoming high for a very small duration and this is known as glitch. So, there will be glitching power dissipation on this output capacitance.

Inputs		Outputs
X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	0

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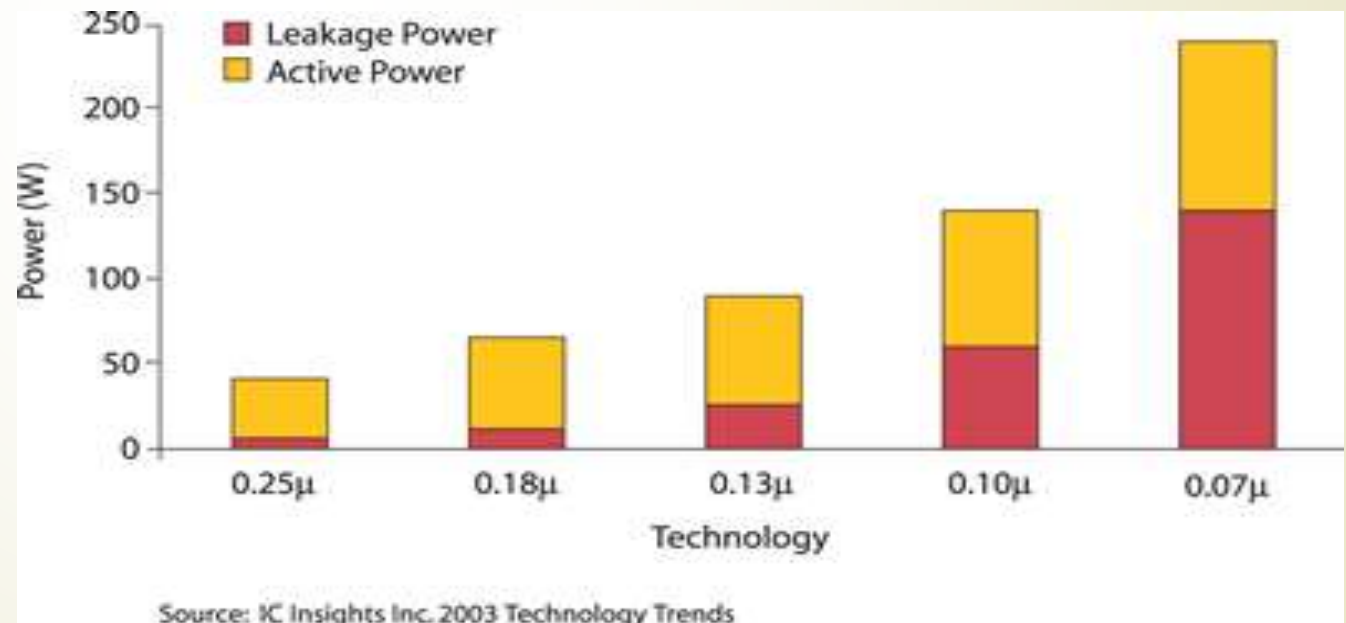


Static Power (Leakage Power)

- ▶ Static power is consumed even when a chip is not switching.
- ▶ Prior to the 90 nm node, leakage power was of concern primarily during sleep mode because it was negligible compared to dynamic power.
- ▶ In nanometer processes with low threshold voltages and thin gate oxides, leakage can account for as much as a third of total active power.

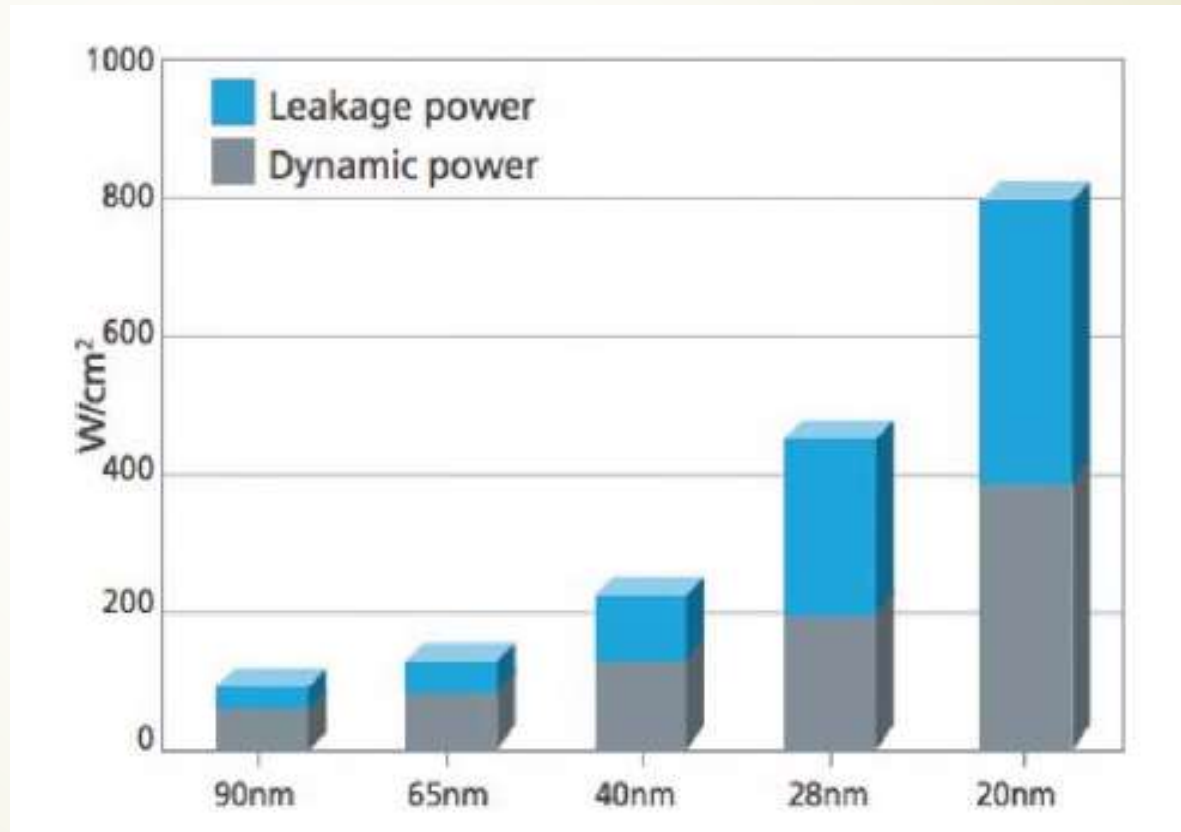
Why Leakage Power is Important

- Leakage power increases at a faster rate than dynamic power in technology generation .
- Leakage power is become a major component of total power dissipation
- So there is need a methodology to reduce leakage power

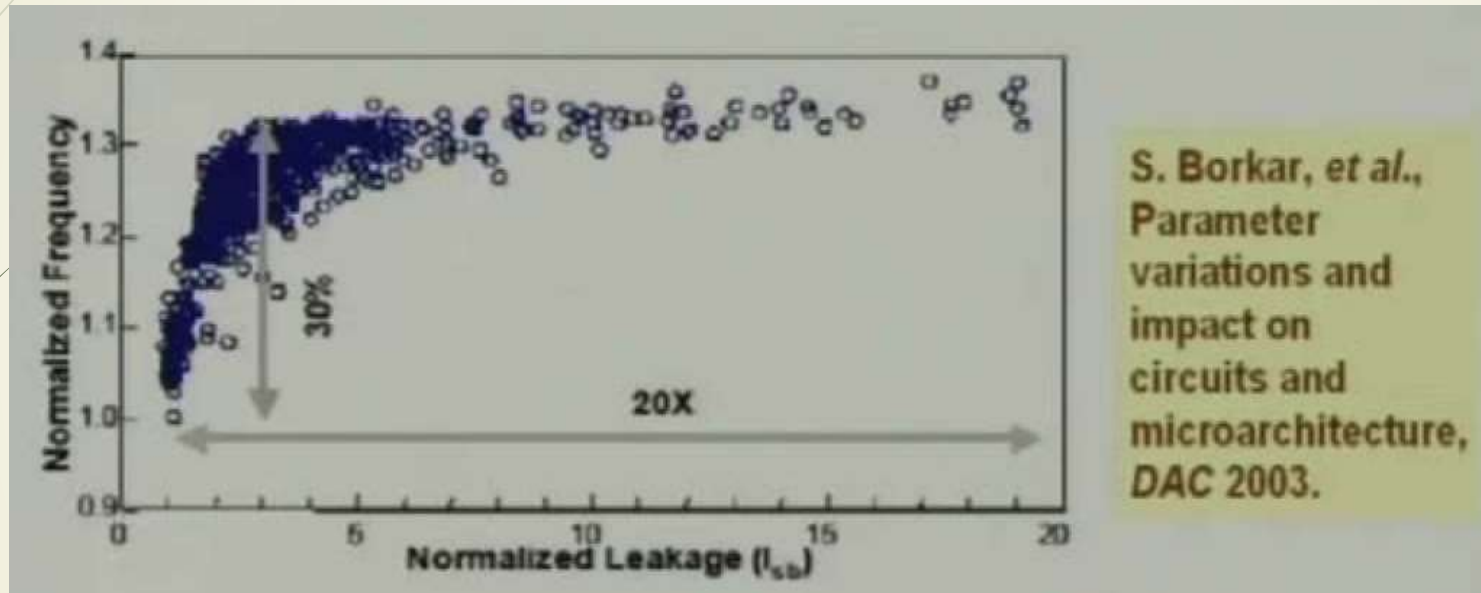


Continue.....

- Power Density



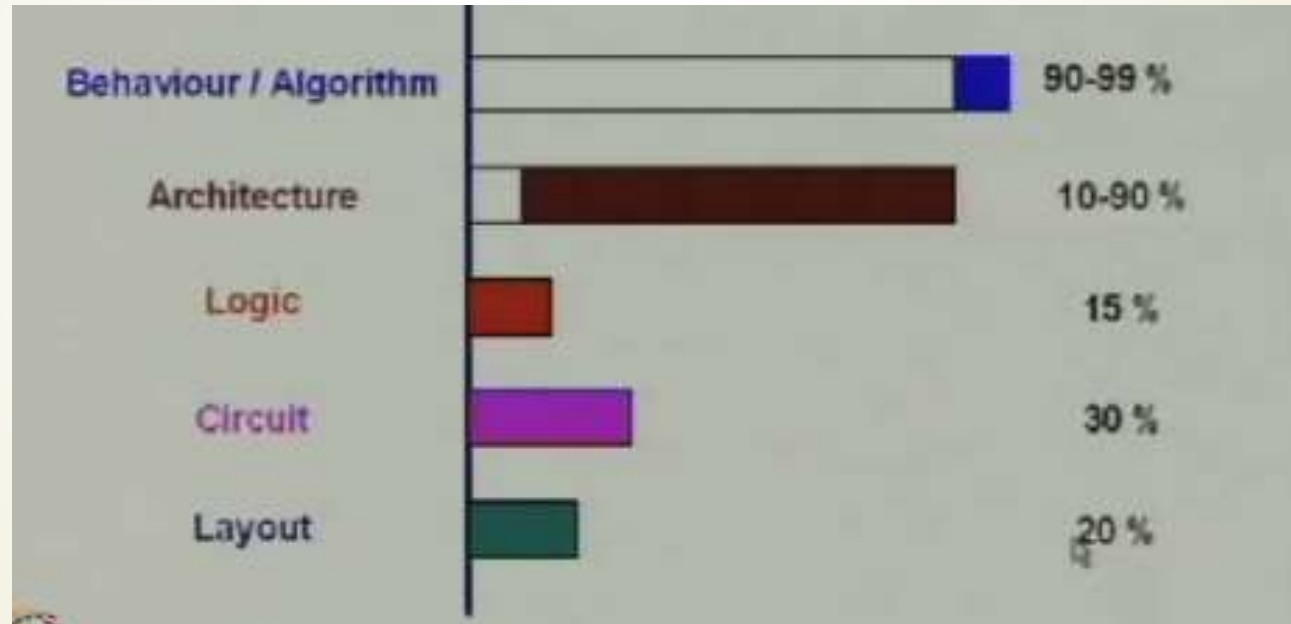
Impact of Process Variation on Leakage and Performance



- 30% delay variation and 20X leakage variation between fast and slow dies reported for 0.18µm CMOS process.
- Low leakage chips with too low frequency must be discarded
- High frequency chips with too high leakage must be discarded

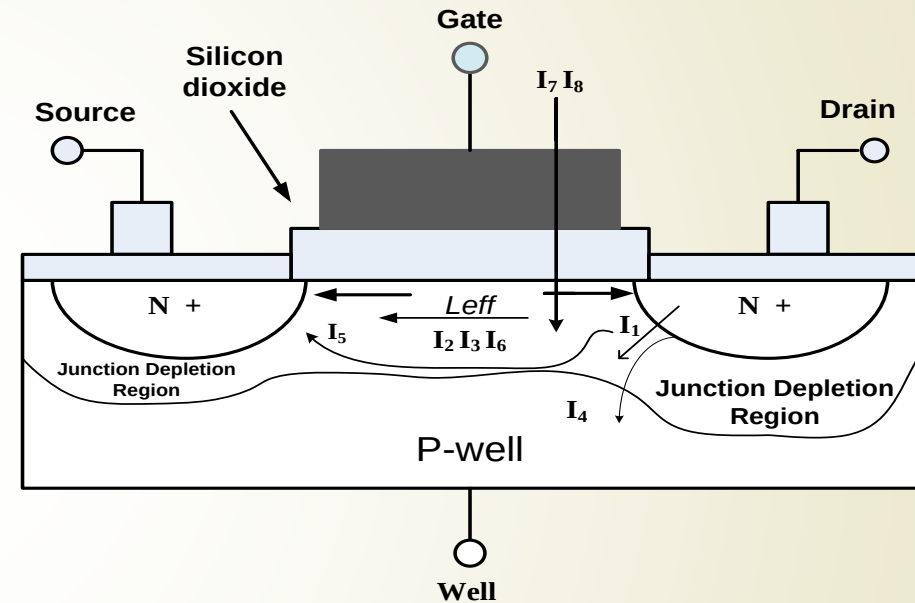
Power Dissipation at different levels

- Power optimization approaches at the high level are significant since research results indicate that higher level of abstraction have greater potential for power reduction



Components of Leakage Power

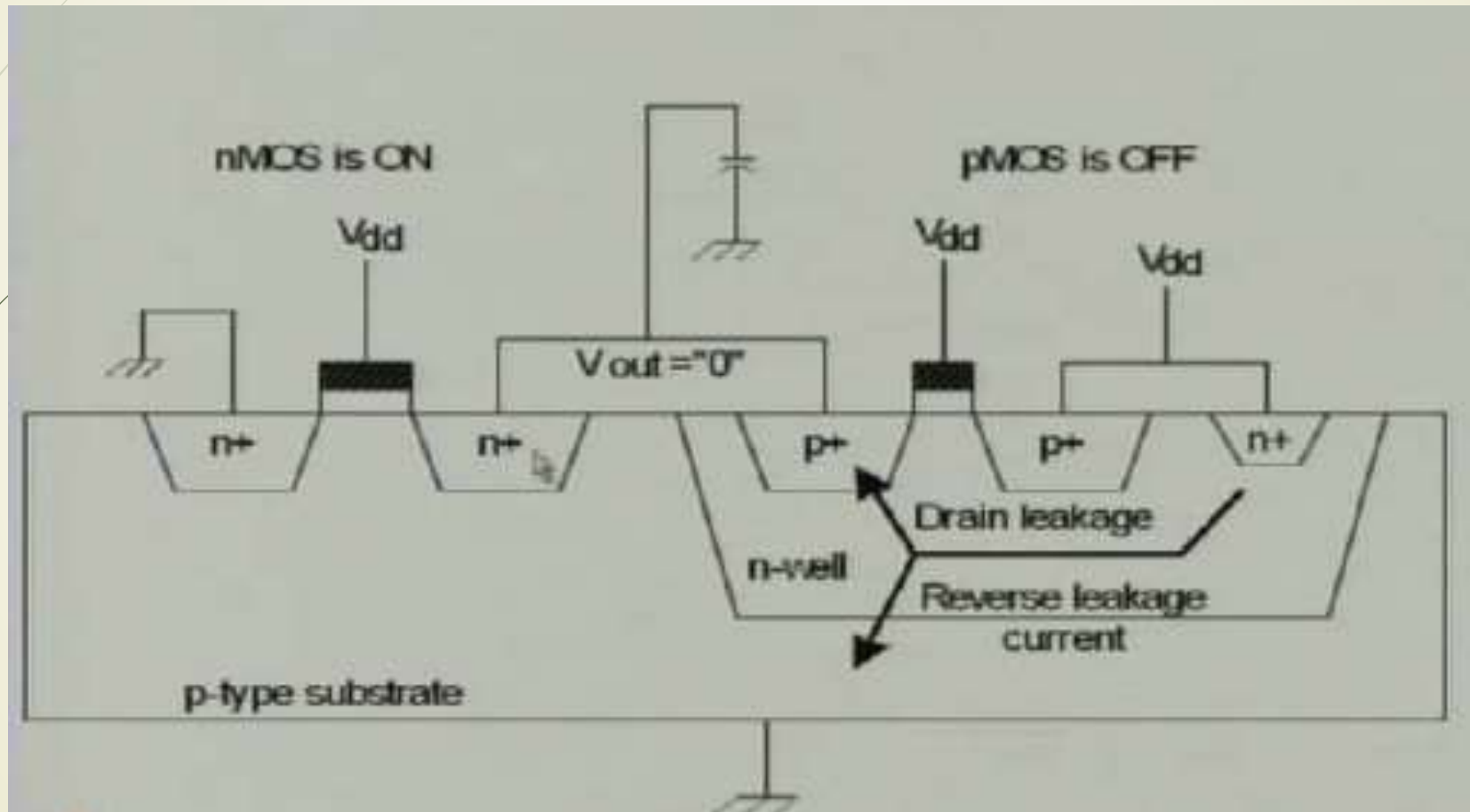
- I_1 = Reverse diode leakage
- I_2 = Sub threshold leakage
- I_3 = Drain Induced barrier lowering
- I_4 = Gate Induced drain leakage
- I_5 = Channel punch through
- I_6 = Narrow Channel effect
- I_7 = Gate oxide tunneling
- I_8 = Hot Carrier Injection



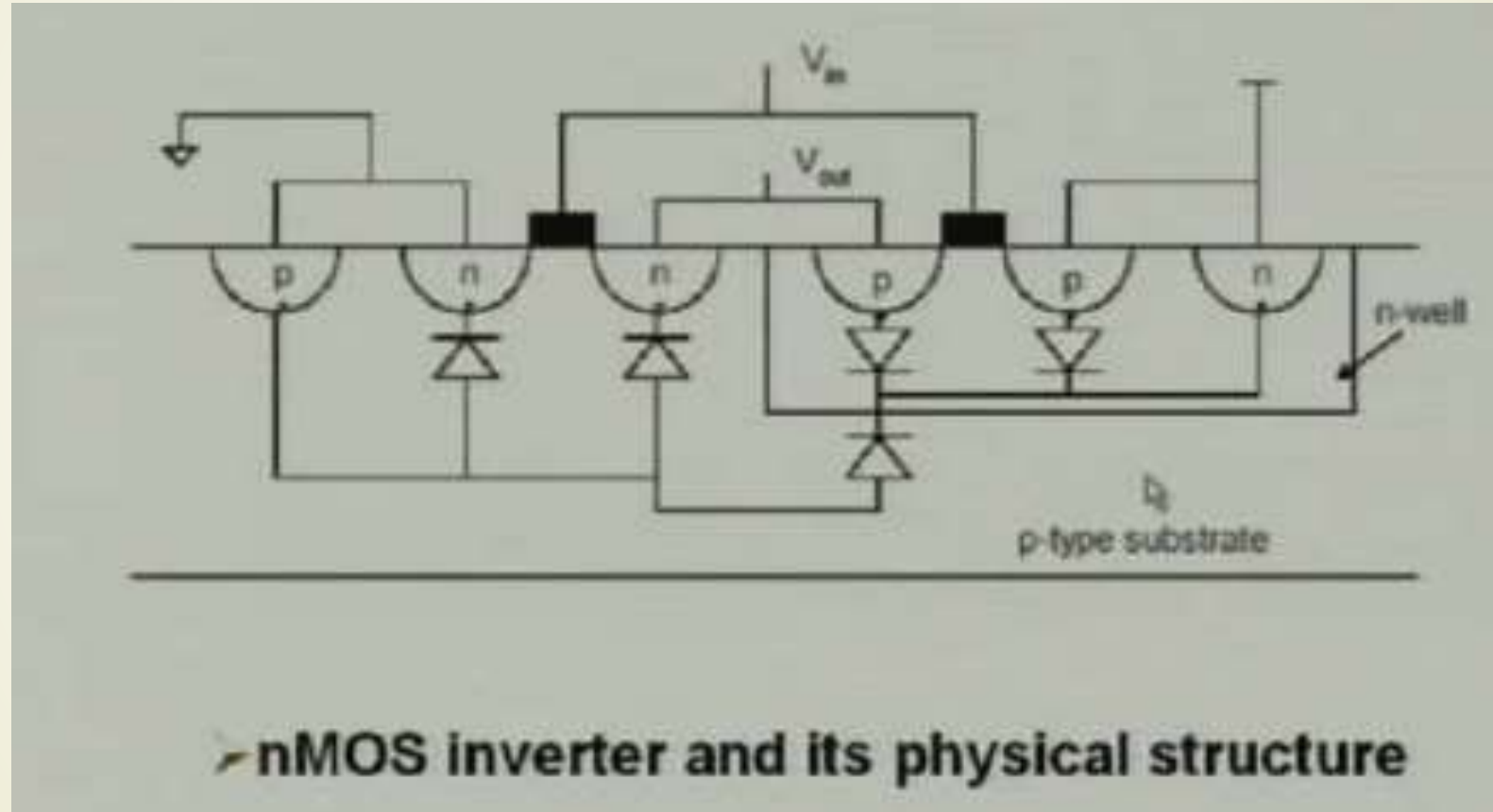
Among all sub threshold leakage current is the dominant leakage current component

“K.Roy , S.mukhopadhyay and H.heimand ‘leakage current mechanism of deep-submicron CMOS circuits
,proc IEEE , vol 91 ,No.2 ,pp-305-327 , feb 01 , 2003”

Reversed Biased Leakage



P-n Junction reverse biased Current



P-n Junction reverse biased Current

$$I_D = I_s(e^{-qv/KT} - 1)$$

Where I_s is the reverse saturation current density

V = diode voltage

q = Charge of an electron

K = Boltzmann constant

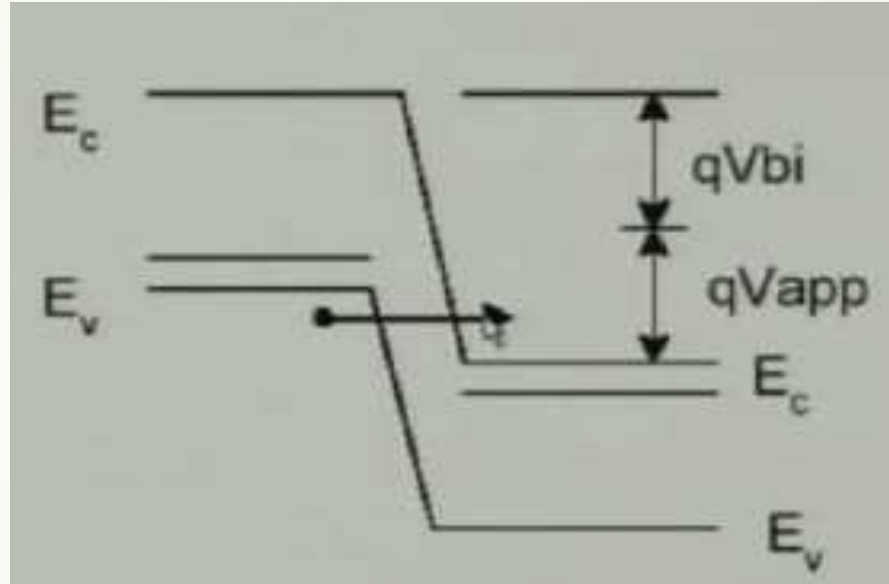
T = Temperature in kelvin

The current is in the range of 0.1nA to 0.5nA per device at room temperature.

The current doubles for every 10°C rise in temperature.

However, this total static power due to diode leakage current for 1 million transistor is substantial.

Band to band tunneling current



- High electric field is across this reverse biased p-n junction diodes causes significant current known as band-to-band tunneling current.
- This current dominates the p-n junction current

Subthreshold Leakage

- Subthreshold leakage current flows when a transistor is supposed to be OFF.

$$I_{\text{sub}} = I_{\text{off}} 10^{\frac{V_{gs} + \eta(V_{ds} - V_{DD}) - k_{\gamma}V_{sb}}{S}}$$

- where I_{off} is the subthreshold current at $V_{gs} = 0$ and $V_{ds} = V_{DD}$. S is the subthreshold
- Slope. I_{off} is a key process parameter defining the leakage of a single OFF transistor. It is low for high V_t Transistor and high for low V_t Transistor. η is the DIBL coefficient.
- K_{γ} is the body effect coefficient



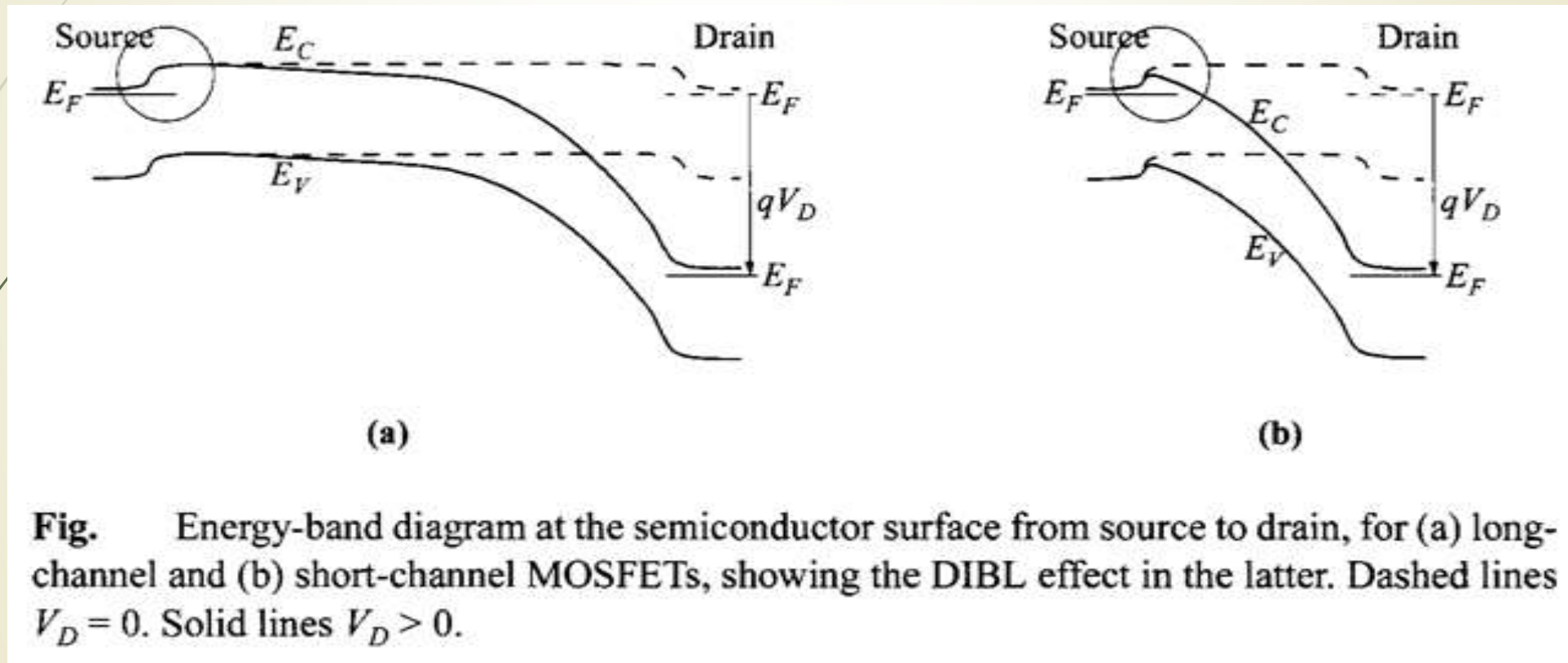
Continue....

- Various mechanism which effects the sub-threshold leakage current are-
 - Drain induced barrier lowering effect.
 - Body effect
 - Narrow width effect
 - Effect of channel length with V_{th} Roll off
 - Effect of temperature

Drain Induced Barrier Lowering (DIBL)

- In long-channel devices, the source and drain are separated enough so that their depletion regions have no effect on the electrostatic potential developed across the channel.
- Hence, for such devices, the V_t is virtually independent of the channel length and drain bias.
- In a short-channel device, however, the source and drain depletion width in the vertical direction and the source drain potential have a strong effect on the band bending over a significant portion of the device.
- Therefore, the V_t and consequently the sub-threshold current of short-channel devices, varies with the drain bias. This effect is referred to as drain induced barrier lowering (DIBL).

DIBL



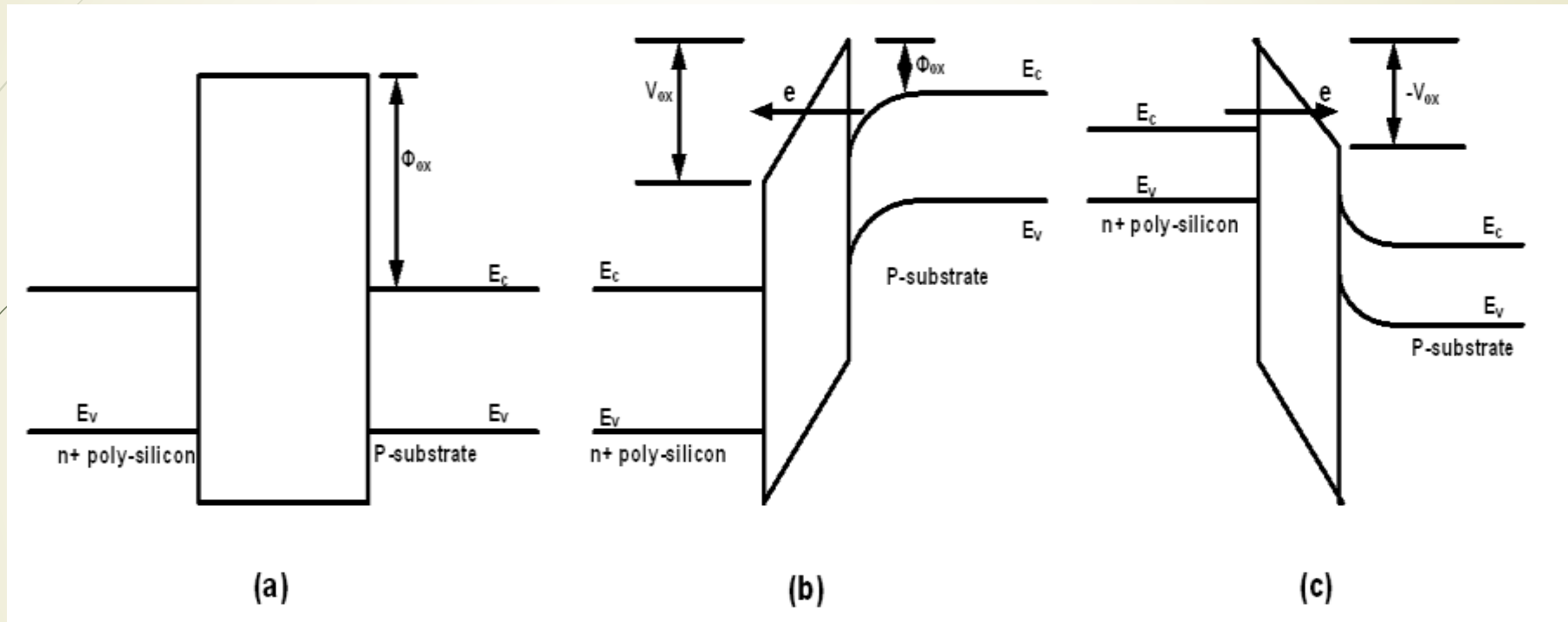
Body Effect

- As a negative voltage is applied to the substrate with respect to the source, the well to source junction the device is reverse biased and bulk depletion region is widened. This leads to an increase in threshold voltage. This effect is known as Body effect.

$$V_T = V_{T0} + \gamma \cdot \left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|} \right)$$

- ϕ_F is the fermi potential.
- γ is the body effect coefficient
- V_{T0} is the threshold voltage at zero bias

Gate oxide tunneling



- **Figure :** Tunneling of electron through an MOS capacitor (a) Energy band diagram at flat-band condition. (b) Energy band diagram with positive gate bias showing tunneling of electron from substrate to gate. (c) Energy-band diagram at negative gate bias showing tunneling of electron from gate to substrate

Gate-Induced Drain Leakage

- Due to a high electric field (V_{dg}) across the oxide, a deep depletion region under the drain overlap region is created, which generates electrons and holes by band-to-band tunneling.
- The resulting drain to body current is called GIDL current. When the negative gate bias is large (V_g gate at zero or negative voltages with respect to drain at V_{dd}), the n+ region under the gate can be depleted and even inverted as shown in Fig.
- As a consequence, minority carriers are emitted in the drain region underneath the gate.
- As the substrate is at a lower potential, the minority carriers at the drain depletion region are swept away laterally to the substrate, creating a GIDL current.
- Thinner oxide thickness and higher V_{dd} enhance the electric field and therefore enhance GIDL current.

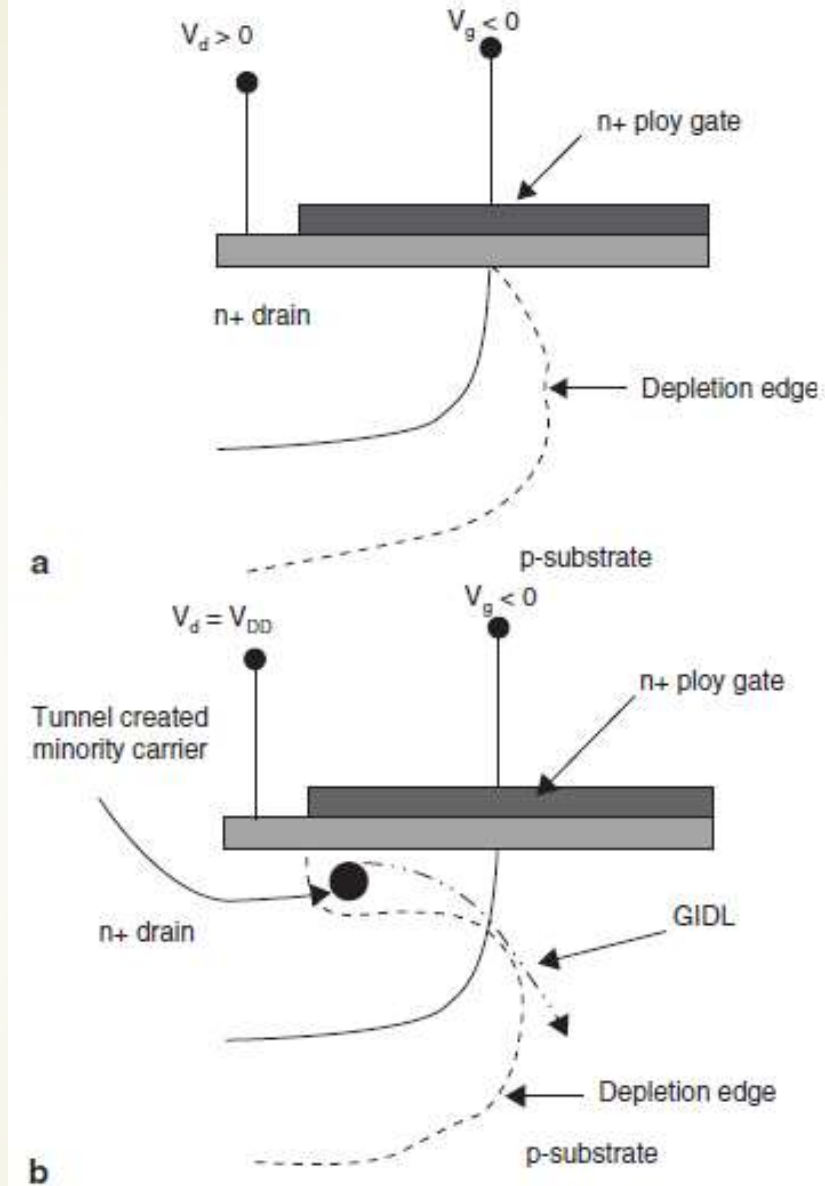


Fig. GIDL effect. GIDL gate-induced drain leakage

V_{th} Roll-Off

- As the channel length is reduced, the threshold voltage of metal–oxide–semiconductor field-effect transistor (MOSFET) decreases.
- This reduction of channel length is known as V_{th} roll-off. Figure shows the reduction of threshold voltage with a reduction in channel length.
- The bulk charge that needs to be inverted by the application of gate voltage is proportional to the area under the trapezoidal region $Q'_B \propto W_{dm} (L + L') / 2$
- The gate voltage has to invert less bulk charge to turn the transistor on leading to more band bending in the Si–SiO₂ interface in the short-channel device compared to long-channel devices.

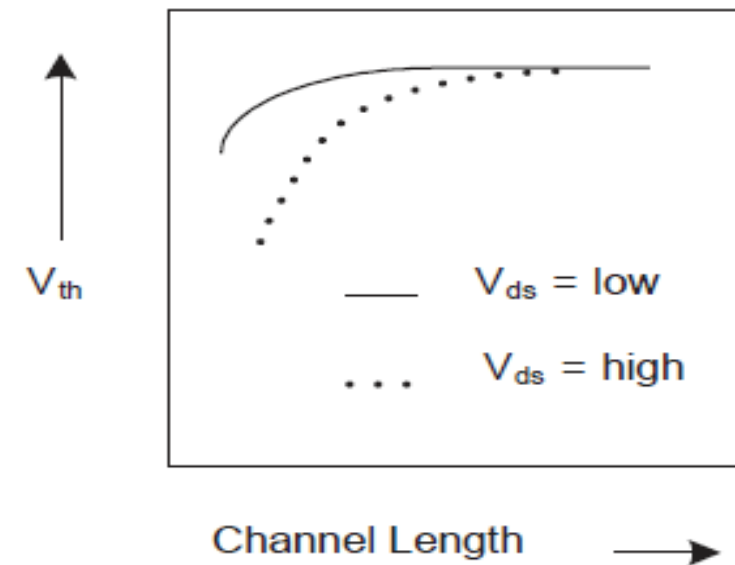


Fig. Threshold voltage roll-off with change in channel length

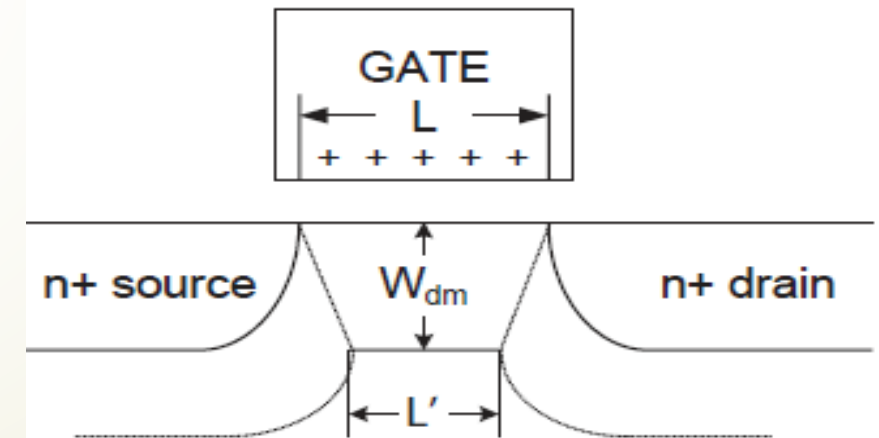


Fig. Schematic diagram for charge-sharing model

Temperature Effect

- There is a strong dependence of threshold voltage on temperature.
- The equation $S_t = 2.3 \frac{kT}{q} \left(1 + \frac{C_{dm}}{C_{ox}} \right)$ indicates that the subthreshold slope changes linearly with temperature.
- For a 0.3- μm technology, S_t varies from 58.2 to 81.9 mV/decade. As the temperature varies from -50 to $+25$ $^{\circ}\text{C}$ in the 0.3- μm technology as shown in Figure
- The I_{OFF} increases from 0.45 to 100 pA, an increase by a factor of 356, for a 20- μm -wide device. The following two parameters are responsible for increases in subthreshold leakage current:
 - (i) S_t increases linearly with temperature.
 - (ii) The threshold voltage decreases with temperature at the rate of 0.8 mV/ $^{\circ}\text{C}$.

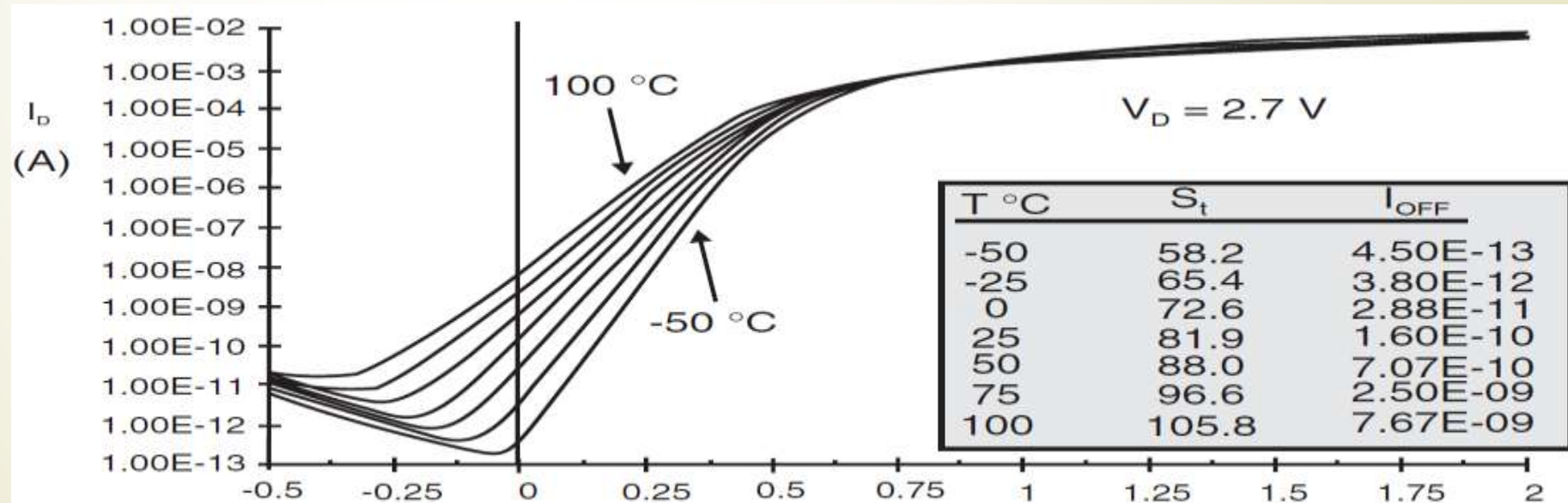


Fig. Variation of drain current with temperature

Hot-Carrier Injection

- ▶ Electrons and holes near the Si-SiO₂ interface can gain sufficient energy due to the high electric field in short-channel MOSFETs and cross the interface potential barrier and enter into the oxide layer as shown in Fig.
- ▶ This phenomenon is known as the hot-carrier injection. Because of the lower effective mass and smaller barrier height (3.1 eV) for electrons compared to holes (4.5 eV), the possibility of an injection due to electrons is more than the holes.

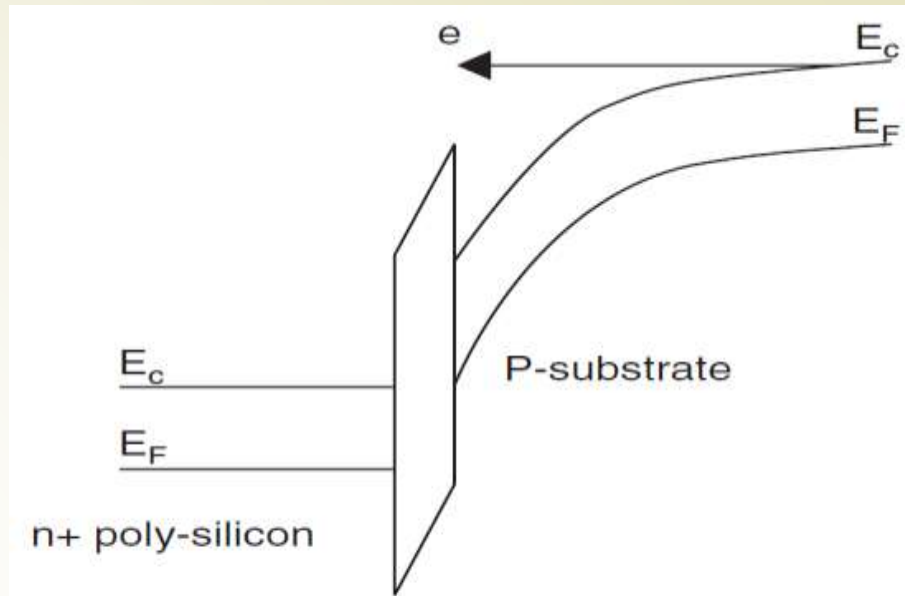
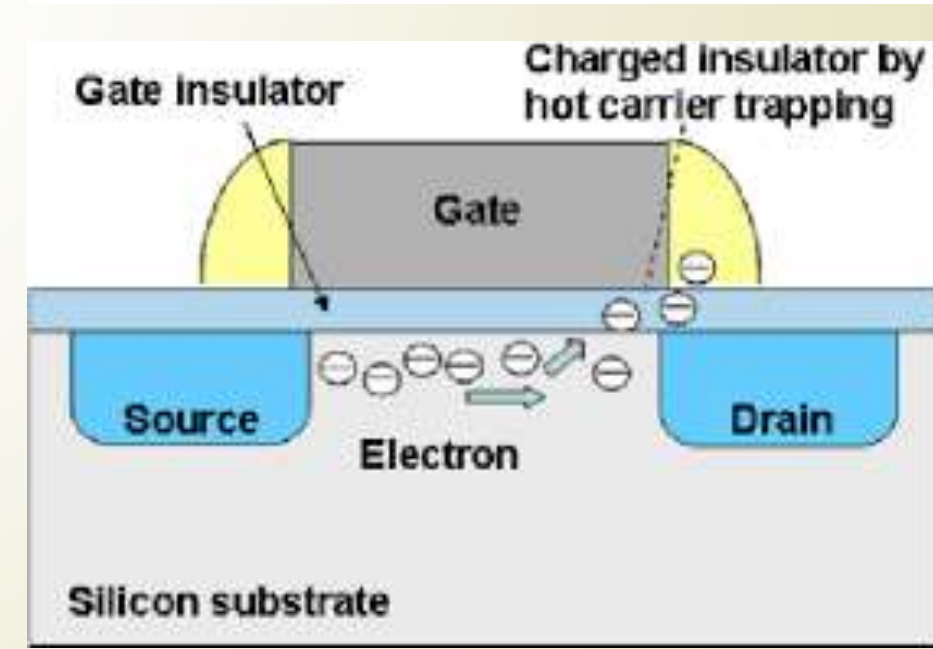
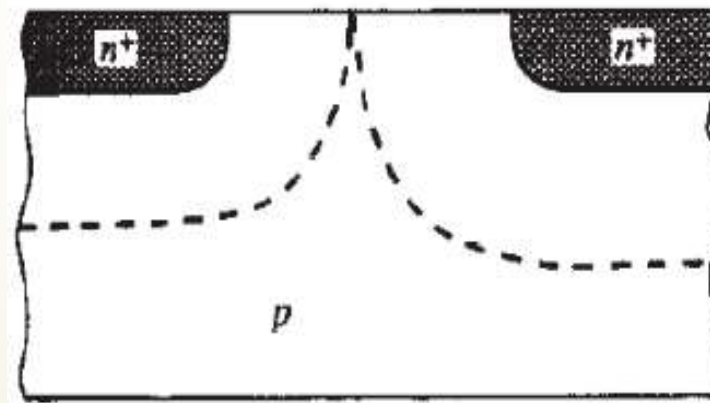


Fig. Injection of hot electrons from substrate to oxide

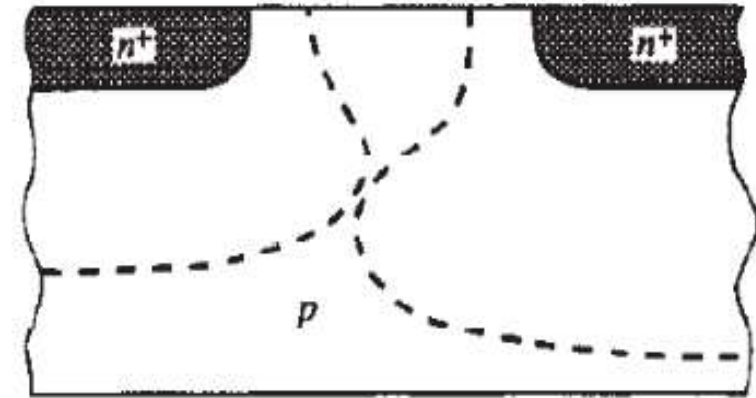


Punch Through

- The drain and source regions are very much close to each other in short channel devices. The depletion region across the drain-substrate and source-substrate junction extends into the channel region.
- The increase in V_{ds} further pushes the junction near to each other. Thus, the overall effect of short channel length and reverse biased substrate-drain junction leads to the merging of the two depletion regions, which cause punch through condition.



(a)



(b)

(a) Surface Punch-through and (b) Bulk Punch-through

Problem

- ▶ A digital system-on-chip in a 1V 65 nm process (with 50 nm drawn channel lengths and $\lambda = 25$ nm) has 1 billion transistors, of which 50 million are in logic gates and the remainder in memory arrays. The average logic transistor width is 12λ and the average memory transistor width is 4λ . Subthreshold leakage for OFF devices is 100 nA/ μm for low-threshold devices and 10 nA/ μm for high-threshold devices. Gate leakage is 5 nA/ μm . Junction leakage is negligible. Memories use low leakage devices everywhere. Logic uses low-leakage devices in all but 5% of the paths that are most critical for performance. Estimate the static power consumption
- ▶ Solution
- ▶ There are $(50 \times 10^6 \text{ logic transistors})(0.05)(12\lambda)(0.025 \mu\text{m}/\lambda) = 0.75 \times 10^6 \mu\text{m}$ of low-threshold devices and $[(50 \times 10^6 \text{ logic transistors})(0.95)(12\lambda) + (950 \times 10^6 \text{ memory transistors})(4\lambda)](0.025 \mu\text{m}/\lambda) = 109.25 \times 10^6 \mu\text{m}$ of high-threshold devices.
- ▶ $I_{\text{sub}} = [(0.75 \times 10^6 \mu\text{m})(100 \text{ nA}/\mu\text{m}) + (109.25 \times 10^6 \mu\text{m})(10 \text{ nA}/\mu\text{m})]/2 = 584 \text{ mA}$. $I_{\text{gate}} = [(0.75 + 109.25) \times 10^6 \mu\text{m}](5 \text{ nA}/\mu\text{m})/2 = 275 \text{ mA}$. $P_{\text{static}} = (584 \text{ mA} + 275 \text{ mA})(1 \text{ V}) = 859 \text{ mW}$.

The Power-Delay Product, or Energy per Operation

- ▶ The power-delay product (PDP) as a quality measure for a logic gate

$$PDP = P_{av}t_p$$

- ▶ The PDP presents a measure of energy, as is apparent from the units (Wsec = Joule). Assuming that the gate is switched at its maximum possible rate of $f_{\max} = 1/(2t_p)$. The power delay product can be written as –

$$PDP = C_L V_{DD}^2 f_{\max} t_p = \frac{C_L V_{DD}^2}{2}$$

- ▶ The PDP stands for the **average energy consumed per switching event**

Energy Delay Product

- The validity of the PDP as a quality metric for a process technology or gate topology is questionable. It measures the energy needed to switch the gate, which is an important property for sure.
- A more relevant metric should combine a measure of performance and energy. The energy-delay product (EDP) does exactly that.

$$EDP = PDP \times t_p = P_{av} t_p^2 = \frac{C_L V_{DD}^2}{2} t_p$$

- It is worth analyzing the voltage dependence of the EDP.
- Higher supply voltages reduce delay, but harm the energy,
- and the opposite is true for low voltages.

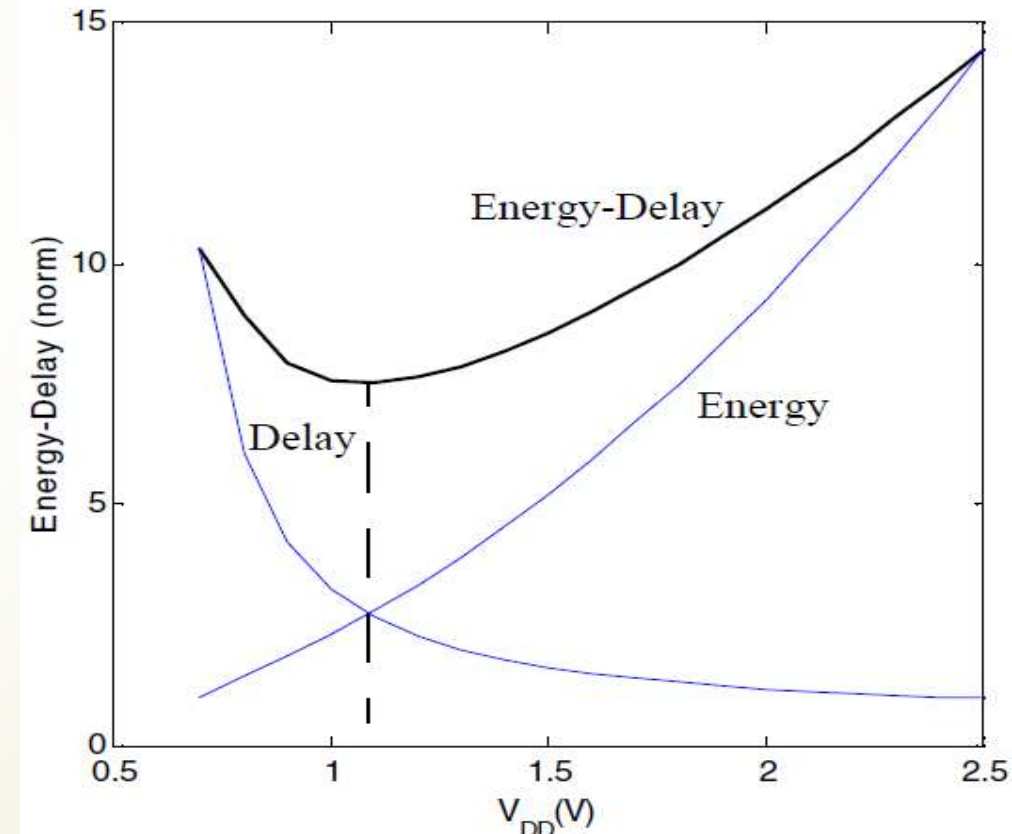


Figure Normalized delay, energy, and energy-delay plots for CMOS inverter



Thank You

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Delay

By

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Talk Outline

- Delay Definitions
- Timing Optimization
- RC Delay
- Problems
- Logical Effort

Delay Definitions

- **Propagation delay time, t_{pd}** = maximum time from the input crossing 50% to the output crossing 50%.
- **Contamination delay time, t_{cd}** = minimum time from the input crossing 50% to the output crossing 50%.
- **Rise time, t_r** = time for a waveform to rise from 20% to 80% of its steady-state value.
- **Fall time, t_f** = time for a waveform to fall from 80% to 20% of its steady-state value.
- **Edge rate, $t_{rf} = (t_r + t_f)/2$**

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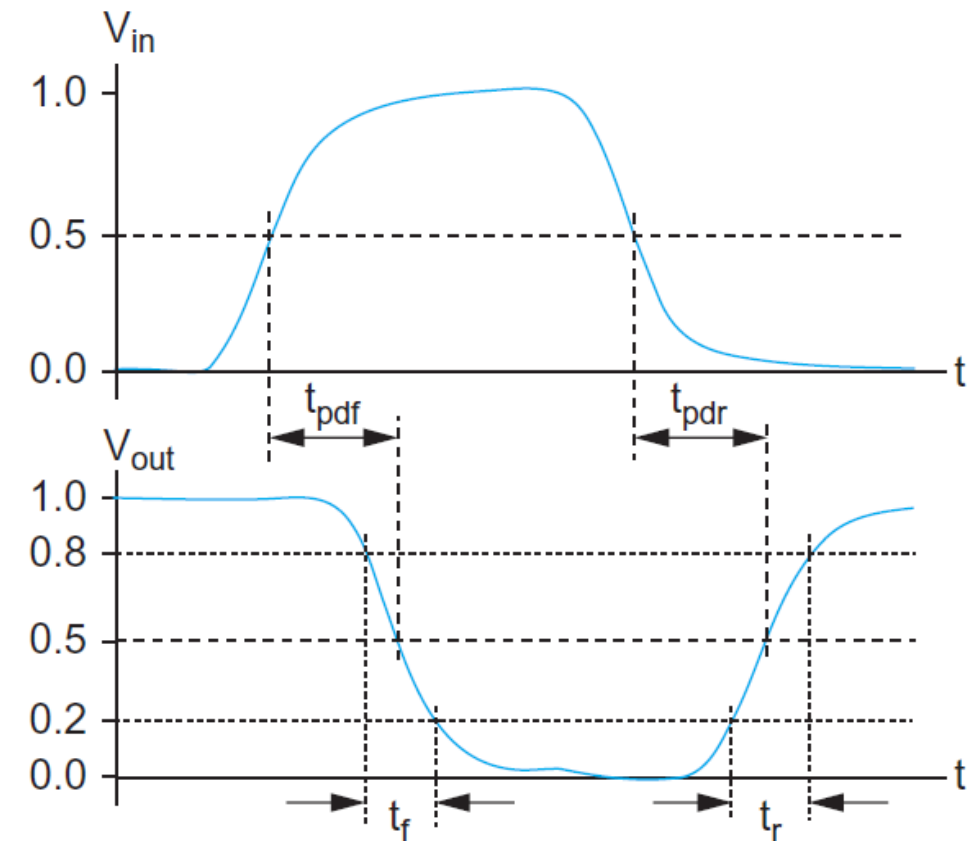


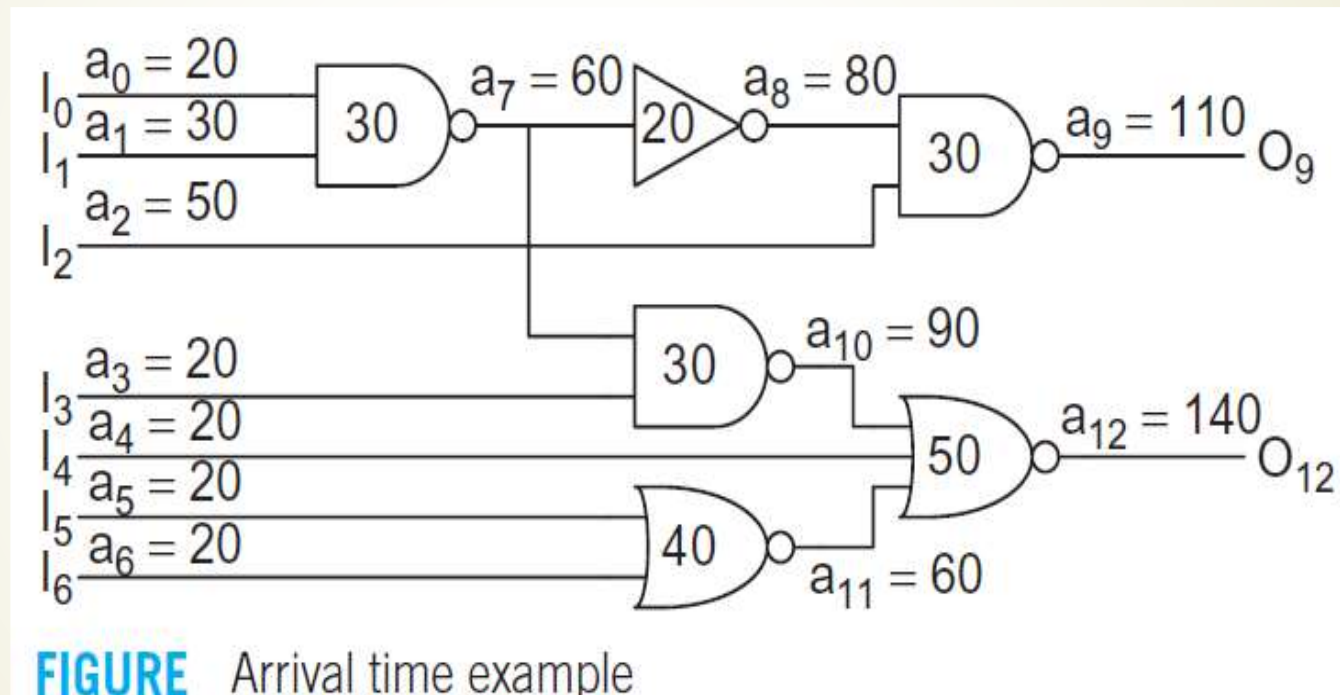
FIGURE Propagation delay and rise/fall times

Continue....

- when an input changes, the output will retain its old value for at least the contamination delay and take on its new value in at most the propagation delay.
- Rise/fall times are also sometimes called **slopes or edge rates**.
- Propagation and contamination delay times are also called **max-time** and **min-time**.
- The gate that charges or discharges a node is called the **driver** and the gates and wire being driven are called the **load**.
- Propagation delay is usually the most relevant value of interest, and is often simply called **delay**.
- The nodes are classified as inputs, outputs, and internal nodes. The user must specify the **arrival time** of inputs and the time data is required at the outputs.
- The timing analyzer computes the arrival times at each node and checks that the outputs arrive by their required time.
- The **slack** is the difference between the required and **arrival times**. **Positive slack** means that the circuit meets timing. **Negative slack** means that the circuit is not fast enough.

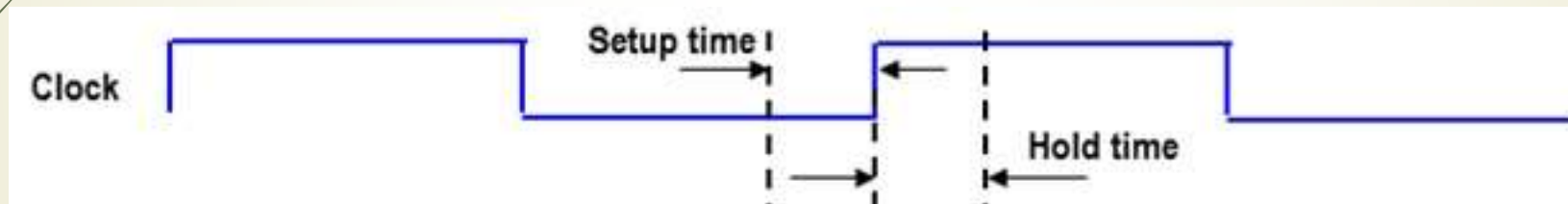
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- **Critical Path** - The **critical path** is the longest **path** in the circuit.



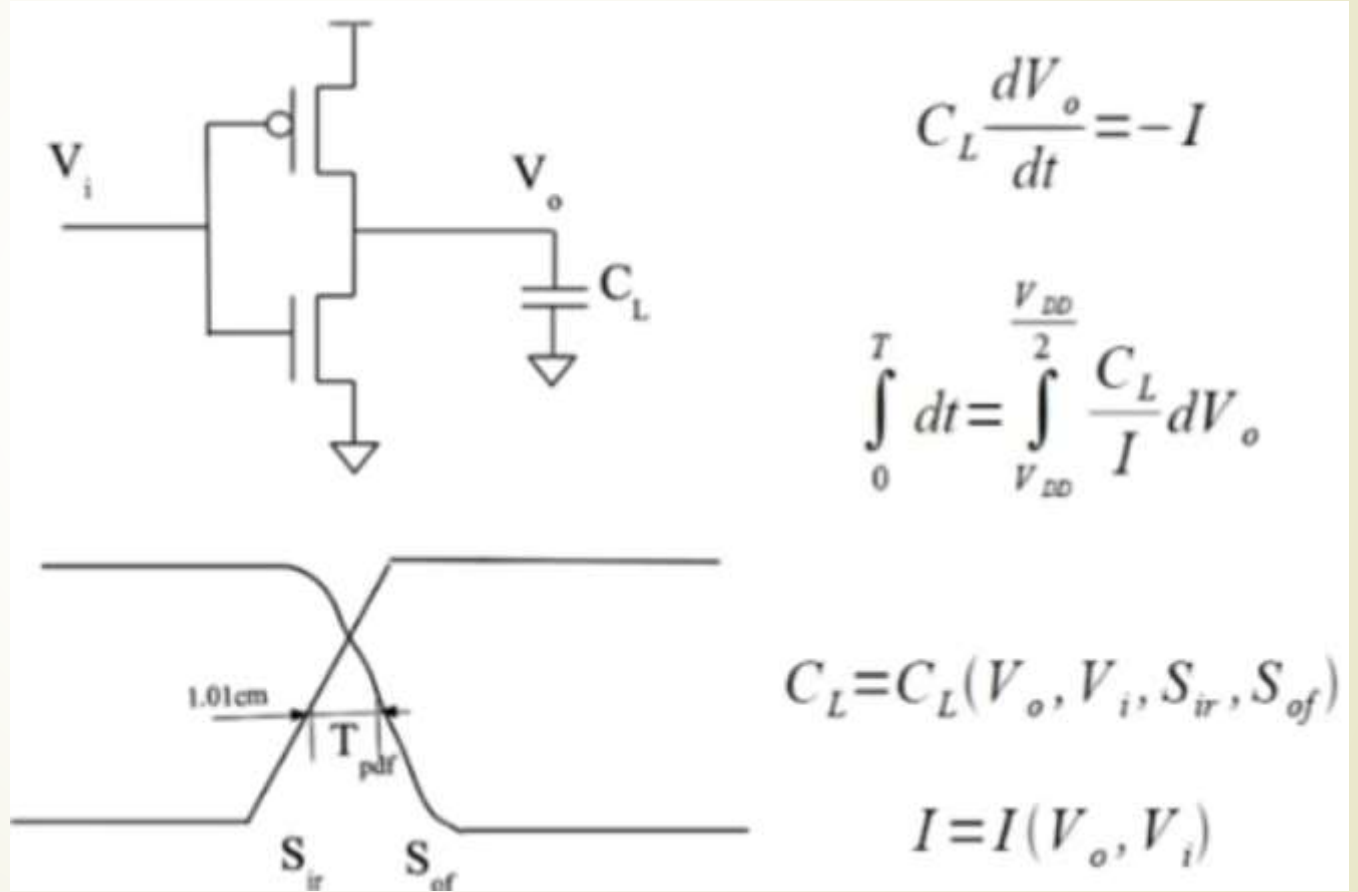
Continue.....

- **Setup Time:** The amount of time the data at the synchronous input (D) must be stable before the active edge of clock.
- **Hold Time:** The amount of time the data at the synchronous input (D) must be stable after the active edge of clock.

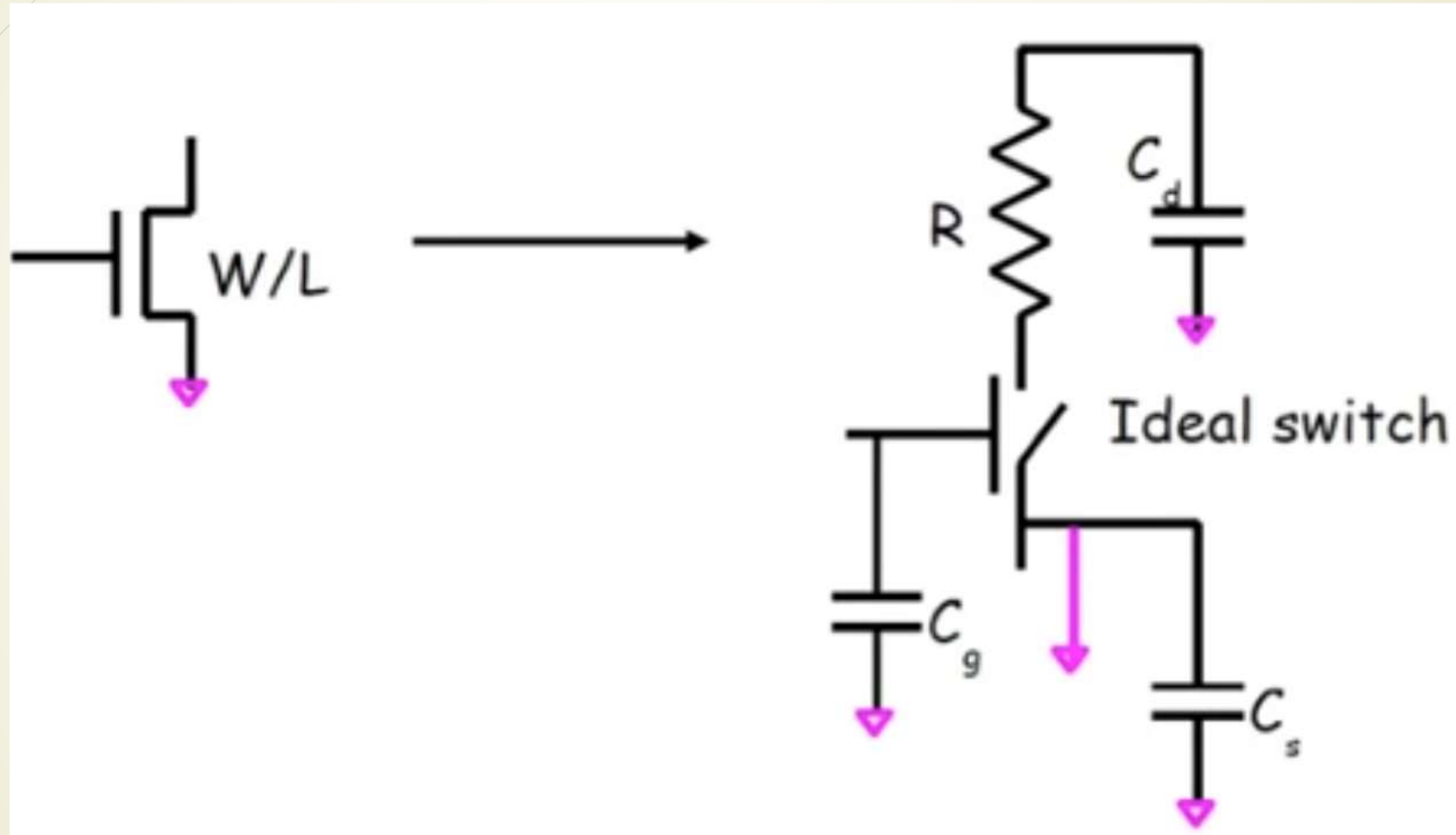


Delay of a single gate

➤ S_{ir} – Input rise & S_{of} – Output Fall



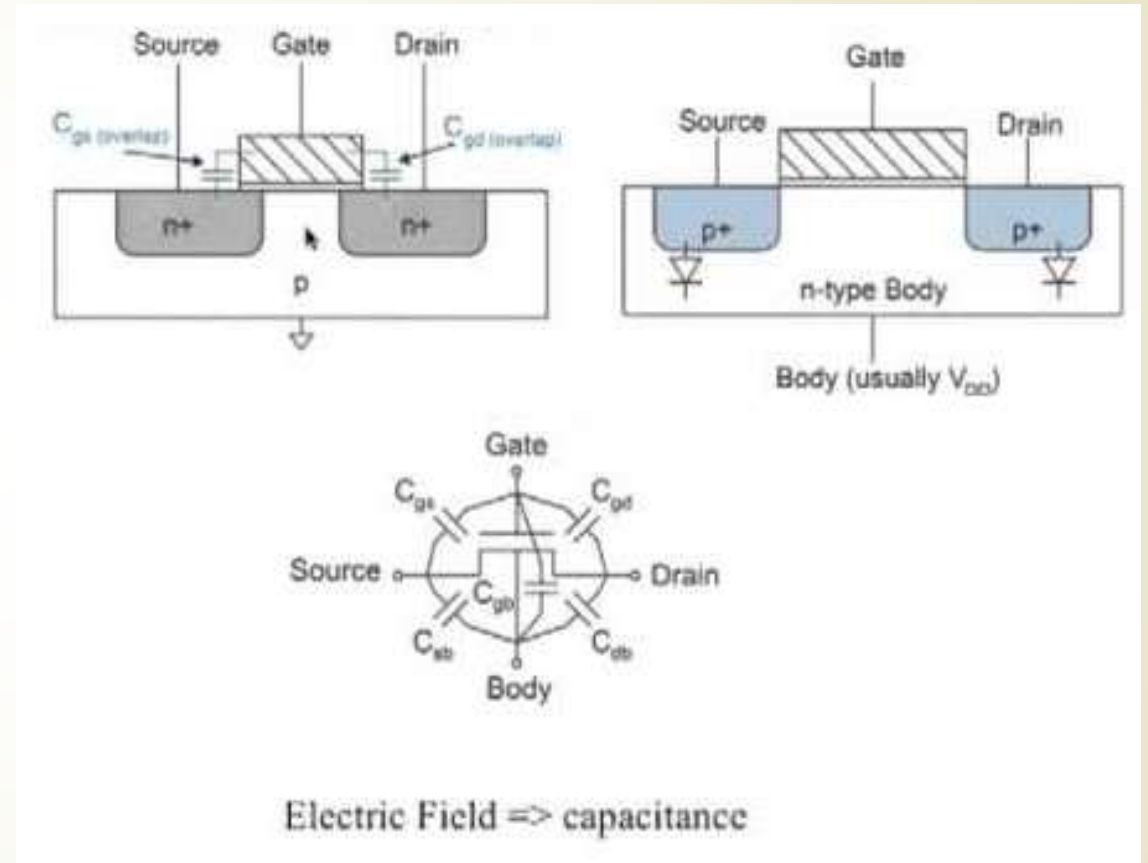
RC Model of a transistor



Capacitances

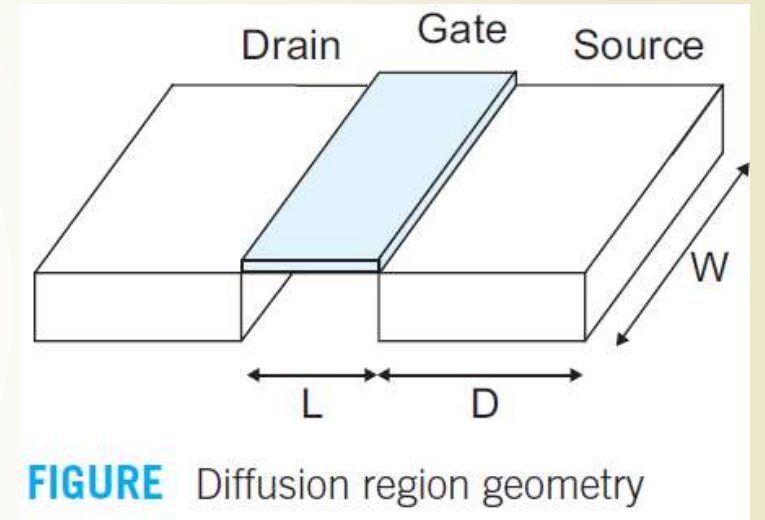
➤ Voltage dependence capacitance

➤ $C = dq/dv$



Simplified Model

- $C_s = AS \times C_{jbs} + PS \times C_{jbsw}$
- $AS = \text{Area of Diffusion region} = D \times W$
- $PS = \text{Perimeter of Diffusion region} = 2W + 2D$
- $C_{jb} = \text{Unit area capacitance to body}$
- $C_{jbsw} = \text{Units of capacitance/length}$



Sizing Impact on R and C

- Figure shows equivalent RC circuit models for nMOS and pMOS transistors of width k with contacted diffusion on both source and drain.
- The pMOS transistor has approximately twice the resistance of the nMOS transistor because holes have lower mobility than electrons.

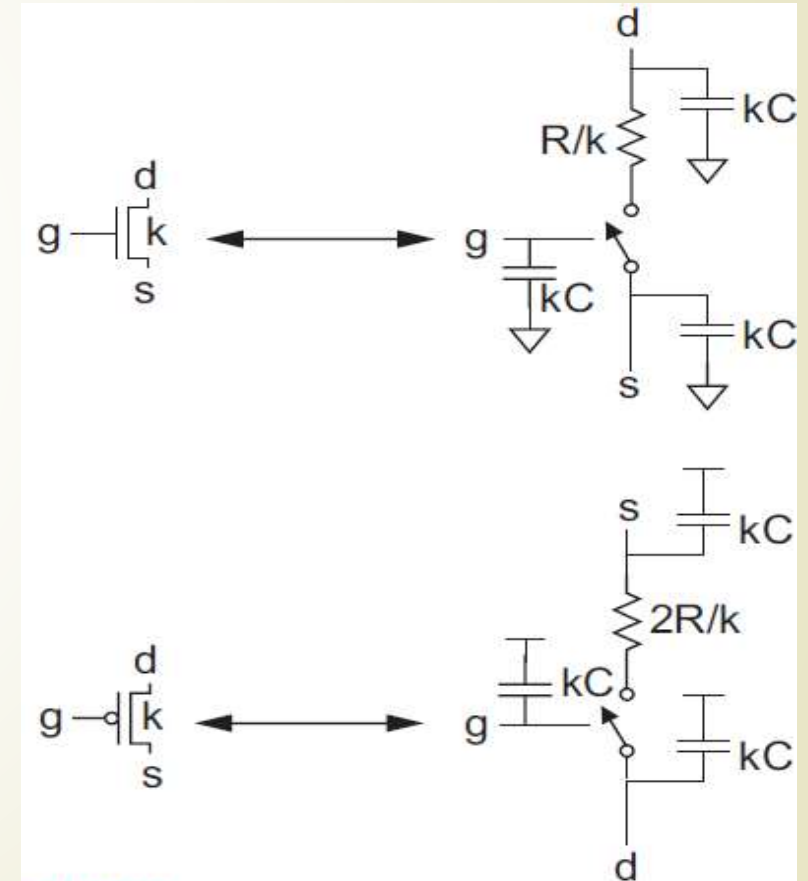
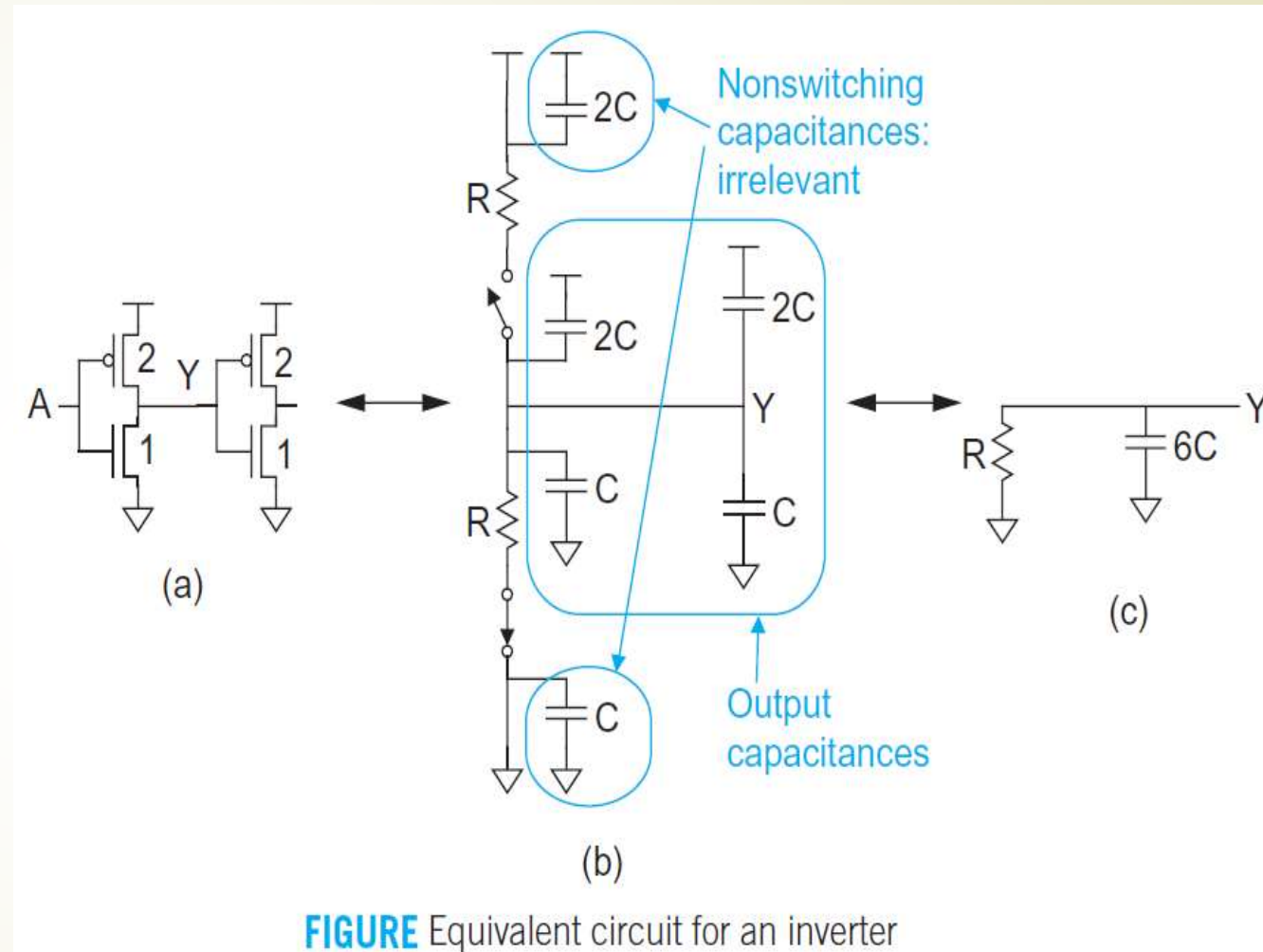


FIGURE
Equivalent circuits for transistors

Equivalent Circuit of Inverter

- Figure shows the equivalent circuit for a fanout-of-1 inverter with negligible wire capacitance.
- The unit inverters of Figure are composed from an nMOS transistor of unit size and a pMOS transistor of twice unit width to achieve equal rise and fall resistance.
- If input A rises, the nMOS transistor will be ON and the pMOS OFF.
- The capacitors shorted between two constant supplies are also removed because they are not charged or discharged

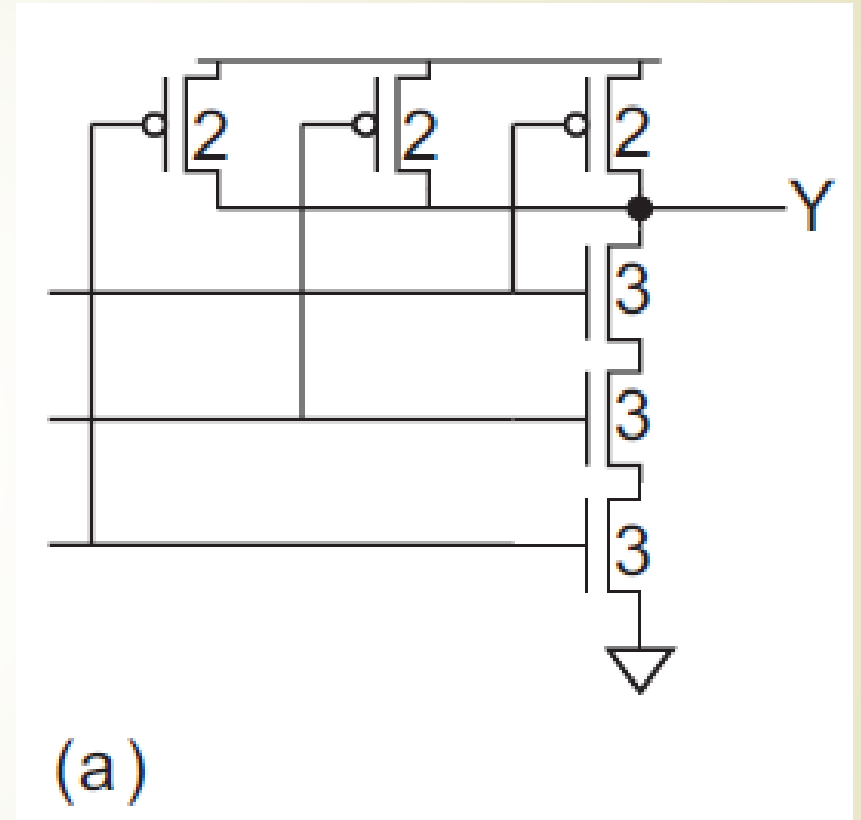


Problem 1

- ▶ **Q)** Sketch a 3-input NAND gate with transistor widths chosen to achieve effective rise and fall resistance equal to that of a unit inverter (R). Annotate the gate with its gate and diffusion capacitances. Assume all diffusion nodes are contacted. Then sketch equivalent circuits for the falling output transition and for the worst-case rising output transition ?

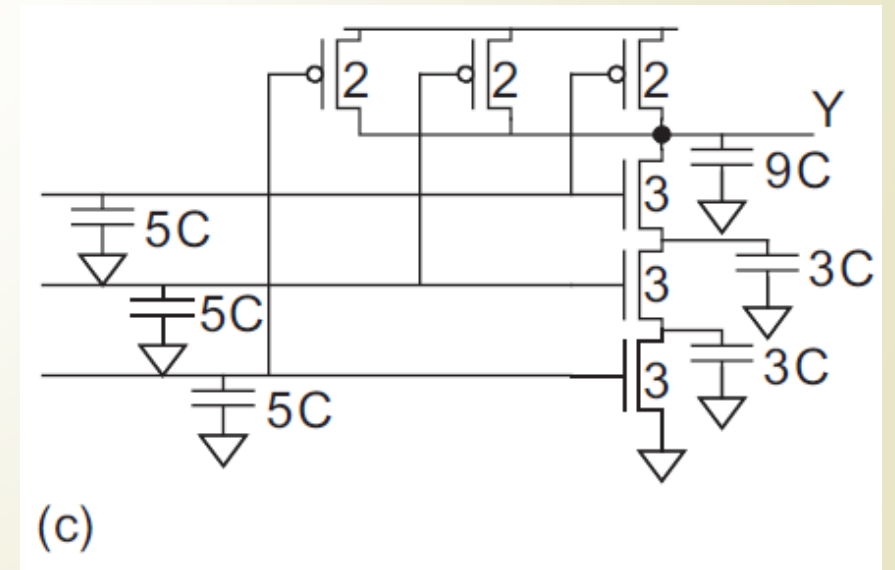
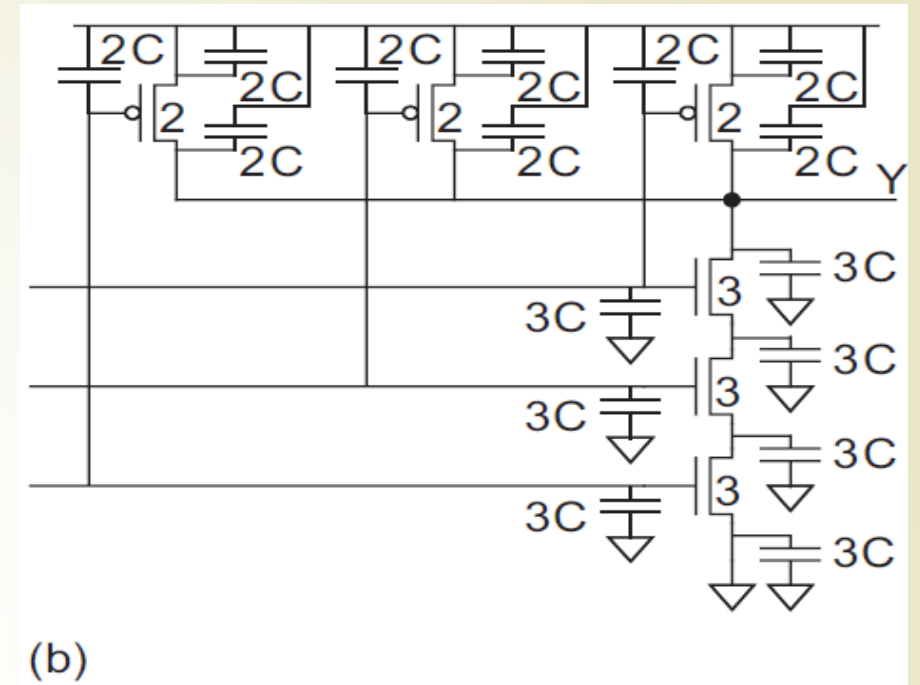
- ▶ **Solution :**

- ▶ The three nMOS transistors are in series so the resistance is three times that of a single transistor. Therefore, each must be three times unit width to compensate. In other words, each transistor has resistance $R/3$ and the series combination has resistance R . The two pMOS transistors are in parallel. In the worst case (with one of the inputs low), only one of the pMOS transistors is ON. Therefore, each must be twice unit width to have resistance R .



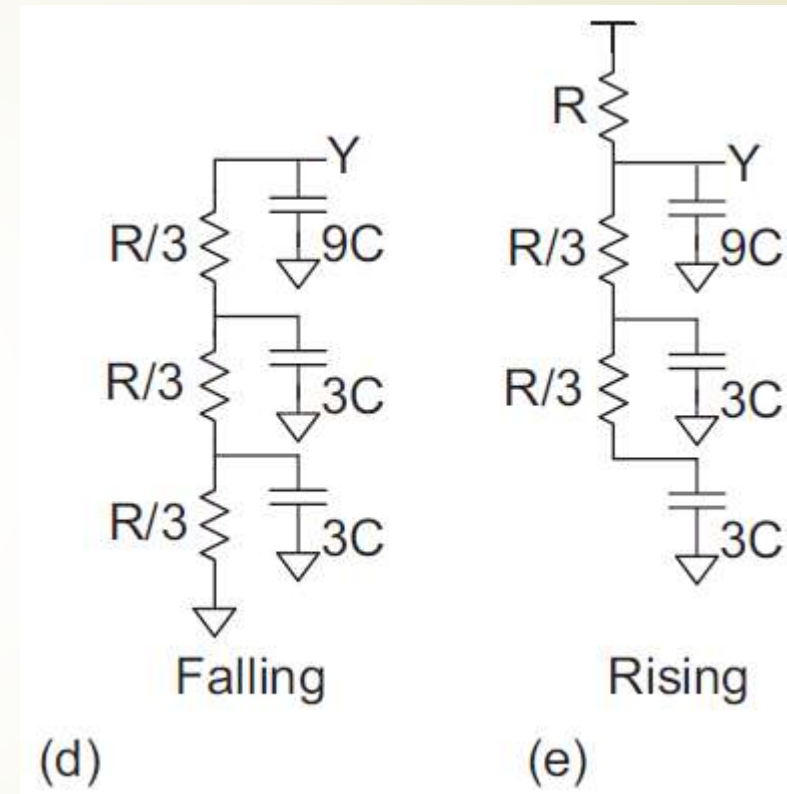
Continue....

- Figure (b) shows the capacitances. Each input presents five units of gate capacitance to whatever circuit drives that input. Notice that the capacitors on source diffusions attached to the rails have both terminals shorted together so they are irrelevant to circuit operation. Figure (c) redraws the gate with these capacitances deleted and the remaining capacitances lumped to ground.



Continue....

- Figure (d) shows the equivalent circuit for the falling output transition. The output pulls down through the three series nMOS transistors. Figure 4.7(e) shows the equivalent circuit for the rising output transition. In the worst case, the upper two inputs are 1 and the bottom one falls to 0. The output pulls up through a single pMOS transistor. The upper two nMOS transistors are still on, so the diffusion capacitance between the series nMOS transistors must also be discharged.



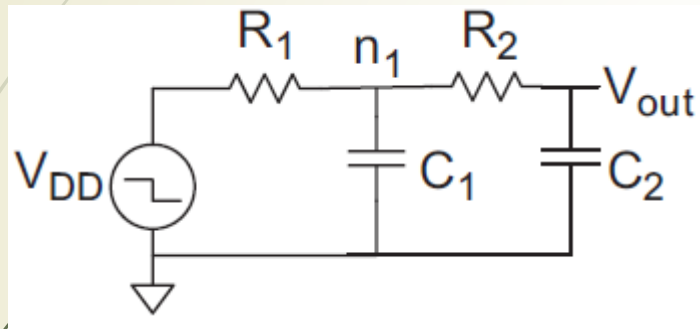
Elmore Delay

- Most circuits of interest can be represented as an *RC tree*, i.e., an RC circuit with no loops. The root of the tree is the voltage source and the leaves are the capacitors at the ends of the branches.
- The Elmore delay model estimates the delay from a source switching to one of the leaf nodes changing as the sum over each node i of the capacitance C_i on the node, multiplied by the effective resistance R_{is} on the shared path from the source to the node and the leaf.

$$t_{pd} = \sum_i R_{is} C_i$$

Problem2

- Q) Compute the Elmore delay for V_{out} in the 2nd order RC system from Figure?

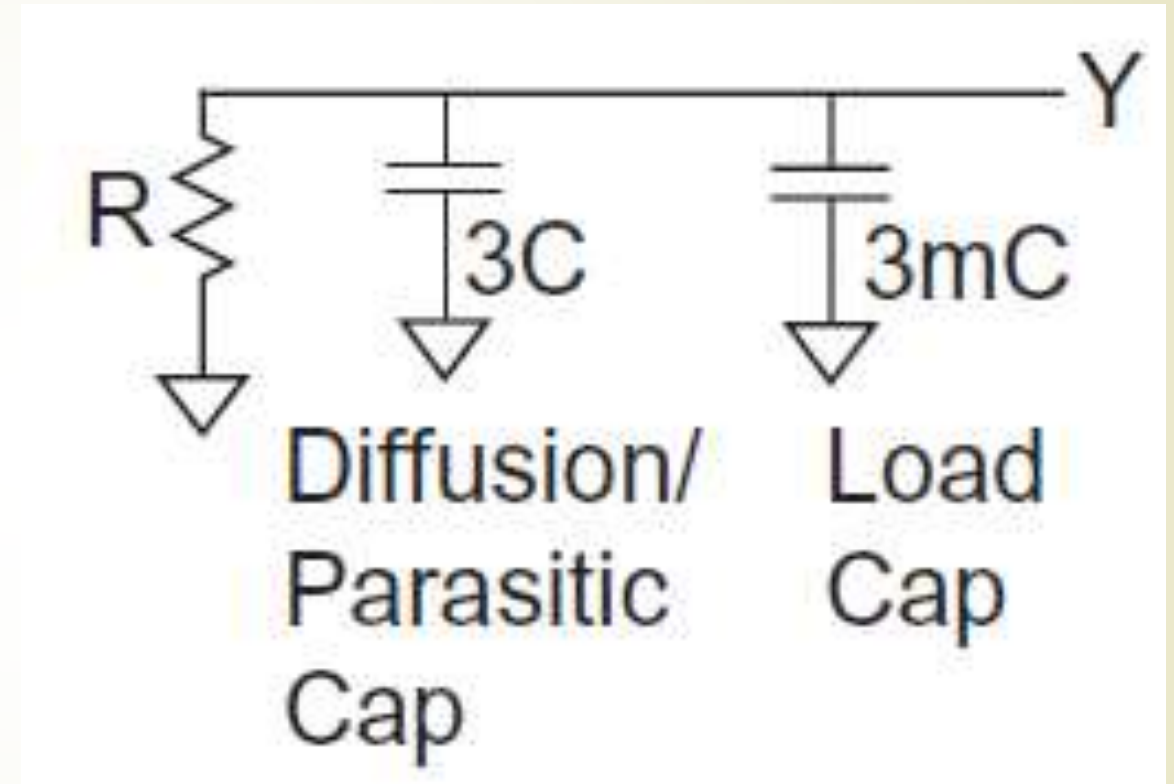


- Solution**

- The circuit has a source and two nodes. At node n_1 , the capacitance is C_1 and the resistance to the source is R_1 . At node V_{out} , the capacitance is C_2 and the resistance to the source is $(R_1 + R_2)$. Hence, the Elmore delay is $t_{pd} = R_1 C_1 + (R_1 + R_2) C_2$

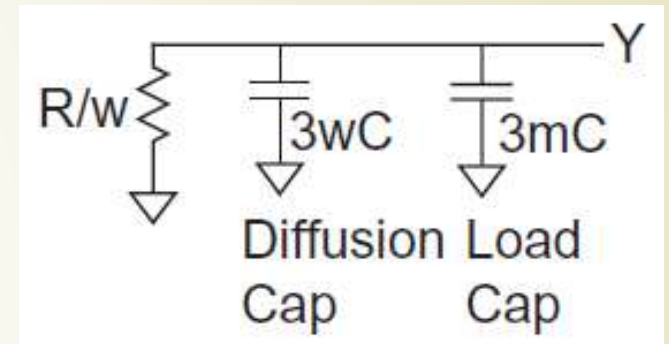
Problem3

- ▶ Q) Estimate **Elmore's delay** for a unit inverter driving m identical unit inverters.?
- ▶ Solution
- ▶ Each load inverter presents $3C$ units of gate capacitance, for a total of $3mC$. The output node also sees a capacitance of $3C$ from the drain diffusions of the driving inverter. This capacitance is called *parasitic* because it is an undesired side-effect of the need to make the drain large enough to contact. The parasitic capacitance is independent of the load that the inverter is driving. Hence, the total capacitance is $(3 + 3m)C$. The resistance is R , so the Elmore delay is $tpd = (3 + 3m)RC$. The equivalent circuit for the rising transition gives the same results.



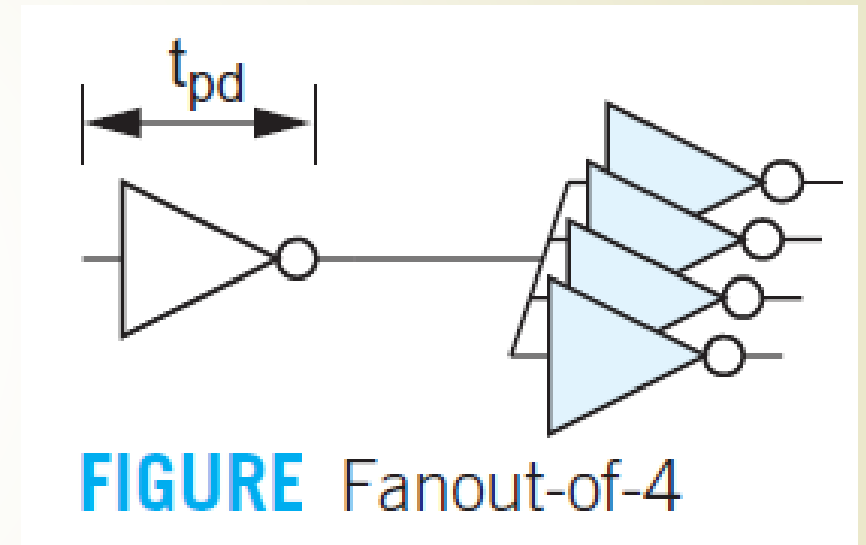
Problem4

- ▶ **Q)** Repeat the previous problem if the driver is w times unit size. Calculate the fan-out of a gate?
- ▶ **Solution**
- ▶ The driver transistors are w times as wide, so the effective resistance decreases by a factor of w . The diffusion capacitance increases by a factor of w . The Elmore delay is **$tpd = ((3w + 3m)C)(R/w) = (3 + 3m/w)RC$** .
- ▶ Define the *fanout* of the gate, h , to be the ratio of the load capacitance to the input capacitance. (Diffusion capacitance is not counted in the fanout.) The load capacitance is $3mC$. The input capacitance is $3wC$. Thus, the inverter has a fanout of $h = m/w$ and the delay can be written as $(3 + 3h)RC$



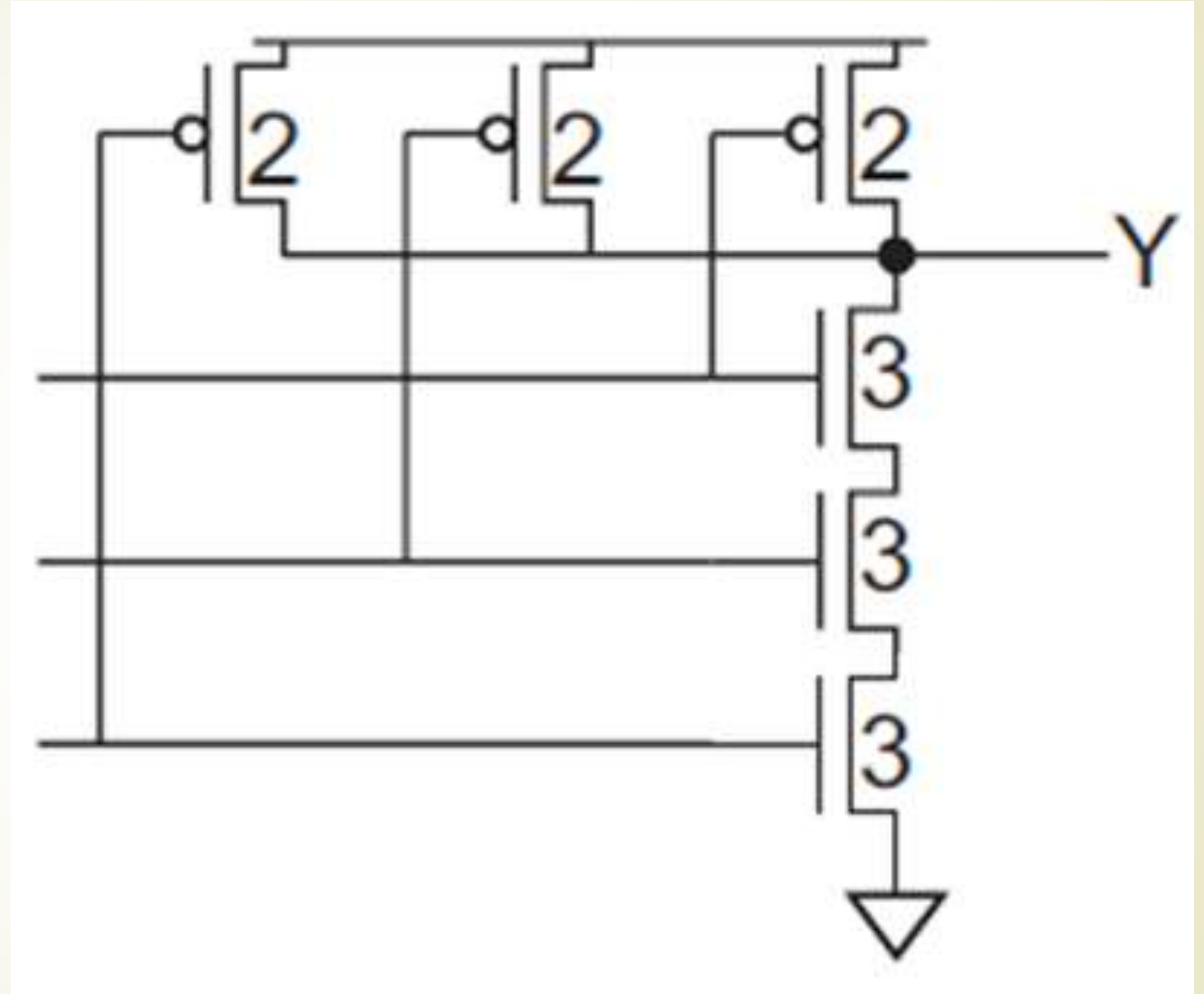
Problem5

- ▶ **Q)** If a unit transistor has $R = 10 \text{ k}\Omega$ and $C = 0.1 \text{ fF}$ in a 65 nm process, compute the delay, in picoseconds, of the inverter in Figure 4.14 with a fanout of $h = 4$?
- ▶ **Solution**
- ▶ The RC product in the 65 nm process is $(10 \text{ k}\Omega)(0.1 \text{ fF}) = 1 \text{ ps}$. For $h = 4$, the delay is $(3 + 3h)(1 \text{ ps}) = 15 \text{ ps}$. This is called the *fanout-of-4 (FO4) inverter delay* and is representative of gate delays in a typical circuit.



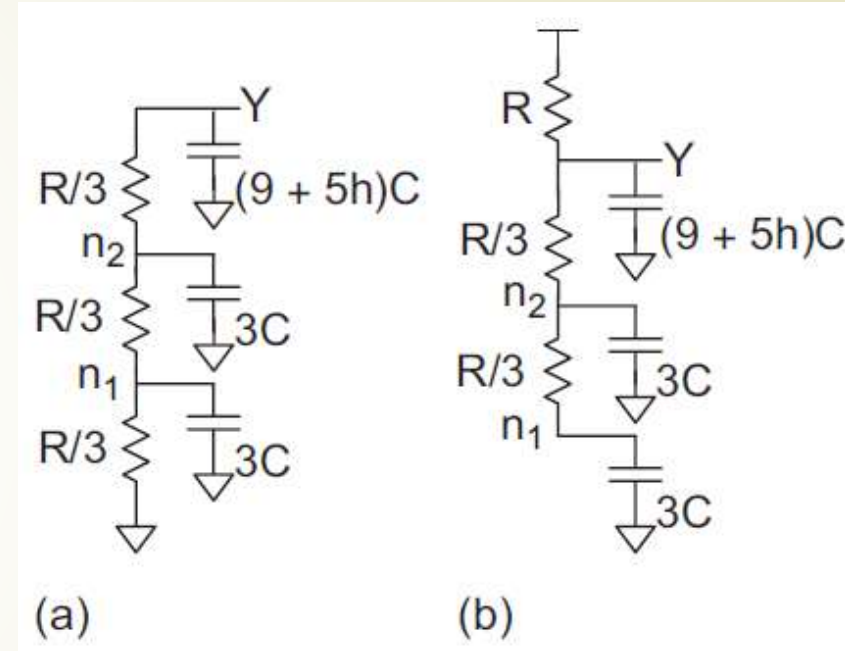
Problem 6

- Q) Estimate t_{pdf} and t_{pdr} for the 3-input NAND gate from given figure (Problem1) if the output is loaded with h identical NAND gates?



Solution

- Each NAND gate load presents 5 units of capacitance on a given input. Figure (a) shows the equivalent circuit including the load for the falling transition. Node $n1$ has capacitance $3C$ and resistance of $R/3$ to ground. Node $n2$ has capacitance $3C$ and resistance $(R/3 + R/3)$ to ground. Node Y has capacitance $(9 + 5h)C$ and resistance $(R/3 + R/3 + R/3)$ to ground. The Elmore delay for the falling output is the sum of these RC products, $tpdf = (3C)(R/3) + (3C)(R/3 + R/3) + ((9 + 5h)C)(R/3 + R/3 + R/3) = (12 + 5h)RC$.
- Figure 4.15(b) shows the equivalent circuit for the falling transition. In the worst case, the two inner inputs are 1 and the outer input falls. Y is pulled up to VDD through a single pMOS transistor. Node Y has capacitance $(9 + 5h)C$ and resistance R to the VDD supply. Node n_2 has capacitance $3C$
- The relevant resistance is only R , not $(R + R/3)$, because the output is being charged only through R . This is what is meant by the resistance on the shared path from the source (VDD) to the node ($n2$) and the leaf (Y). Similarly, node $n1$ has capacitance $3C$ and resistance R . Hence, the Elmore delay for the rising output is $tpdr = (3C)R + (3C)R + (9 + 5h)R = (15 + 5h)RC$.





Thank You

Rohit Lorenzo, VIT-AP

CMOS VLSI Design

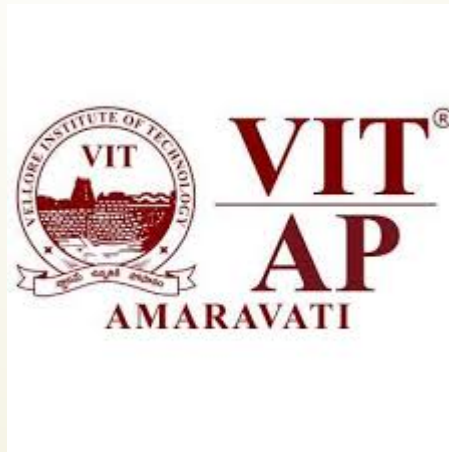
Topic – Logical Effort

By

Dr.Rohit Lorenzo

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Topic – Logical Effort

Talk Outline

- Introduction
- Delay in logic gate
- Multistage Logic Networks
- Example
- Summary

Introduction

- ▶ Chip designer face number of choices like –
 - What is the best topology for a function?
 - How many stage of logic gives best function?
 - How wide should the transistor be?
- ▶ Logical effort is a method to take decisions -
 - Uses a simple model of delay.
 - Helps make rapid comparisons between alternatives.
 - Emphasizes remarkable symmetries.

Delay in a logic gate.

- Express delay in process independent unit.

$$d = \frac{d_{abs}}{\tau}$$

$$\tau = 3RC$$

$\approx$ 12ps in 180nm process

40ps in 0.6 μ m process

- Delay has two components

$$D = f + p$$

- Effort delay $f = gh$ (stage effort) where h is electrical effort

- again has two components.

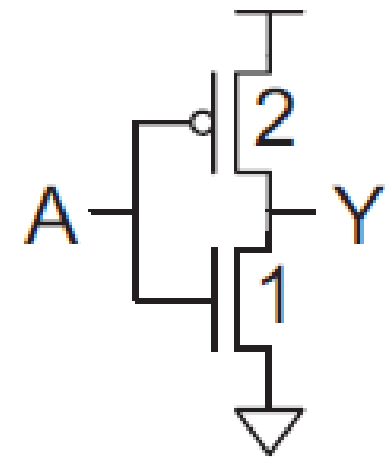
- g : logical effort

- Measure the relative ability to deliver the current

- $g = 1$ for inverter

Computing logical effort

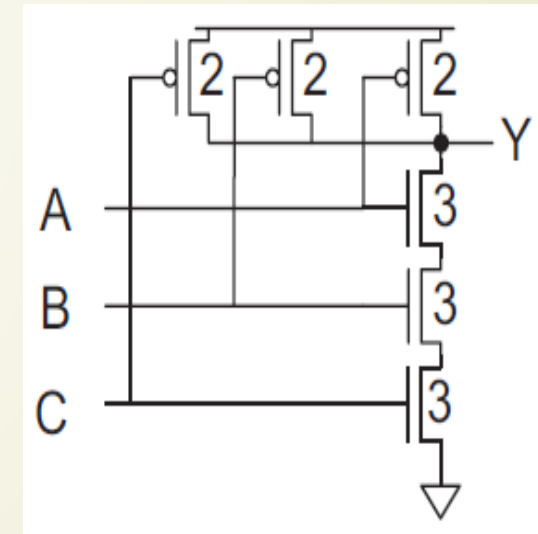
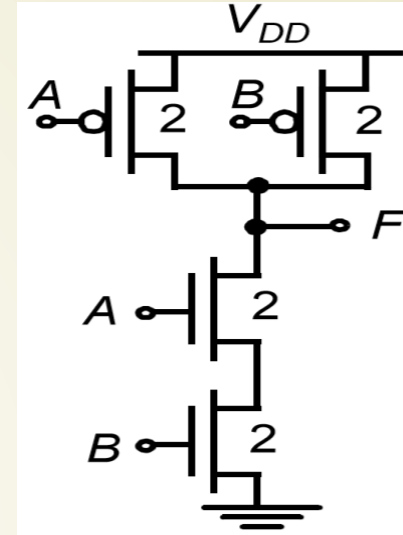
- **Definition:** The Logical effort is the ratio of the input capacitance of the gate to the input capacitance of an inverter delivering the same output current.
- **Inverter**
- Logical Effort of basic gate $g = C_{in, \text{gate}} / C_{in, \text{inverter}}$
- $C_{in, \text{inverter}} = 2\text{unit} + 1\text{Unit} = 3\text{Unit}$



(a) $C_{in} = 3$
 $g = 3/3$

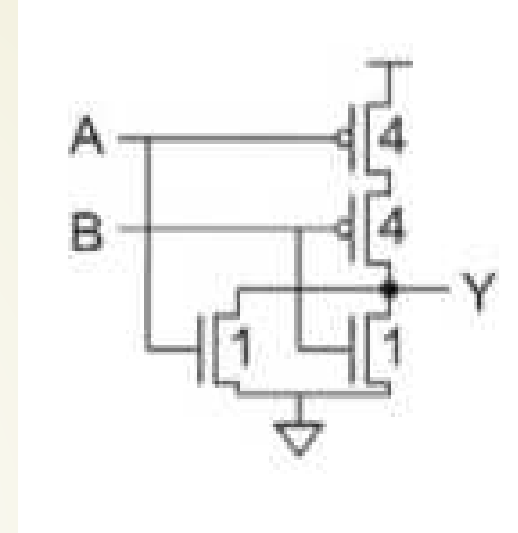
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- ▶ 2 input NAND gate
- ▶ $C_{in} \text{ NAND2} = 2\text{unit(NMOS)} + 2\text{ unit(PMOS)} = 4 \text{ Unit}$
- ▶ The NAND presents 4units of capacitance on each input.
- ▶ $g = C_{in\text{gate}}/C_{in\text{inverter}} = 4/3$
- ▶ 3 Input NAND Gate
- ▶ $C_{in} \text{ NAND3} = 3\text{unit(NMOS)} + 2\text{ unit(PMOS)} = 5 \text{ Unit}$
- ▶ The NAND presents 5units of capacitance on each input.
- ▶ $g = C_{in\text{gate}}/C_{in\text{inverter}} = 5/3$
- ▶ G for NAND Gate
- ▶ $(n+2)/3$, where n is number of inputs



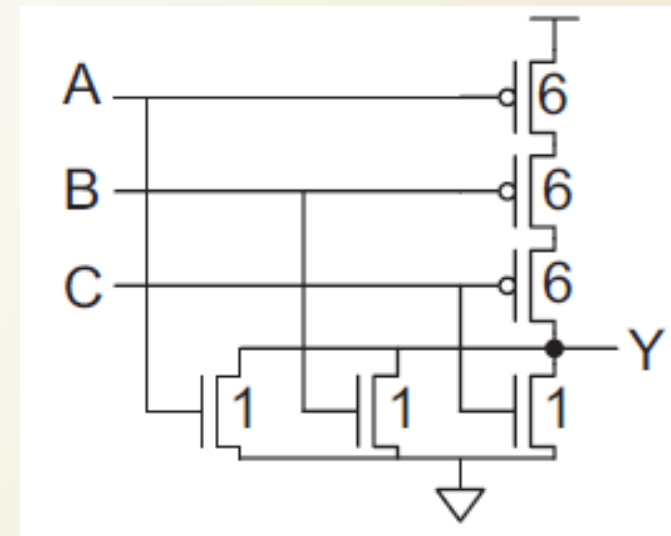
Continue.....

- 2 input NOR gate
- $C_{in} \text{ NOR2} = 1\text{unit(NMOS)} + 4\text{unit(PMOS)} = 5 \text{ Unit}$
- The NOR presents 5units of capacitance on each input.
- $g = C_{in\text{gate}}/C_{in\text{inverter}} = 5/3$



- 3 Input NOR Gate
- $C_{in} \text{ NOR3} = 1\text{unit(NMOS)} + 6\text{unit(PMOS)} = 7\text{Unit}$
- The NOR presents 5units of capacitance on each input.
- $g = C_{in\text{gate}}/C_{in\text{inverter}} = 7/3$

- G for NOR Gate
- $(2n+1)/3$, where n is number of inputs



Logical Effort of Common gate

Gate Type	Number of Inputs				
	1	2	3	4	n
inverter	1				
NAND		$4/3$	$5/3$	$6/3$	$(n + 2)/3$
NOR		$5/3$	$7/3$	$9/3$	$(2n + 1)/3$

Parasitic Delay

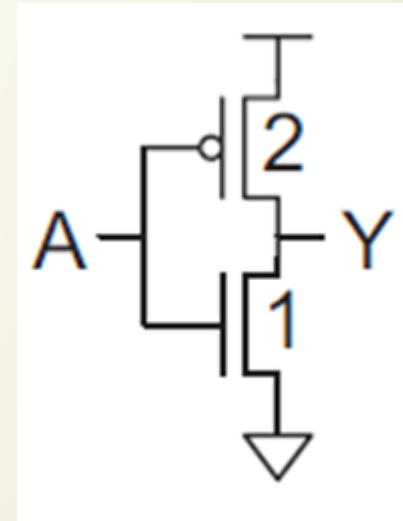
- **P** – parasitic delay
- Represents delay of gate driving no load.
- Set by internal parasitic capacitance.

$$P = \frac{C_{int} Gate}{C_{int} INV}$$

Inverter

$$C_{int} INV = 3$$

$$P = \frac{3}{3} = 1$$



Continue....

► P (NAND2)

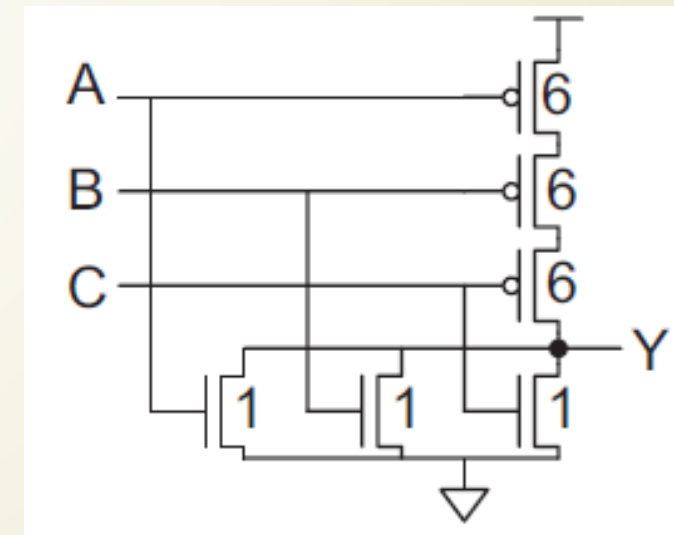
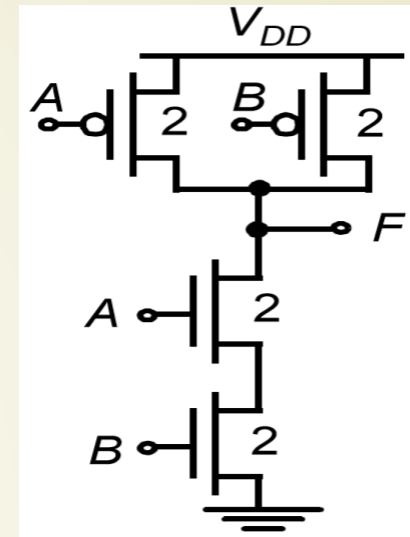
► $C_{in} \text{ NAND2} = 2+2+2 = 6$

► $P = \frac{C_{int} \text{ NAND2}}{C_{int} \text{ INV}} = \frac{6}{3} = 2$

► P (NOR3)

► $C_{in} \text{ NOR3} = 6+1+1+1 = 9$

► $P = \frac{C_{int} \text{ Nor3}}{C_{int} \text{ INV}} = \frac{9}{3} = 3$



Parasitic Delay of Common Gate

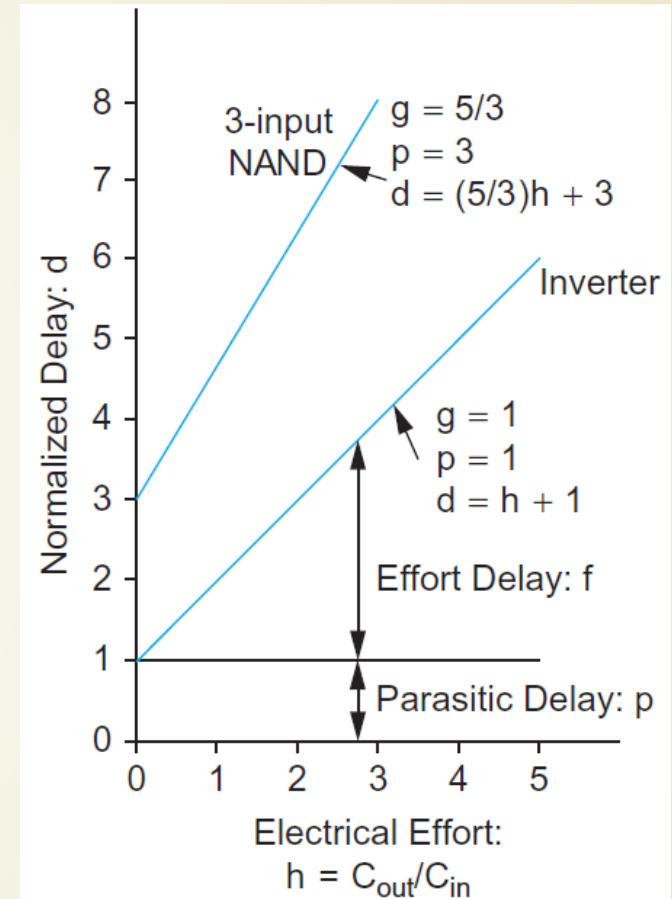
Gate Type	Number of Inputs				
	1	2	3	4	n
inverter	1				
NAND		2	3	4	n
NOR		2	3	4	n

Linear Delay Model

- In general, the normalized delay of a gate can be expressed as
- $d = f + p$
- p is the *parasitic delay*, f is the *effort delay* or *stage effort*
- $f = gh$
- The complexity is represented by the *logical effort* g
- A gate driving h identical copies of itself is said to have a *fanout* or *electrical effort* of h .
- If the load does not contain identical copies of the gate, the electrical effort can be computed as

$$h = \frac{C_{OUT}}{C_{in}}$$

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FIGURE

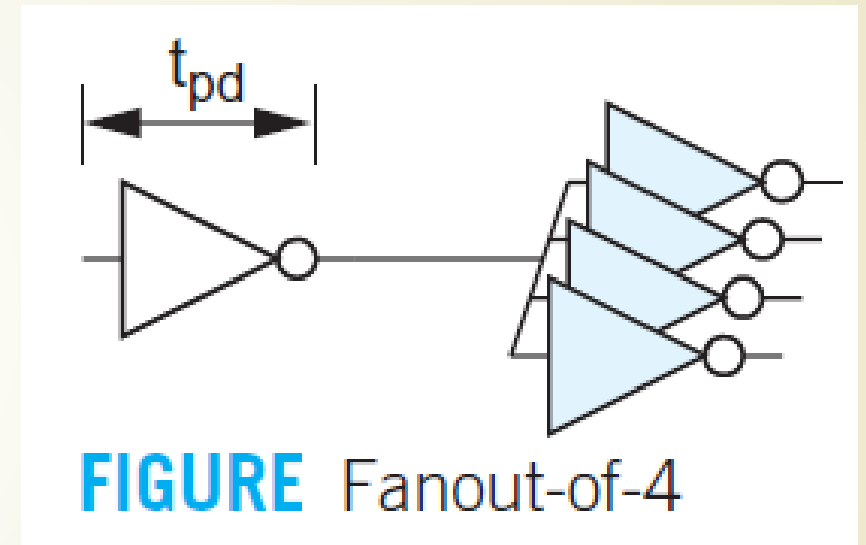
Normalized delay vs. fanout

Problem

- Q) Use the linear delay model to estimate the delay of the fanout-of-4 (FO4) inverter. Assume the inverter is constructed in a 65 nm process with $\tau = 3$ ps?

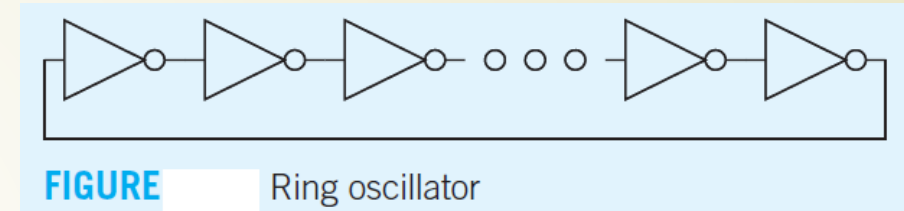
- Solution**

- The logical effort of the inverter is $g = 1$, by definition. The electrical effort is 4 because the load is four gates of equal size. The parasitic delay of an inverter is $p_{\text{inv}} = 1$. The total delay is $d = gh + p = 1 \times 4 + 1 = 5$ in normalized terms, or $t_{\text{pd}} = 15$ ps in absolute terms.



Problem

- Q) A ring oscillator is constructed from an odd number of inverters, as shown in Figure . Estimate the frequency of an N -stage ring oscillator.?



- Solution**
- The logical effort of the inverter is $g = 1$, by definition. The electrical effort of each inverter is also 1 because it drives a single identical load. The parasitic delay is also 1. The delay of each stage is $d = gh + p = 1 \times 1 + 1 = 2$. An N -stage ring oscillator has a period of $2N$ stage delays because a value must propagate twice around the ring to regain the original polarity. Therefore, the period is $T = 2 \times 2N$. The frequency is the reciprocal of the period, $1/4N$.

Logical Efforts in Multi-Stage Logic Network

- Logical effort ensures to get least overall delay by balancing the delay among stages .
- The logical effort along a path compounds by multiplying the logical efforts of all the logic gates
- The *path logical effort* $\mathbf{G}$ can be expressed as the products of the logical efforts of each stage along the path.

$$G = \prod g_i$$

$$H = \frac{C_{\text{out(path)}}}{C_{\text{in(path)}}}$$

- The *path electrical effort* H
- The *path effort* F is the product of the stage efforts of each stage. Recall that the stage effort of a single stage is $f = gh$.

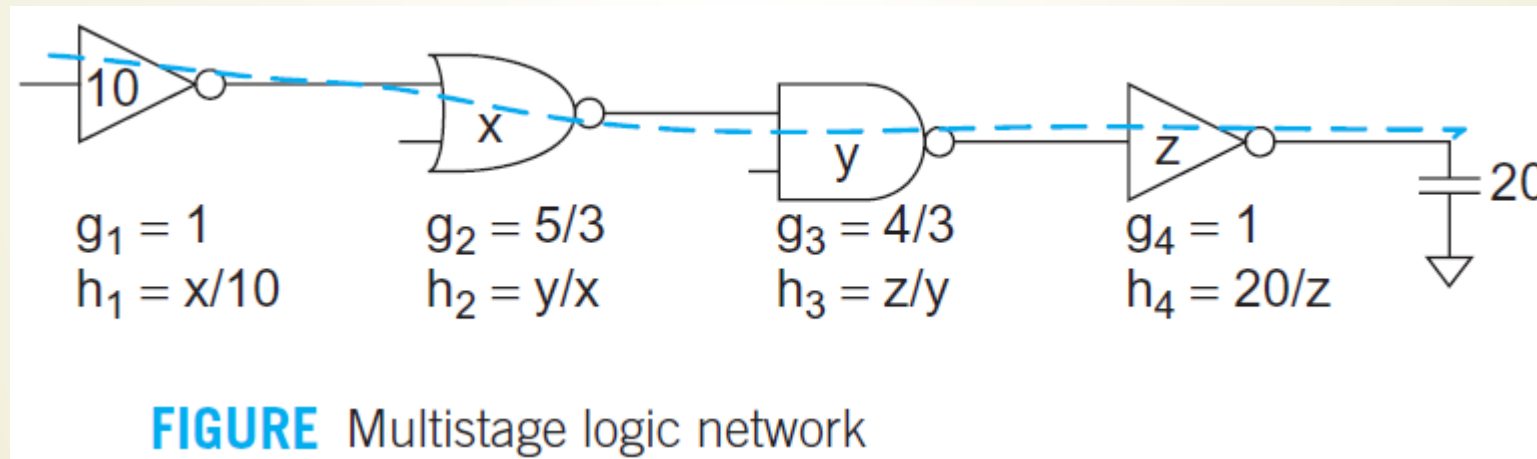
$$F = \prod f_i = \prod g_i h_i$$

- If a path has N stages and each bears the same effort, that effort must be

$$\hat{f} = g_i h_i = F^{1/N}$$

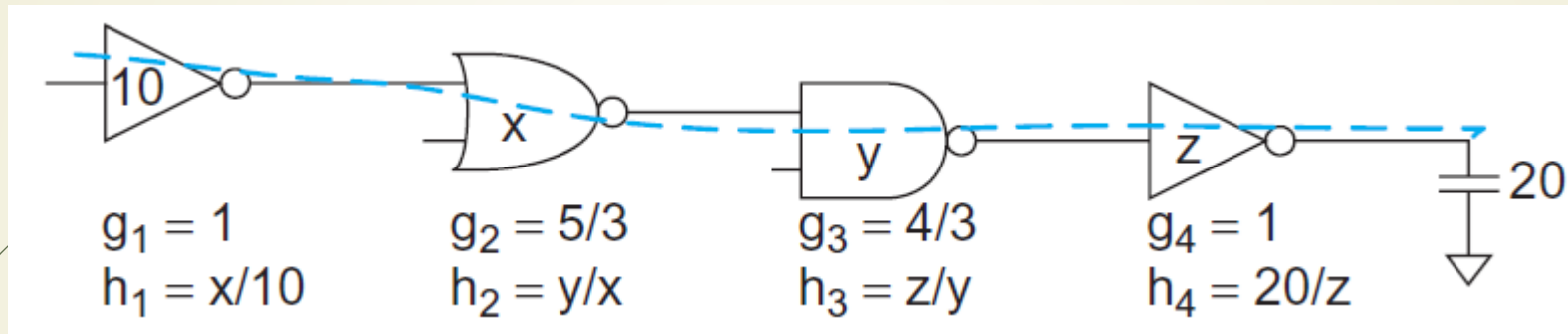
Example

- Figure shows the logical and electrical efforts of each stage in a multistage path as a function of the sizes of each stage. Observe that logical effort is independent of size, while electrical effort depends on sizes.



Problem

- Q) Calculate the size of each gate?



Solution

$$H = \frac{C_{out}}{C_{in}} = \frac{20}{10}$$

$$G = g_1 \times g_2 \times g_3 \times g_4 = 1 \times \frac{5}{3} \times \frac{4}{3} \times 1 = \frac{20}{9}$$

Continue....

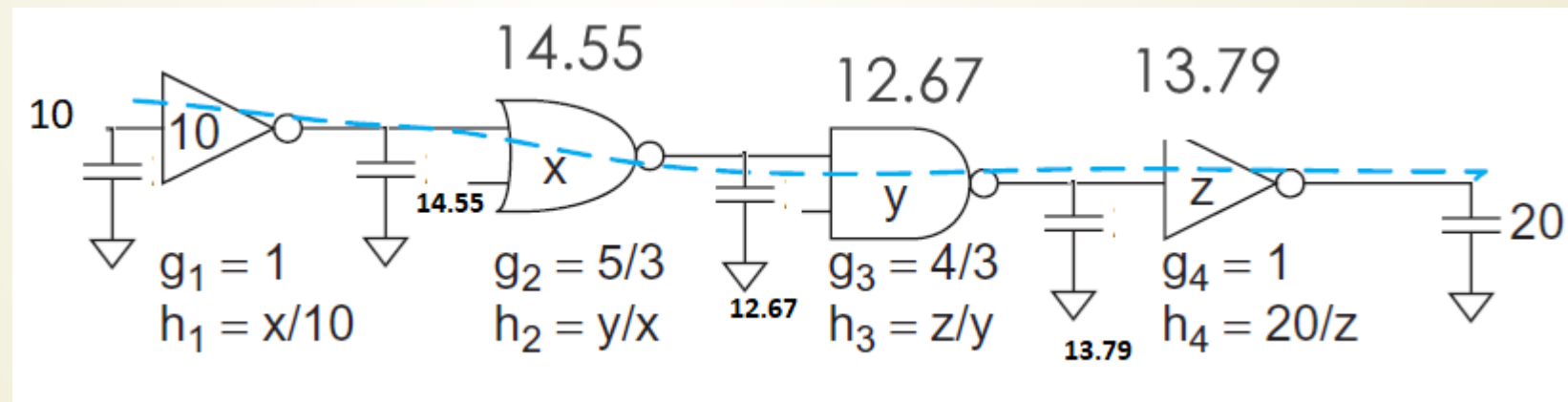
- $P = P_1 + P_2 + P_3 + P_4 = 1 + 2 + 2 + 1 = 6$
- $D = f + P$
- $D = GH + P = \frac{20}{9} \times 2 + 6 = 10.44$
- $F = GH = \frac{20}{9} \times 2 = 4.44$
- $N = 4$
- $F = f^N$
- $f = 1.451$

Continue....

- $f_4 = g_4 \times h_4$
- $g_4 = 1$, $f_4 = 1.451$, $h_4 = \frac{f_4}{g_4} = \frac{1.451}{1}$
- $h_4 = \frac{20}{Z}$, $Z = \frac{20}{1.45} = 13.79$
- -----
- $f_3 = g_3 \times h_3$
- $g_3 = \frac{4}{3}$, $f_3 = 1.451$, $h_3 = \frac{Z}{Y}$
- $Y = 12.67$

Continue...

- $f_2 = g_2 \times h_2$
- $g_2 = \frac{5}{3}, f_2 = 1.451, h_2 = \frac{Y}{X}$
- $X = 14.55$



Branching Effort

- ▶ This *branching effort* b is the ratio of the total capacitance seen by a stage to the capacitance on the path.

$$b = \frac{C_{\text{onpath}} + C_{\text{offpath}}}{C_{\text{onpath}}}$$

- ▶ The *path branching effort* B is the product of the branching efforts between stages.

$$B = \prod b_i$$

- ▶ we can define the path effort F as the product of the logical, electrical, and branching efforts of the path

$$F = GBH$$

Continue...

- ▶ The path delay **D** is the sum of the delays of each stage. It can also be written as the sum of the path effort delay **D_F** and path parasitic delay **P**.

$$\begin{aligned} D &= \sum d_i = D_F + P \\ D_F &= \sum f_i \\ P &= \sum p_i \end{aligned}$$

- ▶ If a path has N stages and each bears the same effort, that effort must be

$$\hat{f} = g_i h_i = F^{1/N}$$

- ▶ the minimum possible delay of an N -stage path with path effort F and path parasitic delay P is

$$D = NF^{1/N} + P$$

Continue...

- ▶ $f = gh$
- ▶ $h = \frac{C_{out_i}}{C_{in_i}}$
- ▶ Combining gives us the *capacitance transformation* formula to find the best input capacitance for a gate given the output capacitance it drives.

$$C_{in_i} = \frac{C_{out_i} \times g_i}{\hat{f}}$$

Summary

Term	Stage expression	Path expression
Logical effort	g	$G = \prod g_i$
Electrical effort	$h = C_{out}/C_{in}$	$H = C_{path-out}/C_{path-in}$
Branching effort	—	$B = \prod b_i$
Effort	$f = gh$	$F = GBH = \prod f_i$
Effort delay	f	$D_F = \sum f_i$
		minimized when $f_i = F^{1/\hat{N}}$
Number of stages	1	$\hat{N}$ (Table 1.3)
Parasitic delay	p	$P = \sum p_i$
Delay	$d = f + p$	$D = D_F + P$

Path branch

Q Calculate path effort (F) of a given circuit?

$$G = 1$$

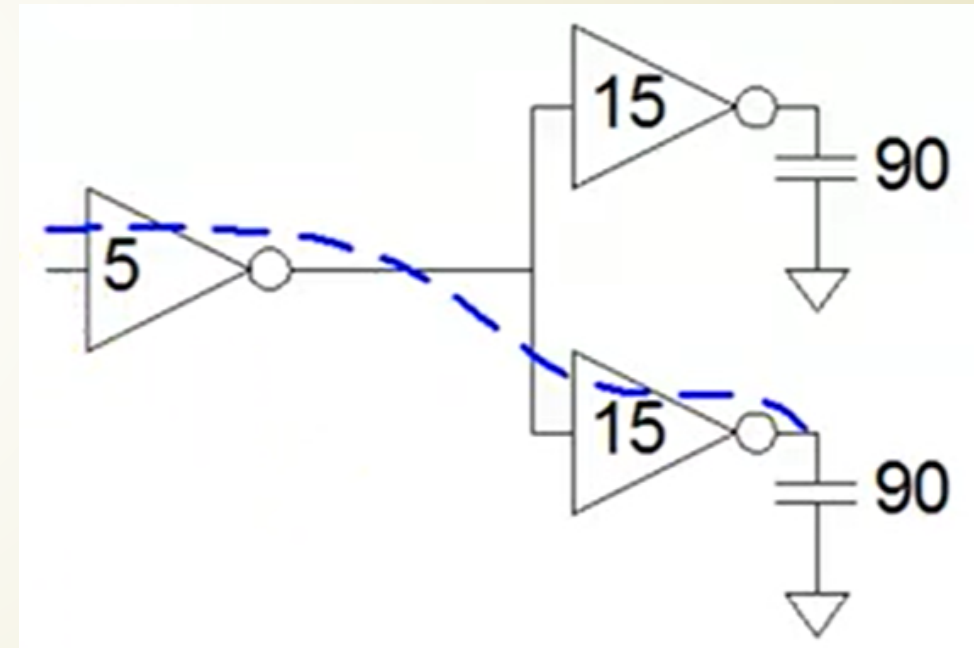
$$H = 90 / 5 = 18$$

$$GH = 18$$

$$h_1 = (15 + 15) / 5 = 6$$

$$h_2 = 90 / 15 = 6$$

$$F = g_1 g_2 h_1 h_2 = 36 = 2GH$$

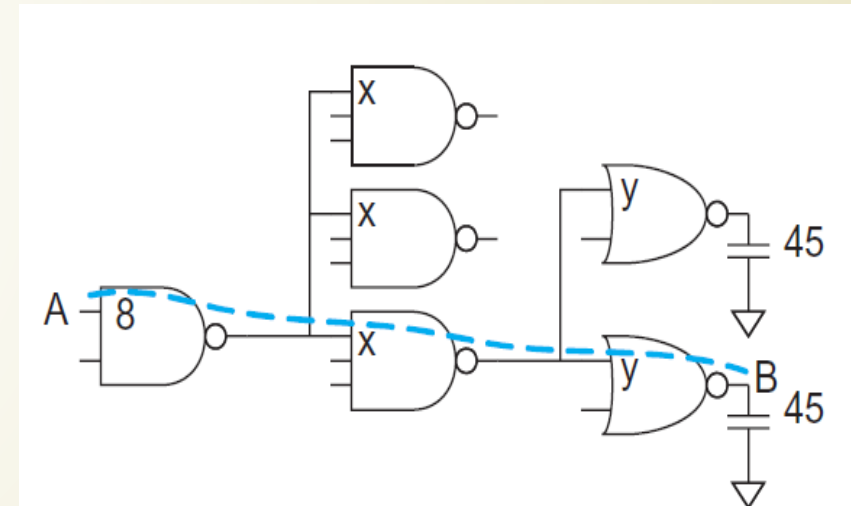


Problem

- Q) a) Estimate the minimum delay of the path from A to B
b) Choose the transistor size to achieve this delay.

Solution (a)

Logical Effort	$G = (4/3) * (5/3) * (5/3) = 100/27$
Electrical Effort	$H = 45/8$
Branching Effort	$B = 3 * 2 = 6$
Path Effort	$F = GBH = 125$
Best Stage Effort	$\hat{f} = \sqrt[3]{F} = 5$
Parasitic Delay	$P = 2 + 3 + 2 = 7$
Delay	$D = 3 * 5 + 7 = 22$



Continue...

- ▶ **Solution b**

- ▶ The gate sizes are computed with the capacitance transformation from working backward along the path

- ▶ $y = 45 \times (5/3)/5 = 15$

- ▶ $x = (15 + 15) \times (5/3)/5 = 10$

Problem

- Q) A ring oscillator is constructed from an odd number of inverters, as shown in Figure . Estimate the frequency of an N -stage ring oscillator by using path logical effort method ?

- Solution**

- Logical effort $G = 1$
- Electrical effort $H = 1$
- Branching effort $B = 1$
- Path effort $F = GBH = 1$
- Parasitic delay $P = 1$
- Delay $D = F + P = GH + P = 2$
- $$F = \frac{1}{2ND} = \frac{1}{2 \times 2 \times N} = \frac{1}{4N}$$

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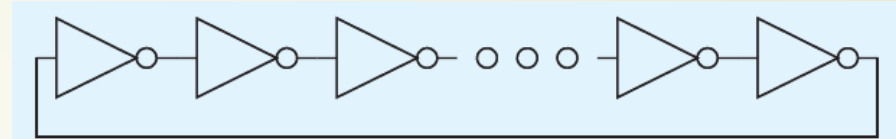
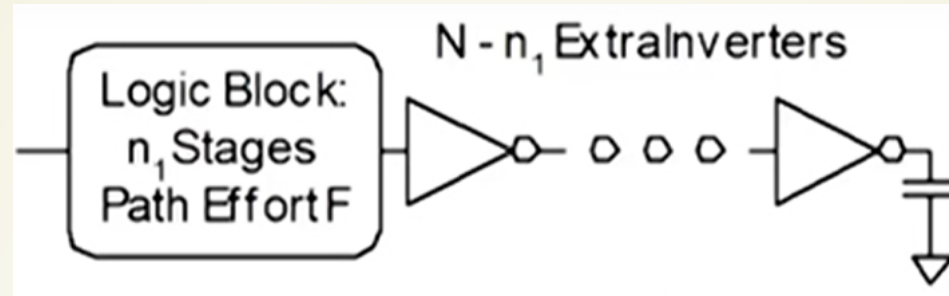


FIGURE Ring oscillator

Derivation for least possible delay for a logic block



- Consider the circuit containing n number of stages, n_1 is the number of stages are in the logic block and $N - n_1$ are the extra inverters. Least delay for the logic path will be

$$\hat{D} = NF^{1/N} + \left(\sum_{i=1}^{n_1} p_i \right) + (N - n_1)p_{inv}$$

- The first term is the delay obtained by distributing effort equally among the stage. The second term is the parasitic delay of the logic stages, and the third term is the parasitic delay of the inverters

Best stage effort

- Differentiating this expression with respect to N and setting the result to zero, we obtain:

$$\frac{\partial \hat{D}}{\partial N} = -F^{1/N} \ln(F^{1/N}) + F^{1/N} + p_{inv} = 0$$

- Let us define N is the number of stages and $\rho = F^{1/N}$ be the effort borne by each stages. The solution of the equation can be expressed as

$$p_{inv} + \rho(1 - \ln \rho) = 0$$

- Equation has no closed for solution
- For $p_{inv} = 1$ the value of $\rho = 3.59$.
- Therefore, the best stage effort will be nearly 4.

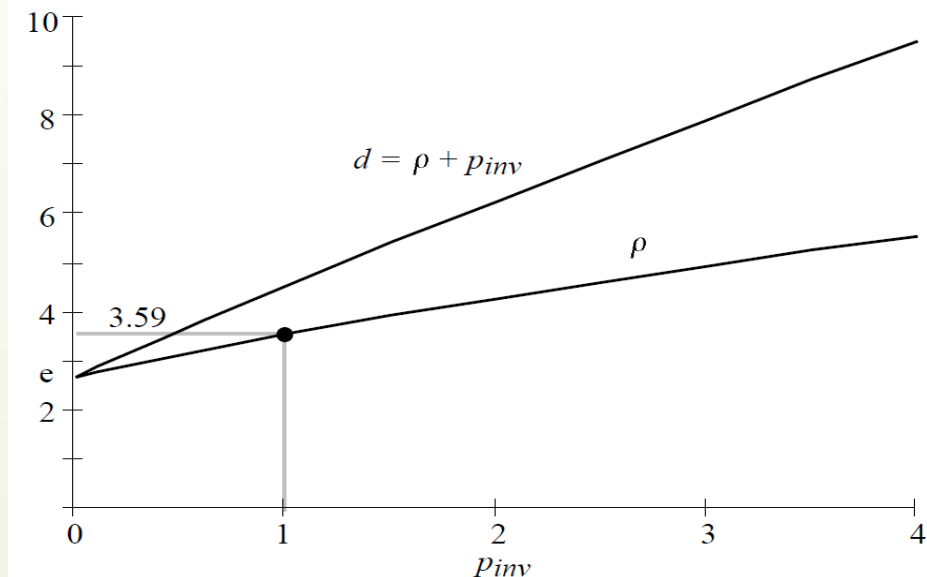


Figure Best effort per stage, ρ , and corresponding best stage delay $\rho + p_{inv}$, as a function of p_{inv} .

Asymmetric logic gates

we will analyze an example in which the logical effort differs for different inputs.

For example $x = ab\bar{c}$

AND –OR-INVERT

The logical effort per unit input

A $\longrightarrow \frac{4 + 2}{3} = 2$

B $\longrightarrow \frac{4 + 2}{3} = 2$

C $\longrightarrow \frac{4 + 1}{3} = \frac{5}{3}$

Total Logical effort = $2 + 2 + \frac{5}{3} = \frac{17}{3}$

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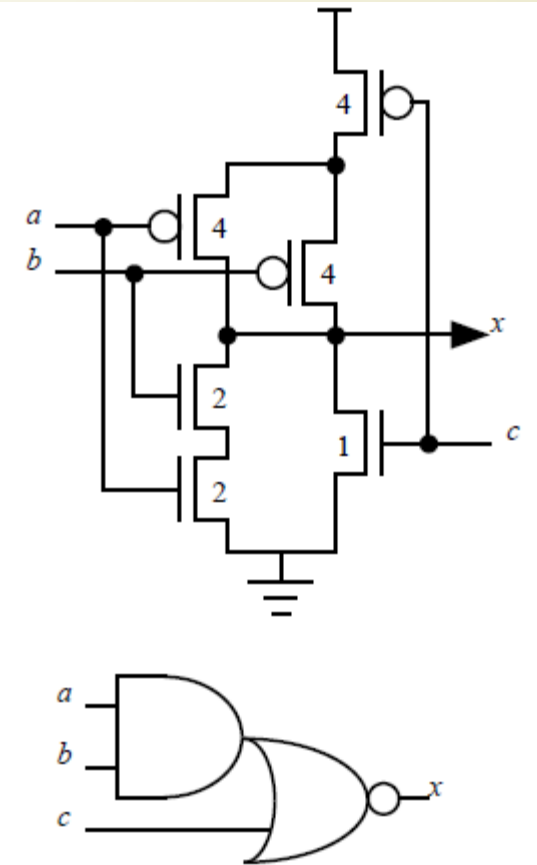


Figure An asymmetric and-or-invert gate.



Thank You

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